

AM69 Processor Starter Kit

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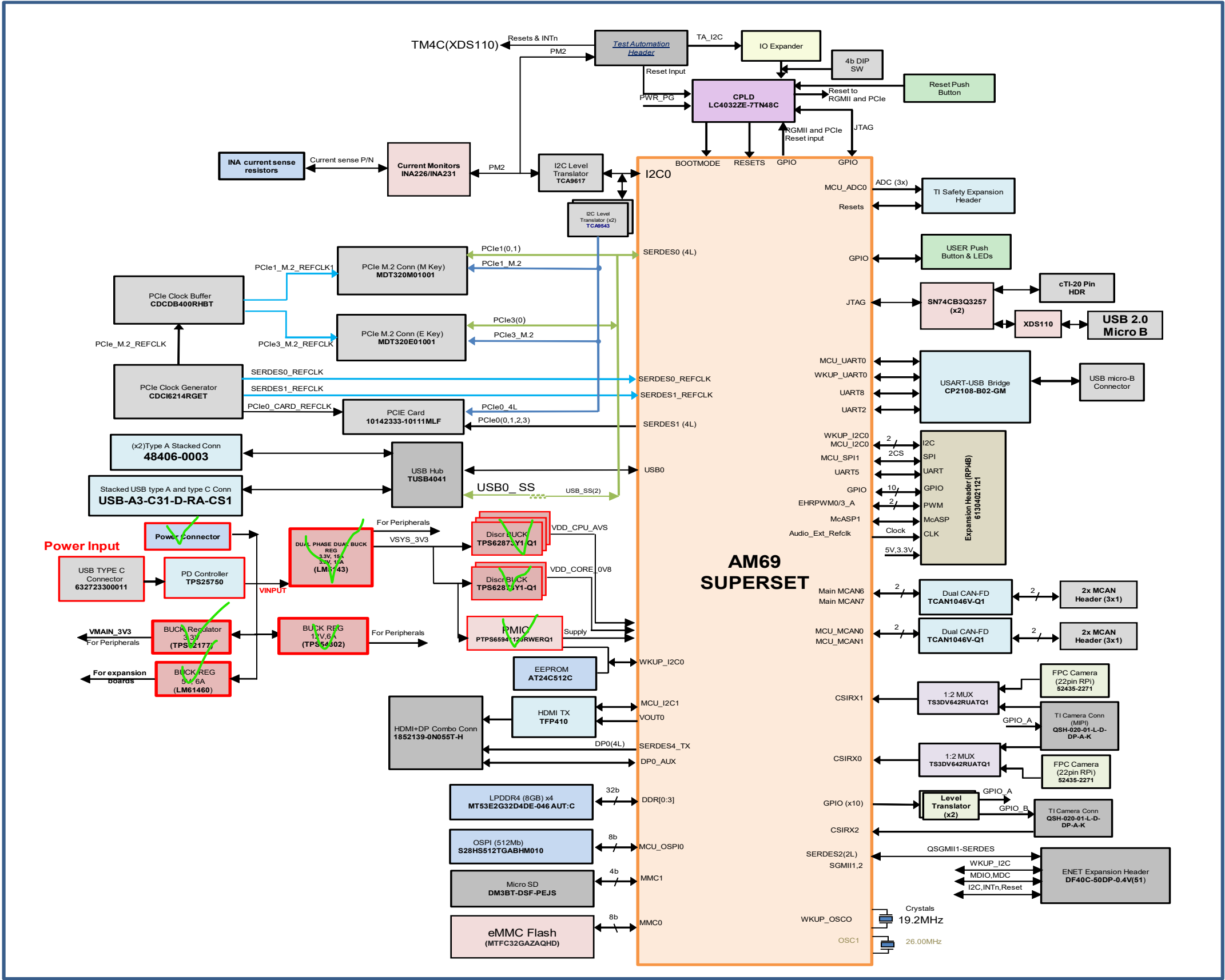
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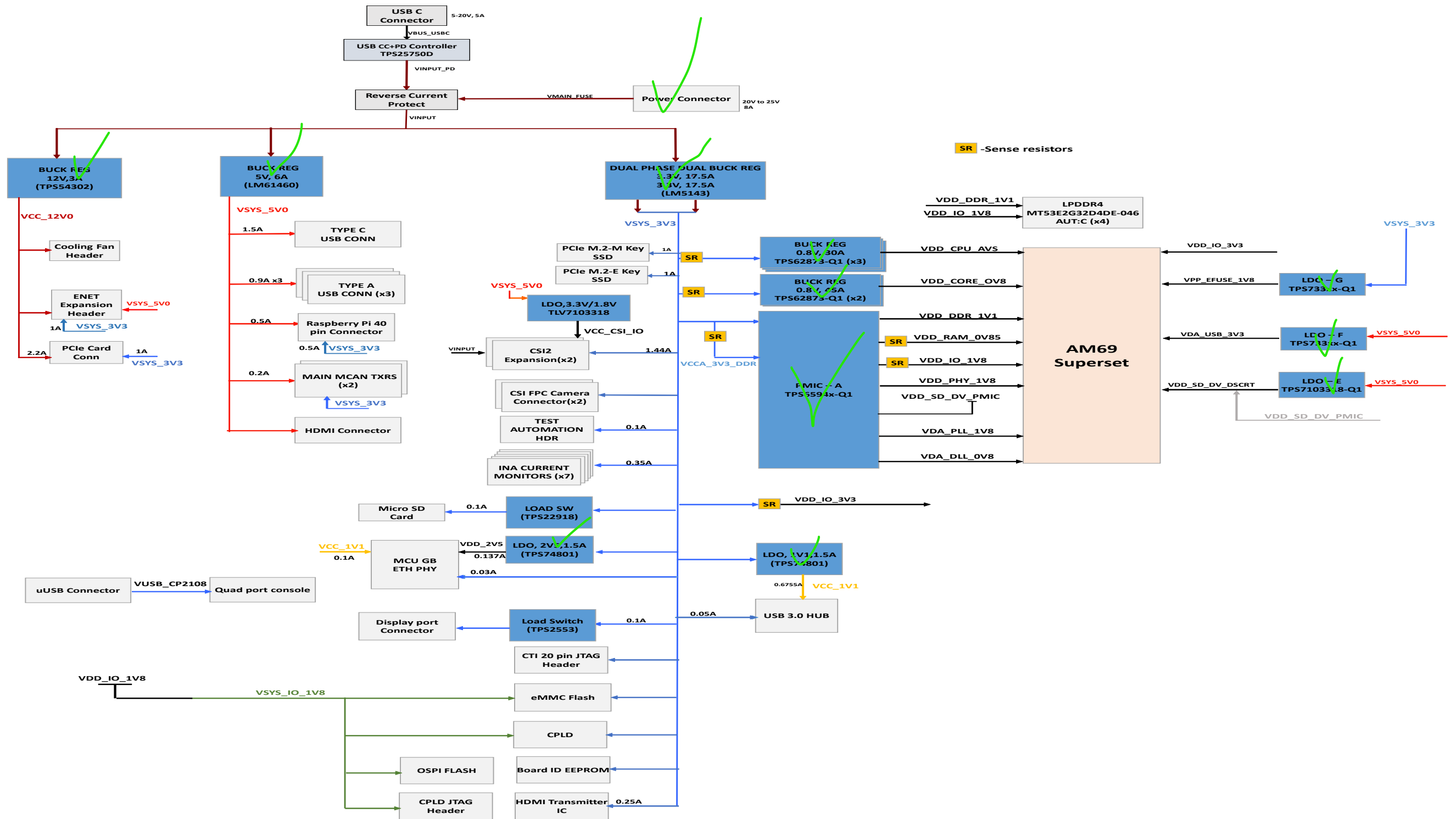
REVISION HISTORY

REV #	DATE	DESCRIPTION OF CHANGES	AUTHOR	REVIEWED BY	APPROVED BY
E2	21 OCT 2022	Taken AM69 SK Rev E1 design as reference and Added CDCI6214 clock generators for PCIe devices	Mistral Design Team		
	2 NOV 2022	1. Replaced U113 with clock buffer CDCDB400 as a clock source for PCIe M key and E key. 2. Updated U50 connection to provide resistor mux for CDCI1_OE2/OE3 and CDCI1_OE1/OE4 signals from U112	Mistral Design Team		
E2A	23 FEB 2023	Added ECN information for TIVA automation rework.	Mistral Design Team		
E2B	13 MAR 2023	1.Updated L1 L6 part number to XAL7070-122MEC 2.Updated U76 U78 part number to BSZ019N03LSATMA1	Mistral Design Team		
E3	16 MAR 2023	1.Updated L1 L6 part number to XAL7070-102MEC 2.Updated U76 U78 part number to CSD18563Q5A 3.Updated U75 U81 part number to CSD18543Q3A. 4. C36, R48,C42, R50 values have been modified. 5.TA_I2C_SCL , TA_I2C_SDA have been swapped with PM2_SCL, PM2_SDA. 6.Updated FB1, L28, R130, R368,R347,R267 to 0E for EMC compliance. 7.Made C801, C448, C441,C111 DNI.	Mistral Design Team		

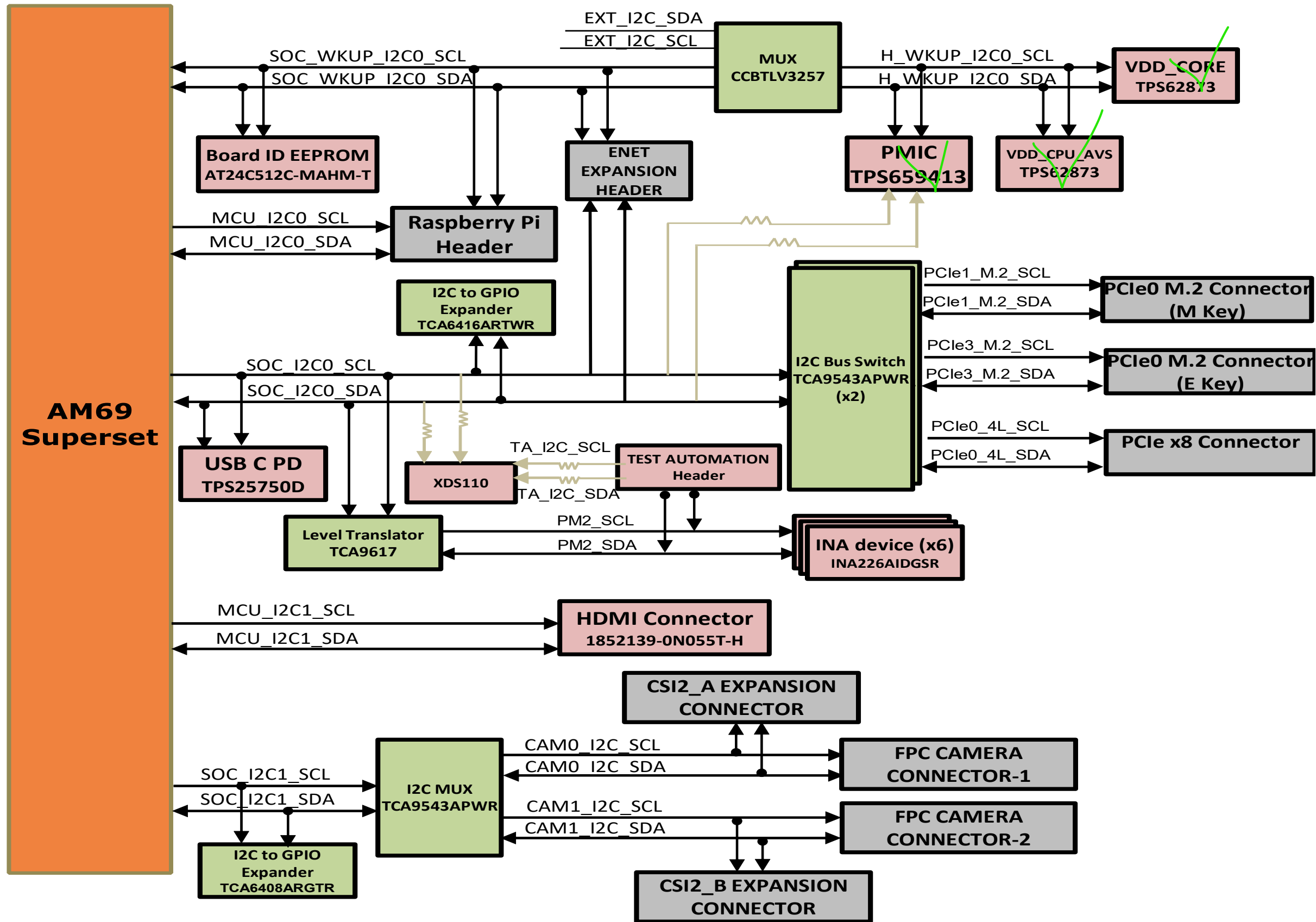
SYSTEM BLOCK DIAGRAM



AM69 SK POWER FLOW DIAGRAM



I2C TREE



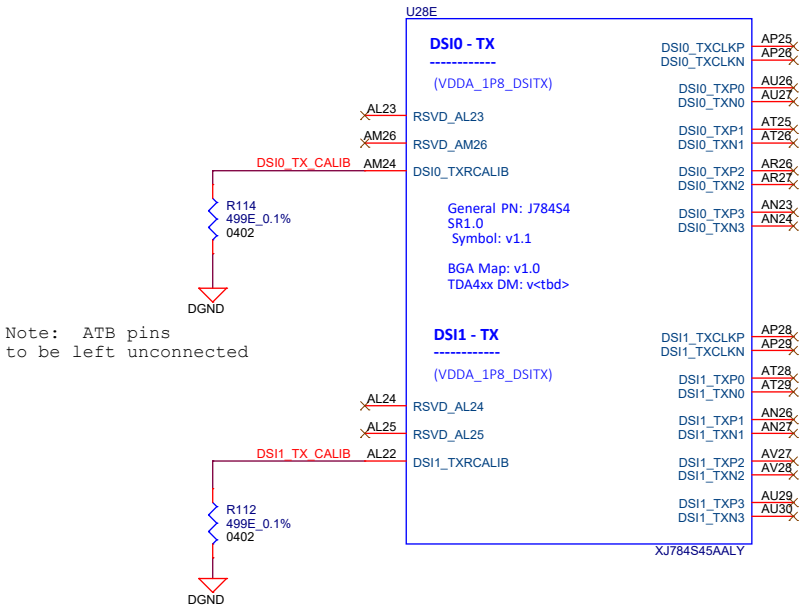
I2C TABLE

AM69 SK I2C Slave Address Table			
SOC I2C Port	Device Description	Part#	I2C Address
WKUP_I2C0	PMIC	TPS6594133ARWERQ1	0x48, 0x49, 0x4A & 0x4B
	VDD_CPU_AVS High-Current Power Stage A	TPS62873Y1QWRXSRQ1	0x40
	VDD_CORE_OV8 High-Current Power Stage B	TPS62873Y1QWRXSRQ1	0x43
	Raspberry Pi Header	61304021121	
	Board ID EEPROM	AT24C512C-MAHM-T	0x51
	ENET Expansion Header	171446-1109	0x57
MCU_I2C0	Raspberry Pi Header	61304021121	
MAIN_I2C0	Test Automation Header	687140183622	
	INA226 device for VCCA_3V3_CORE	INA226AIDGSR	0x45
	INA226 device for VCCA_3V3_CPU_AVS	INA226AIDGSR	0x4F
	INA226 device for VCCA_3V3_DDR	INA226AIDGSR	0x4D
	INA226 device for VDD_RAM_OV85	INA226AIDGSR	0x46
	INA226 device for VDD_IO_3V3	INA226AIDGSR	0x41
	INA226 device for VDD_IO_1V8	INA226AIDGSR	0x40
	PCIe_M.2_Interface M Key	MDT320M01001	
	PCIe_M.2_Interface E Key	MDT320E01001	
	Ext Power Measurement Header	61300311121	
	PCIe Card Slot	10018783-10202TLF	
	USB C PD Controller	TPS25750D	0x20
	Level Translator-1	TCA9543APWR	0x71
	Level Translator-2	TCA9543APWR	0x72
	GPIO Expander	TCA6416ARTWR	0x21
	PMIC	TPS6594133ARWERQ1	
	ENET Expansion Header	171446-1109	0x77
MCU_I2C1	HDMI Connector	1852139-0N055T-H	
MAIN_I2C1	FPC Camera Connector 1	52435-2271	
	FPC Camera Connector 2	52435-2271	
	CSI2_A Expansion Connector	QSH-020-01-L-D-DP-A-K	
	CSI2_B Expansion Connector	QSH-020-01-L-D-DP-A-K	
	GPIO Expander	TCA6408ARGTR	0x21
	Level Translator	TCA9543APWR	0x70

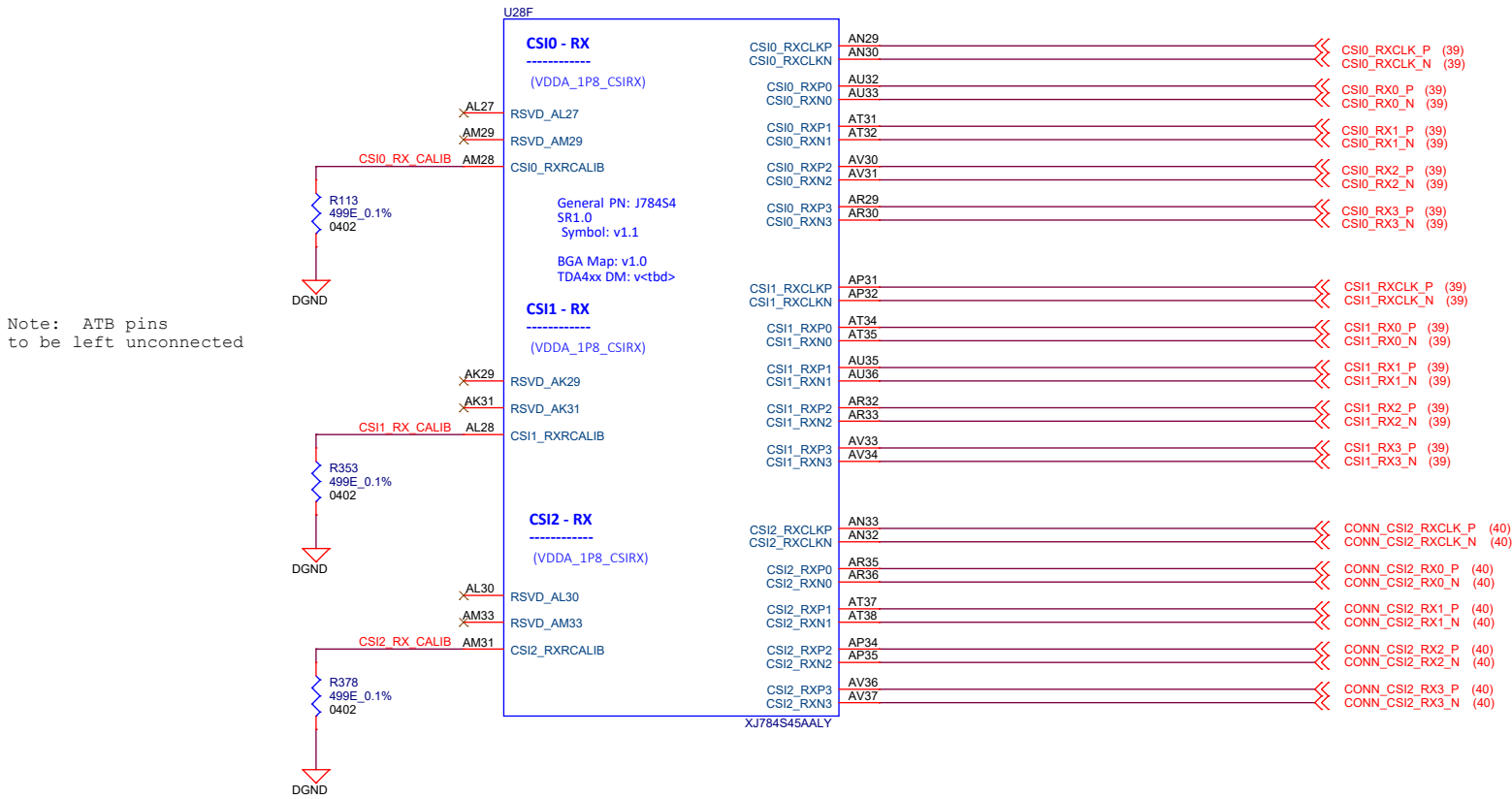
GPIO MAPPING TABLE

GPIO Mapping						
WKUP Domain						
J7AHP Mapping		Net Name	Input/Output	Default	State	Usage
Package Signal Name	GPIO					
MCU_OSPiO_CSn1	WKUP_GPIO0_28	EN_EFUSE_VPP	O	PD	Active High	Enable for VPP_EFUSE_1V8 LDO
MCU_OSPiO_CSn2	WKUP_GPIO0_29	CPLD_TMS/CSiB_EXP_GPIO1	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_CLK	WKUP_GPIO0_31	CPLD_TCK/CSiB_EXP_GPIO2	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_LBCLKO	WKUP_GPIO0_32	CSi_EXP_GPIO_1	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_DQS	WKUP_GPIO0_33	CPLD_TDO/CSiB_EXP_GPIO3	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_D0	WKUP_GPIO0_34	CPLD_TDI/CSiB_EXP_GPIO4	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_D1	WKUP_GPIO0_35	CSi_EXP_GPIO_5	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_D2	WKUP_GPIO0_36	CSi_EXP_GPIO_2	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_D3	WKUP_GPIO0_37	CSi_EXP_GPIO_3	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_CSn0	WKUP_GPIO0_38	CSi_EXP_GPIO_4	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_OSPi1_CSn1	WKUP_GPIO0_39	CSiB_EXP_GPIO5	IO	NA	NA	GPIO signal for CSi2 EXPANSION Connector
MCU_SPiO_CLK	WKUP_GPIO0_54	WKUP_GPIO0_54	O	Bootmode	Active High	Select line for CPLD's Mux
MCU_SPiO_D0	WKUP_GPIO0_55	USER_LED1	O	Bootmode	Active High	User LED
MCU_SPiO_D1	WKUP_GPIO0_69	SYS_MCU_PWRDN	O	Bootmode	Active High	System Power Down ('0' - normal operation '1' - system power down)
MCU_SPiO_CS0	WKUP_GPIO0_70	WKUP_GPIO0_70	IO	NA	NA	GPIO signal for FPC camera Connector
WKUP_GPIO0_10	WKUP_GPIO0_10	HDMI_LS_OE	O	PU	Active High	Level Shifter Output Enable for HDMI
WKUP_GPIO0_14	WKUP_GPIO0_14	HDMI_PDn	O	Bootmode	Active Low	Power Down Signal for HDMI
WKUP_GPIO0_49	WKUP_GPIO0_49	WKUP_GPIO0_49	IO	NA	NA	GPIO signal for 40 pin Expansion Header
PMIC_POWER_EN1	WKUP_GPIO0_88	WKUP_GPIO0_88	IO	NA	NA	GPIO signal for FPC camera Connector
WKUP_GPIO0_56	WKUP_GPIO0_56	WKUP_GPIO0_56	IO	NA	NA	GPIO signal for 40 pin Expansion Header
WKUP_GPIO0_57	WKUP_GPIO0_57	WKUP_GPIO0_57	IO	NA	NA	GPIO signal for 40 pin Expansion Header
MCU_ADC1_AIN0	WKUP_GPIO0_79	SOC_INT1z	I	PU	Active Low	Test Automation INT signal
MCU_ADC1_AIN1	WKUP_GPIO0_80	SOC_INT2z	I	PU	Active Low	Test Automation INT signal
MCU_ADC1_AIN2	WKUP_GPIO0_81	MCU_RGMII_INT#	I	PU	Active Low	Interrupt Signal from RGMII
MCU_ADC1_AIN3	WKUP_GPIO0_82	SOC_WAKE	I	PU	Active High	SOC wake signal from Reset Button
MCU_ADC1_AIN4	WKUP_GPIO0_83	PMIC_INTn	I	PU	Active Low	PMIC interrupt signal
MCU_ADC1_AIN5	WKUP_GPIO0_84	ENET1_EXP_INTB	I	NA	NA	Interrupt Signal from ENET Expansion Header
MCU_ADC1_AIN6	WKUP_GPIO0_85	IO_EXP_I2C0_INTB	O	NA	NA	I2C0 Interrupt Signal to ENET Expansion Header
WKUP_GPIO0_66	WKUP_GPIO0_66	WKUP_GPIO0_66	IO	PU	NA	GPIO signal for 40 pin Expansion Header
WKUP_GPIO0_67	WKUP_GPIO0_67	WKUP_GPIO0_67	IO	NA	NA	GPIO signal for 40 pin Expansion Header
Main Domain						
EXTINTn	GPIO0_0	HDMI_HPD	I	NA	Active High	HDMI hot plug detect signal
MCAN13_TX	GPIO0_3	GPIO0_3	IO	NA	NA	GPIO signal for 40 pin Expansion Header
MCAN13_RX	GPIO0_4	DPO_3V3_EN	O	PD	Active High	Enable signal for Display port Current Limiter
MCAN1_TX	GPIO0_27	GPIO0_27	IO	NA	NA	GPIO signal for 40 pin Expansion Header
MCASP0_AXR8	GPIO0_36	GPIO0_36	IO	NA	NA	GPIO signal for 40 pin Expansion Header
ECAP0_IN_APWM_OUT	GPIO0_49	SEL_SDIO_3V3_1V8n	O	PU	Active Low	One of Enable signal for VDD_SD_DV
GPIO Expander						
Port No	GPIO	I2C	Input/Output	Default	State	Usage
P0	CSi_VIO_SEL	MAIN_I2C1 Address : 0x21 Part No - TCA6408ARGTR	O	PD	Active High	Enable signal for Camera IO supply
P1	CSi_MUX_SEL_2		O	PD	Active High	Select lines for CSi mux
P2	CSi2_RSTz		O	PD	Active Low	Reset signal for CSi Expansion Connector
P3	IO_EXP_CAM0_GPIO1		IO	NA	NA	GPIO signals for FPC Camera Connector
P4	IO_EXP_CAM1_GPIO1		IO	NA	NA	GPIO signals for FPC Camera Connector
P00	BOARDID_EEPROM_WP	MAIN_I2C0 Address : 0x21 Part No - TCA6416ARTWR	O	PD	Active High	Board ID EEPROM Write Protect
P01	CAN_STB		O	PD	Active High	Stand By Input for CAN Transceiver
P02	GPIO_uSD_PWR_EN		O	PU	Active High	One of Enable signal for Micro SD Load Switch
P03	IO_EXP_MCU_RGMII_RST#		O	NA	Active Low	MCU_RGMII Resetz signal to CPLD
P04	IO_EXP_PClE0_4L_PERST#		O	NA	Active Low	PCle 4 lane Resetz signal to CPLD
P05	IO_EXP_PClE1_M.2_RTsz		O	NA	Active Low	PCle M Key Resetz signal to CPLD
P06	IO_EXP_PClE3_M.2_RTsz		O	NA	Active Low	PCle E Key Resetz signal to CPLD
P07	PM_INA_BUS_EN		O	PU	Active High	Enable signal for PM2 I2C lines
P10	ENET1_EXP_PWRDN		O	PU	Active High	Power Down Signal for Enet Expansion Header
P11	EXP1_ENET_RSTz		O	NA	Active Low	Reset Signal for Enet Expansion Header
P12	ENET1_I2CMUX_SEL		O	NA	Active High	I2C mux select Signal for Enet Expansion Header
P13	PCle0_CLKREQ#		I	PU	Active Low	PCle Card Clock request Signal
P14	PCle1_M.2_CLKREQ#		I	PU	Active Low	PCle M Key Clock request Signal
P15	PCle3_M2_CLKREQ#		I	PU	Active Low	PCle E Key Clock request Signal
P16	CDCi1_OE2/OE3		IO	PU	NA	GPIO signal CDCi Clock Generator
P17	CDCi1_OE1/OE4		IO	PU	NA	Output Enable for CDCi Clock Generator
P0	SW_CPLD_CONTROL_IN1	TEST AUTOMATION I2C Address : 0x20 Part No - TCA6408ARGTR	O	NA	NA	CPLD Switch Control Signals for Bootmode Logic
P1	SW_CPLD_CONTROL_IN2		O	NA	NA	CPLD Switch Control Signals for Bootmode Logic
P2	SW_CPLD_CONTROL_IN3		O	NA	NA	CPLD Switch Control Signals for Bootmode Logic
P3	SW_CPLD_CONTROL_IN4		O	NA	NA	CPLD Switch Control Signals for Bootmode Logic

DSI



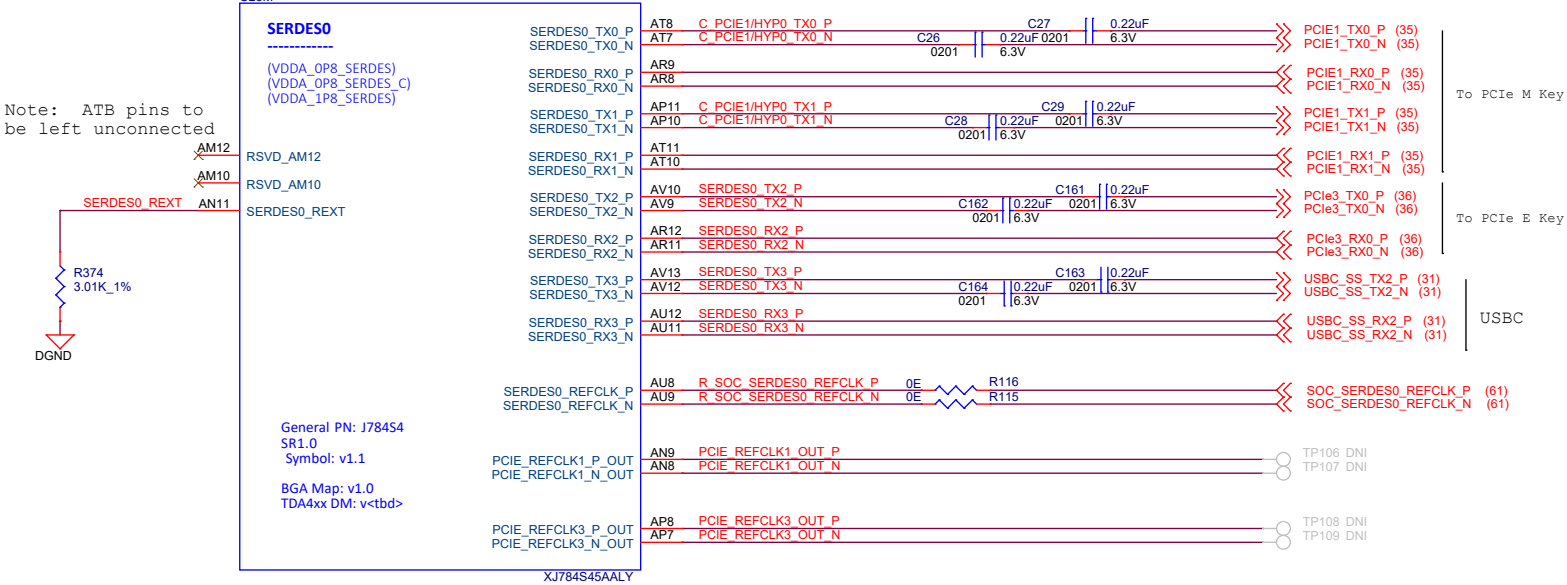
CSI



SERDES0

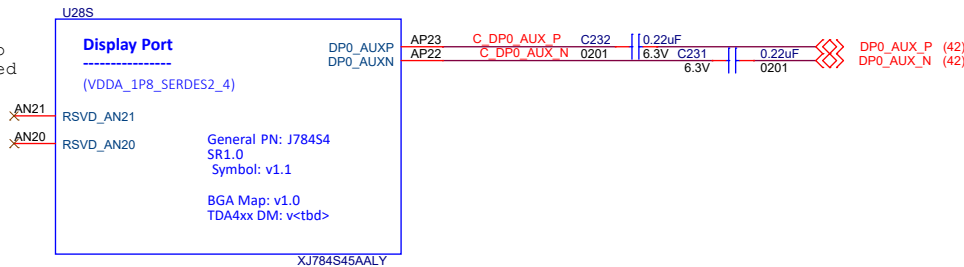
Note: Place DC blocking caps near PCIe Connector

Note: ATB pins to be left unconnected



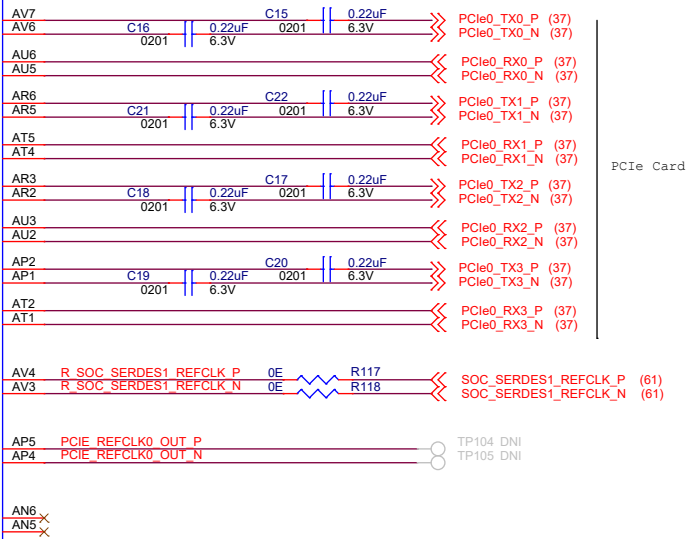
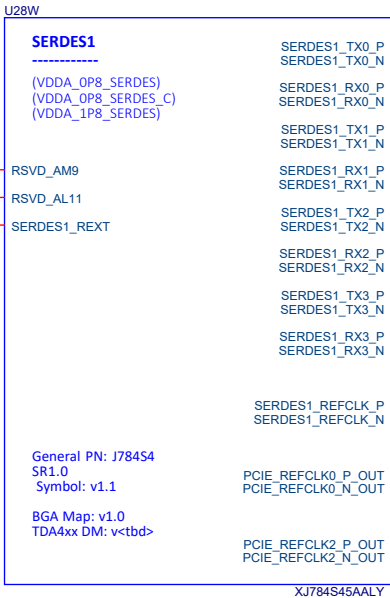
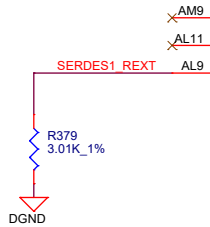
DP_AUX

Note: ATB pins to be left unconnected



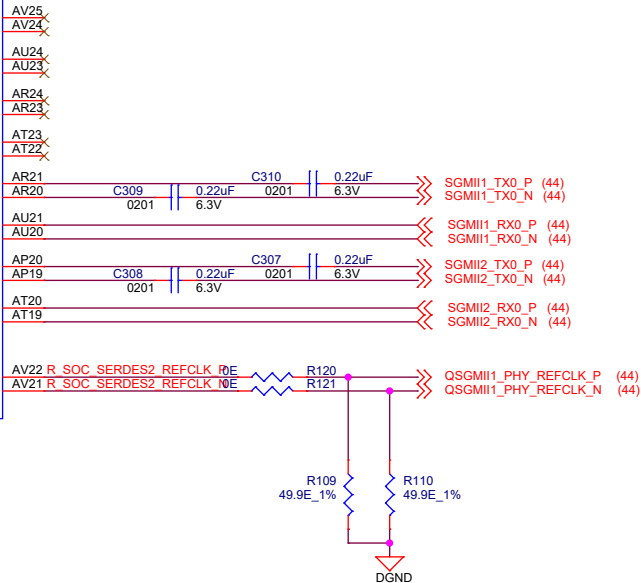
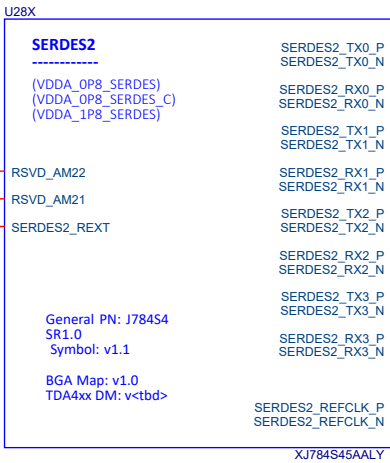
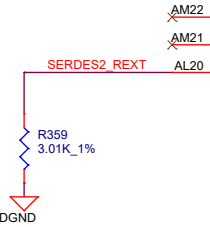
SERDES1

Note: ATB pins to be left unconnected



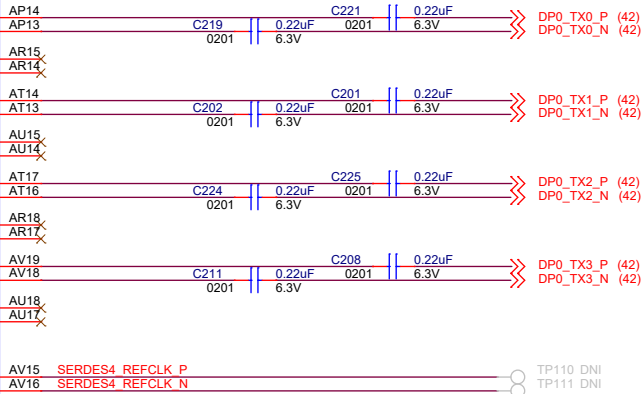
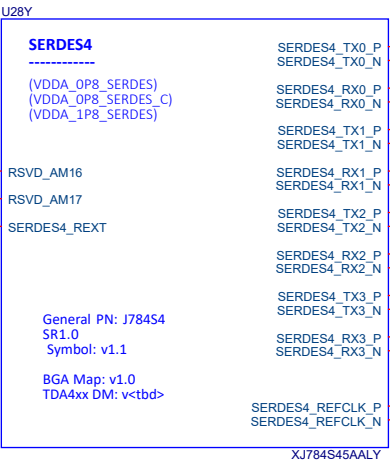
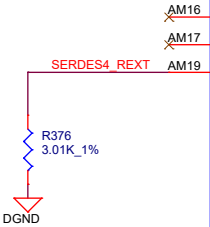
SERDES2

Note: ATB pins to be left unconnected



SERDES4

Note: ATB pins to be left unconnected



Project :
AM69 Edge AI Kit



Title SERDES INTERFACE-2

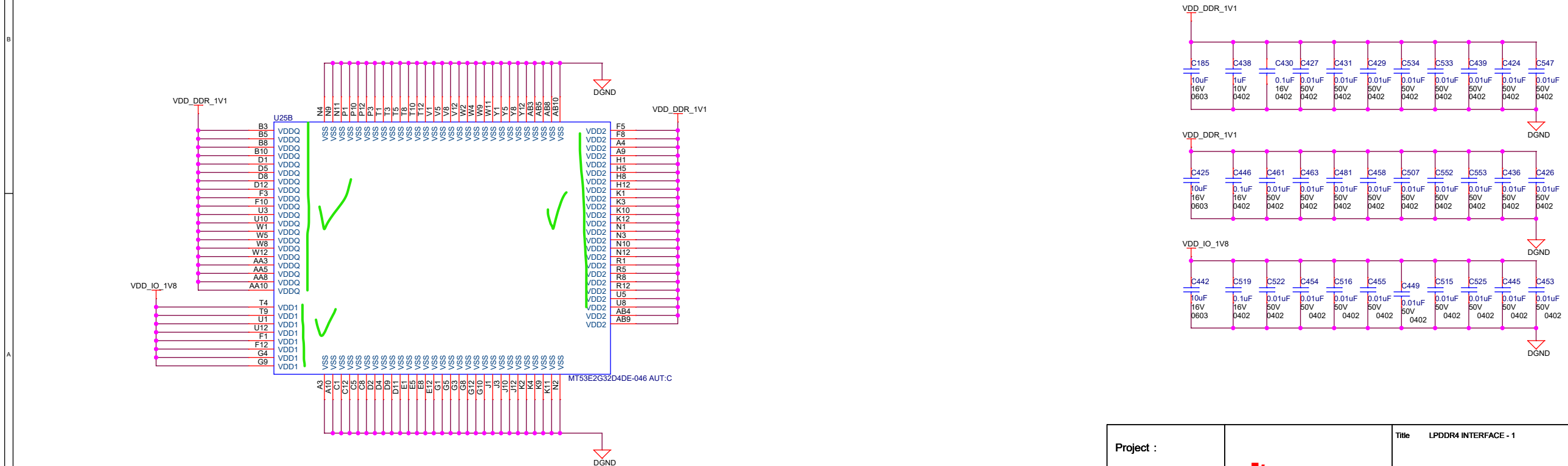
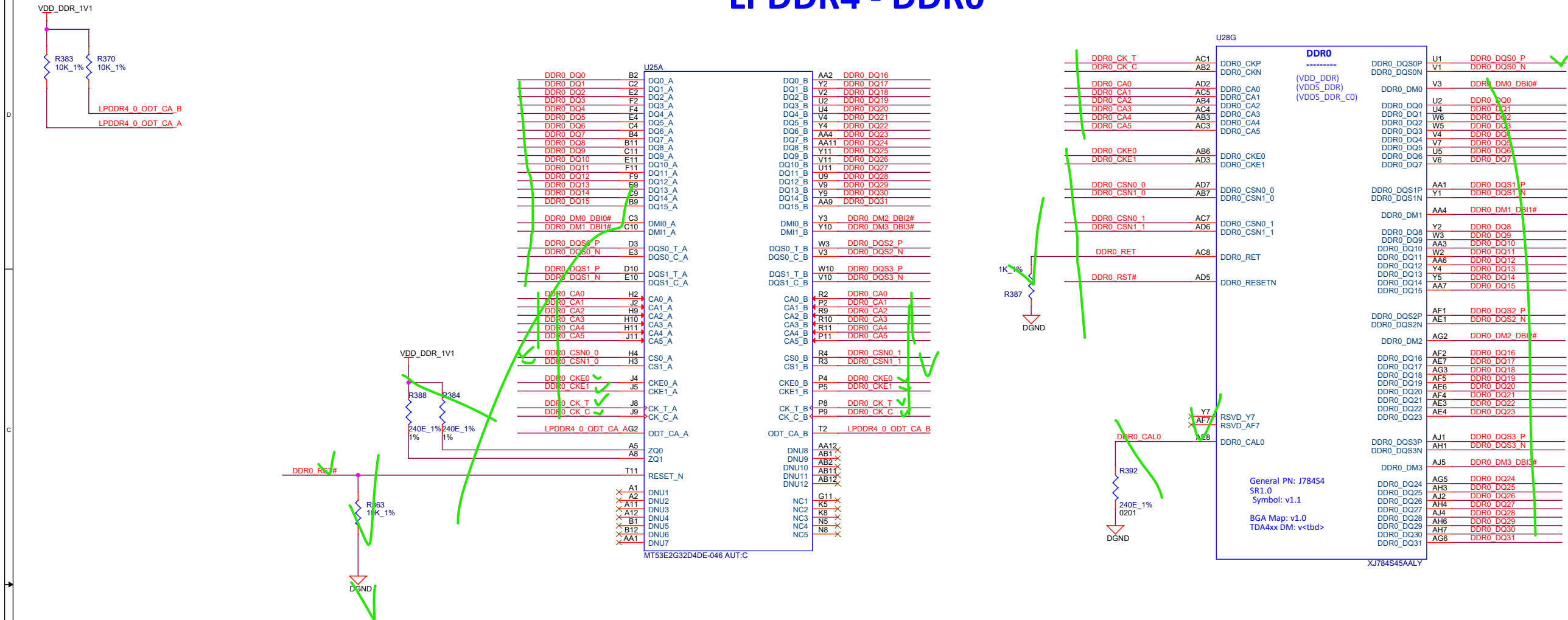
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Rev E3

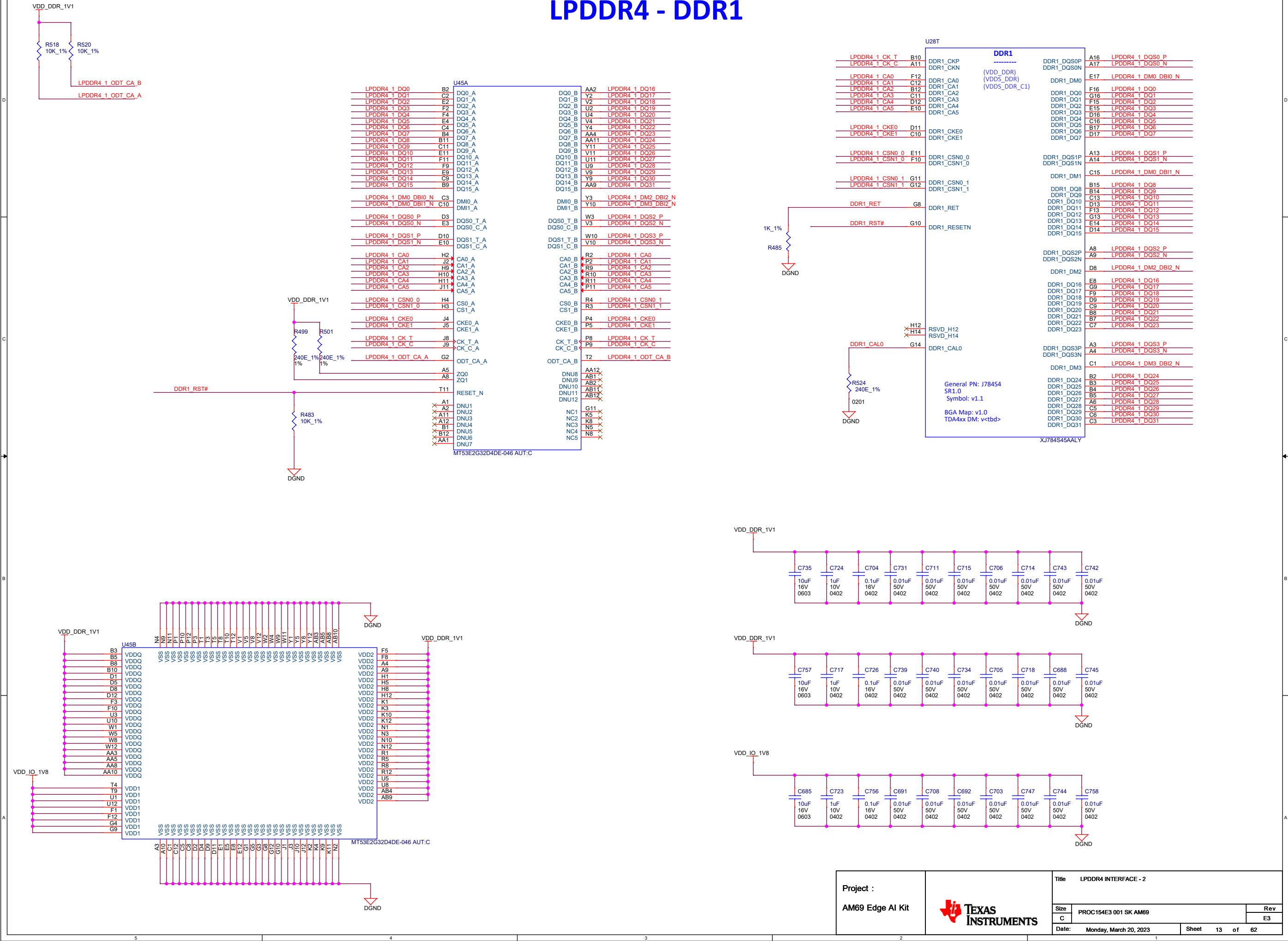
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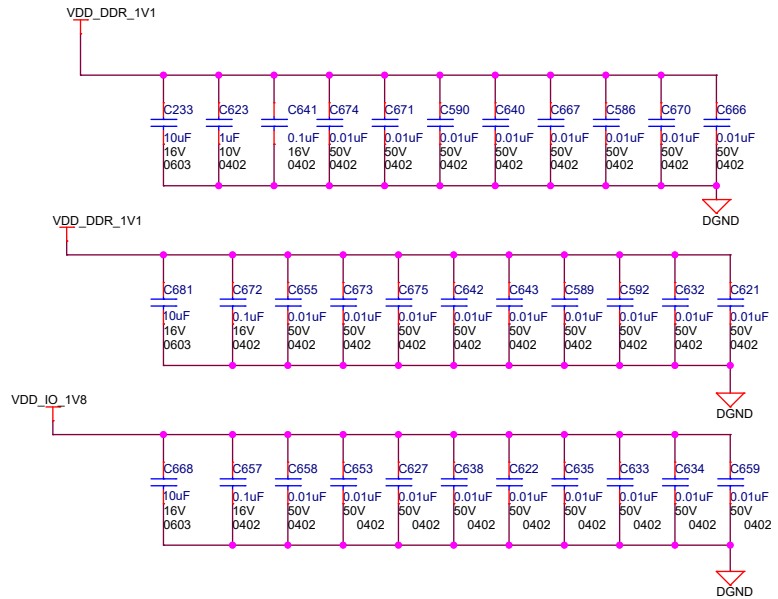
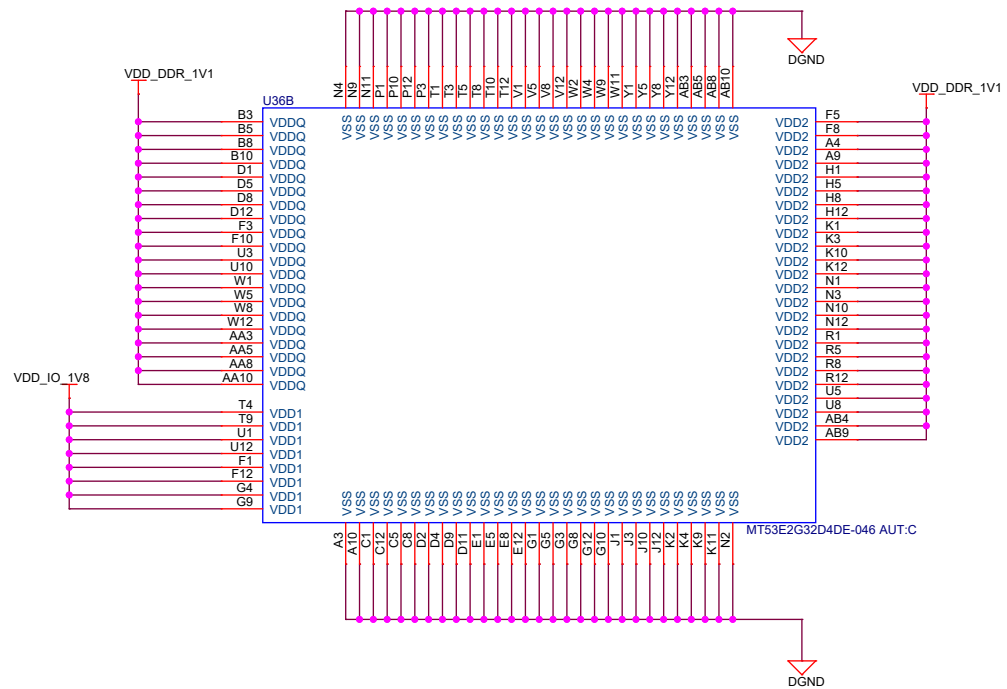
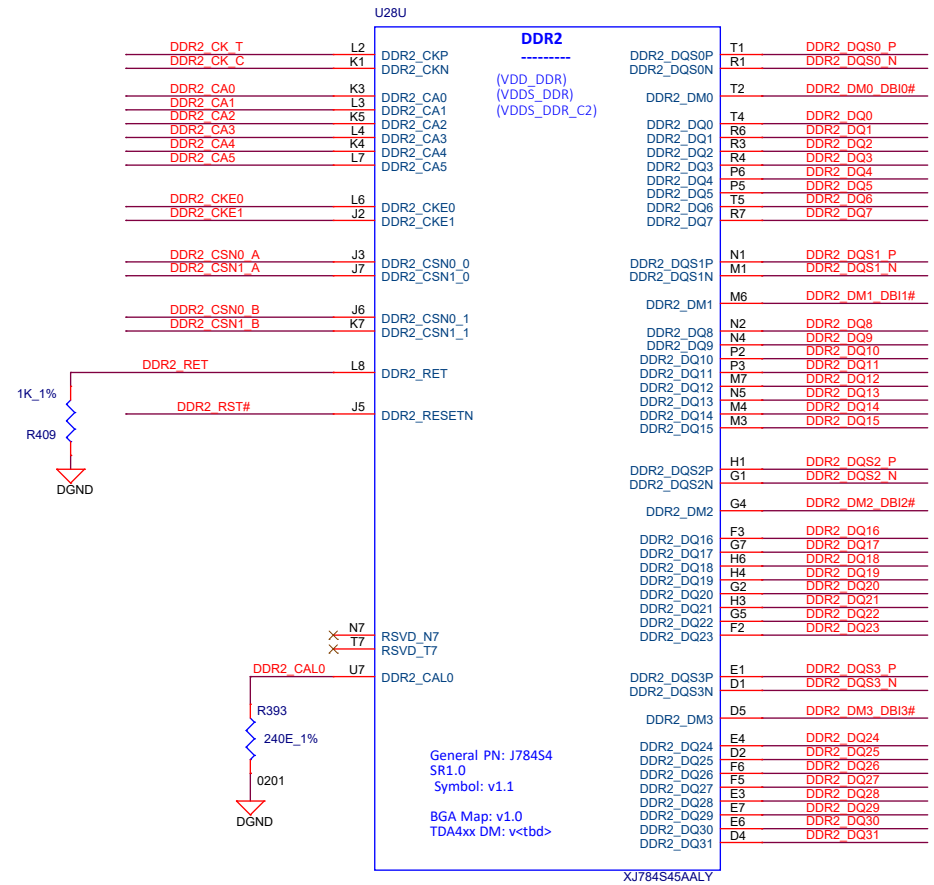
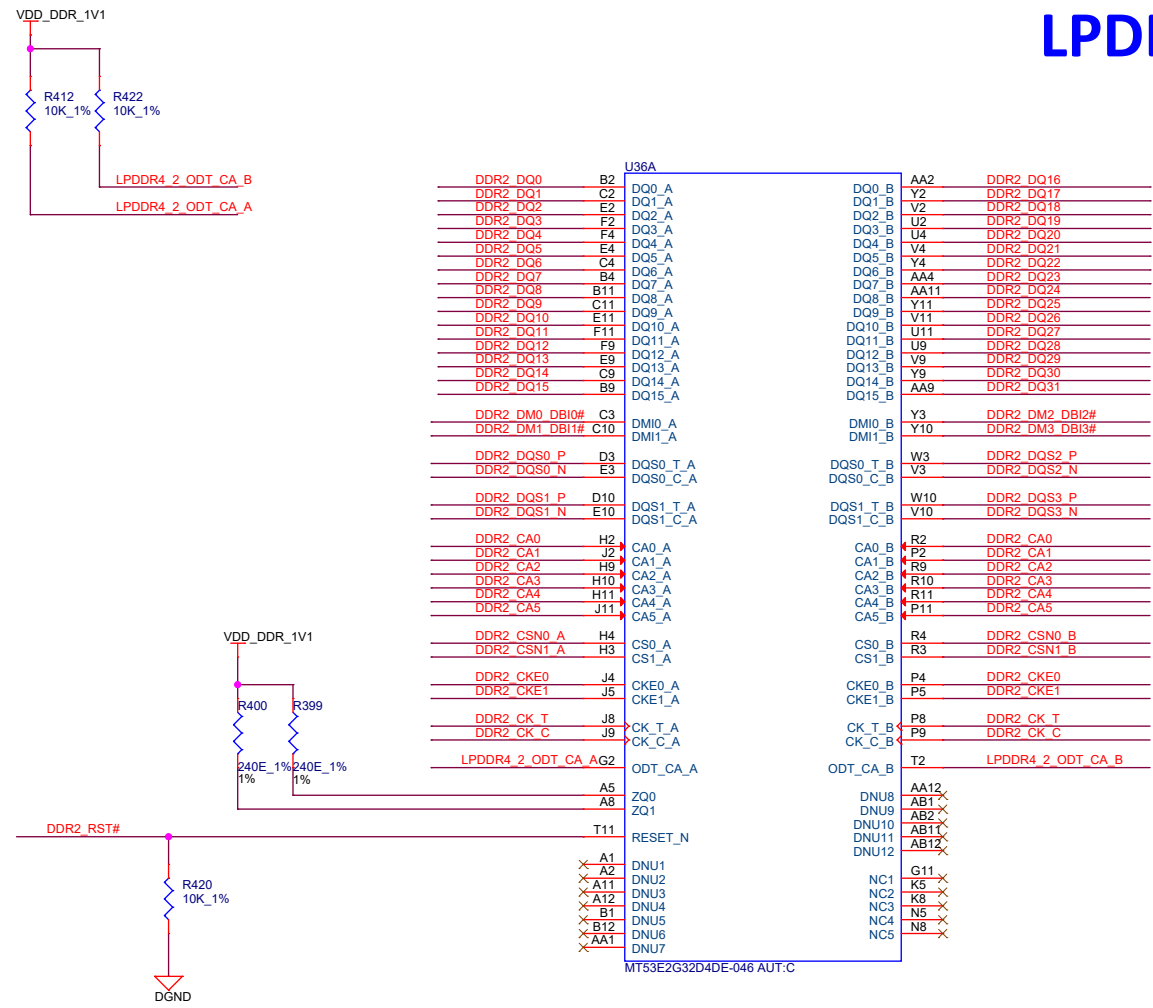
LPDDR4 - DDR0



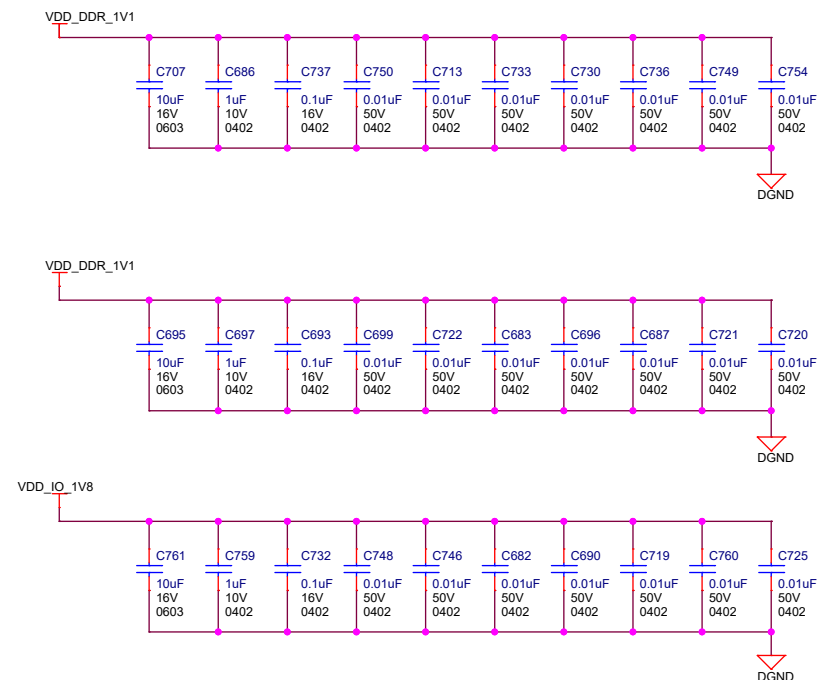
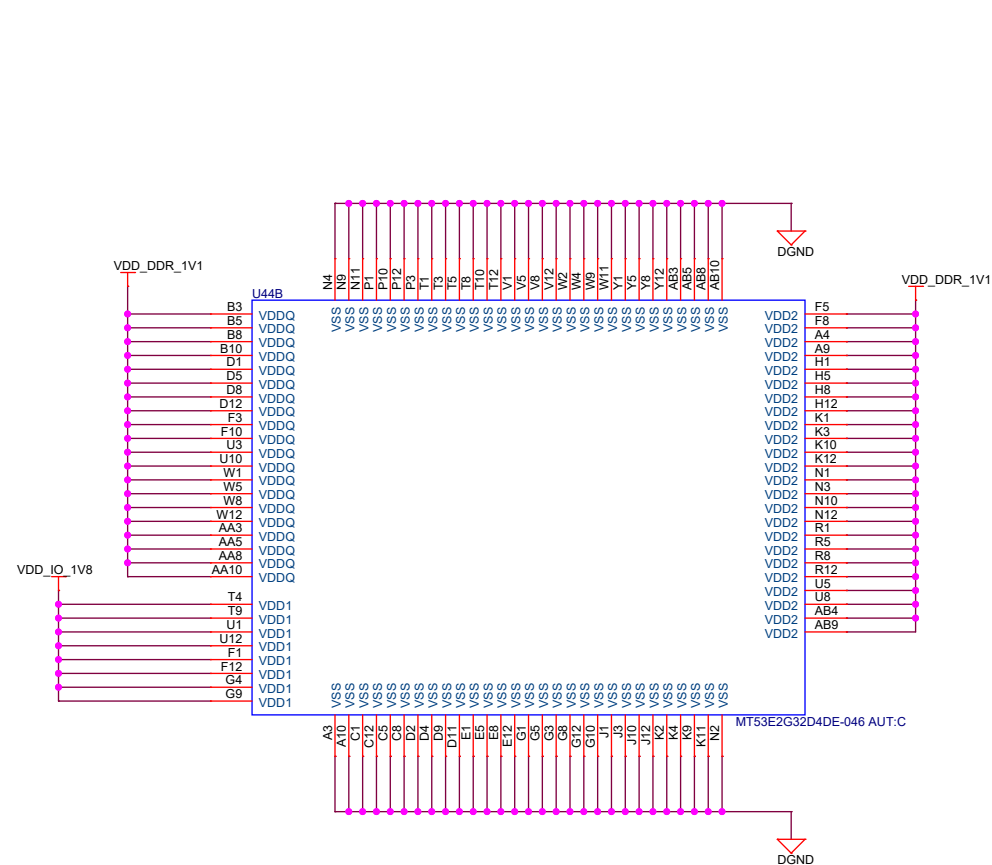
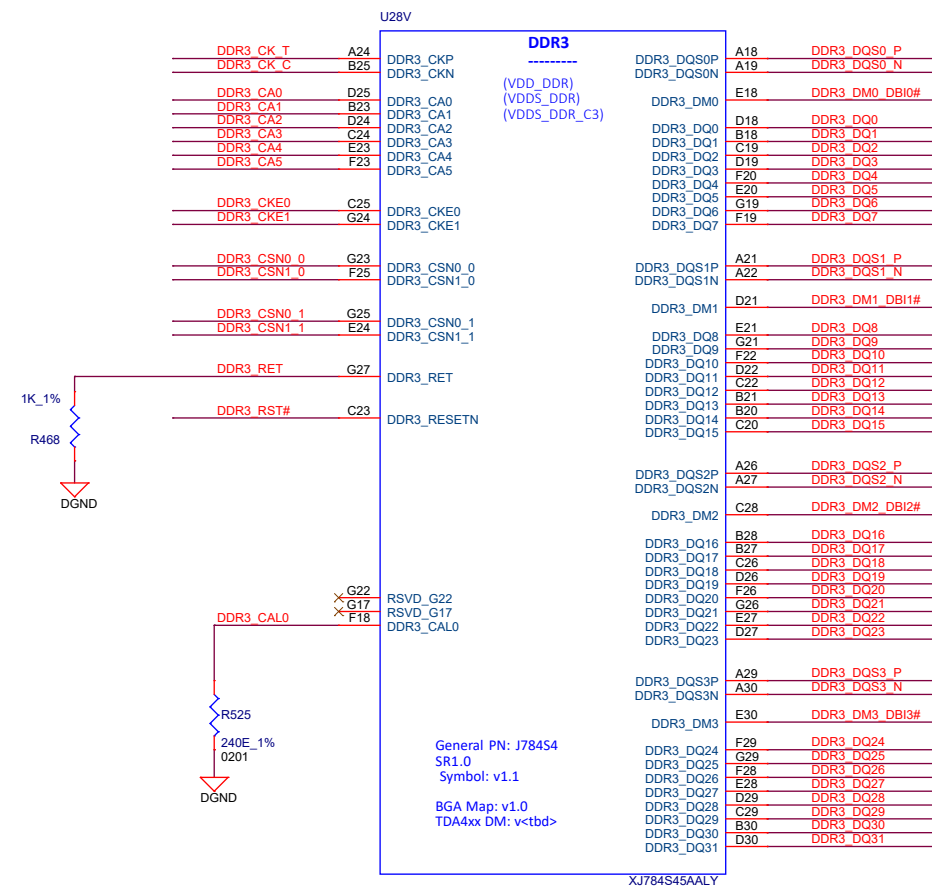
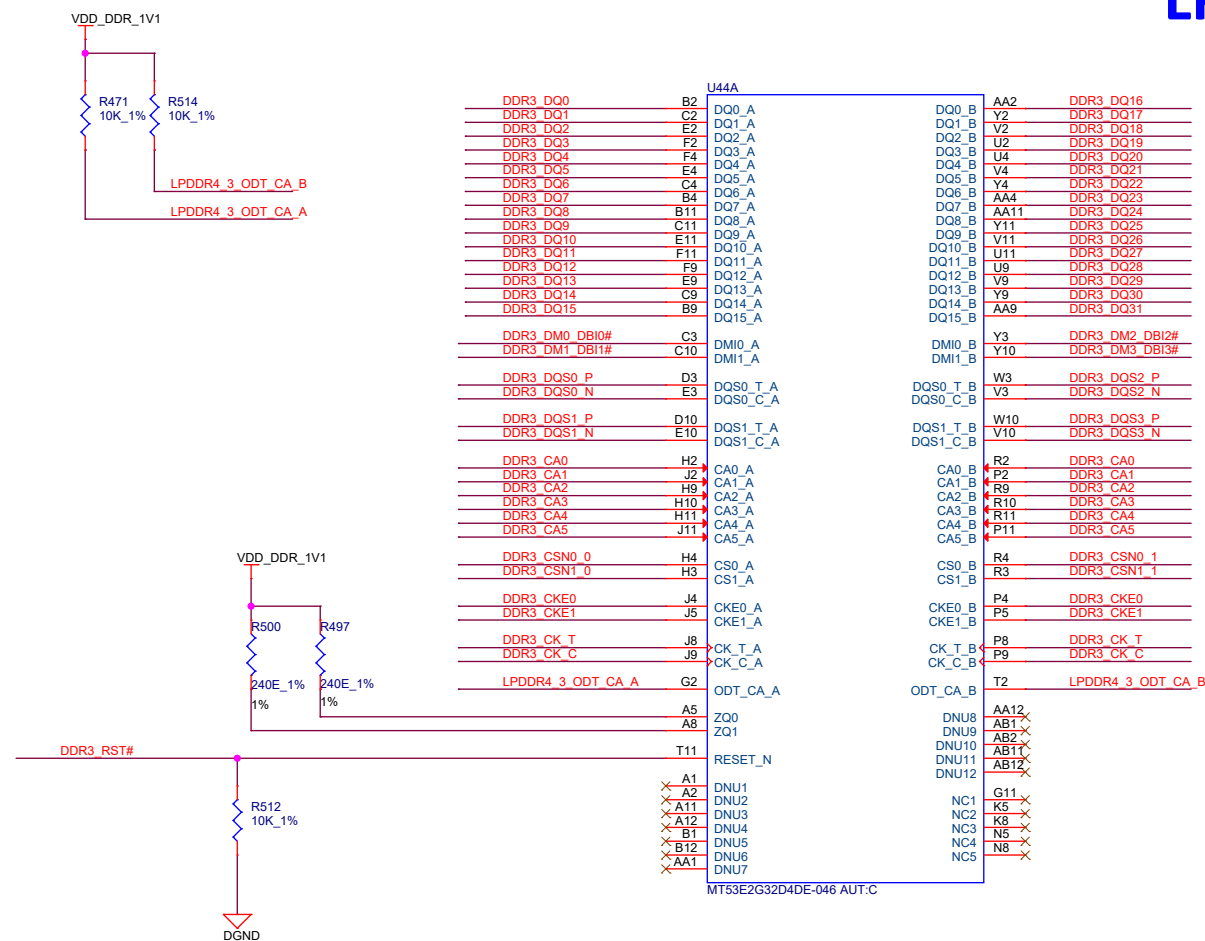
LPDDR4 - DDR1



LPDDR4 - DDR2

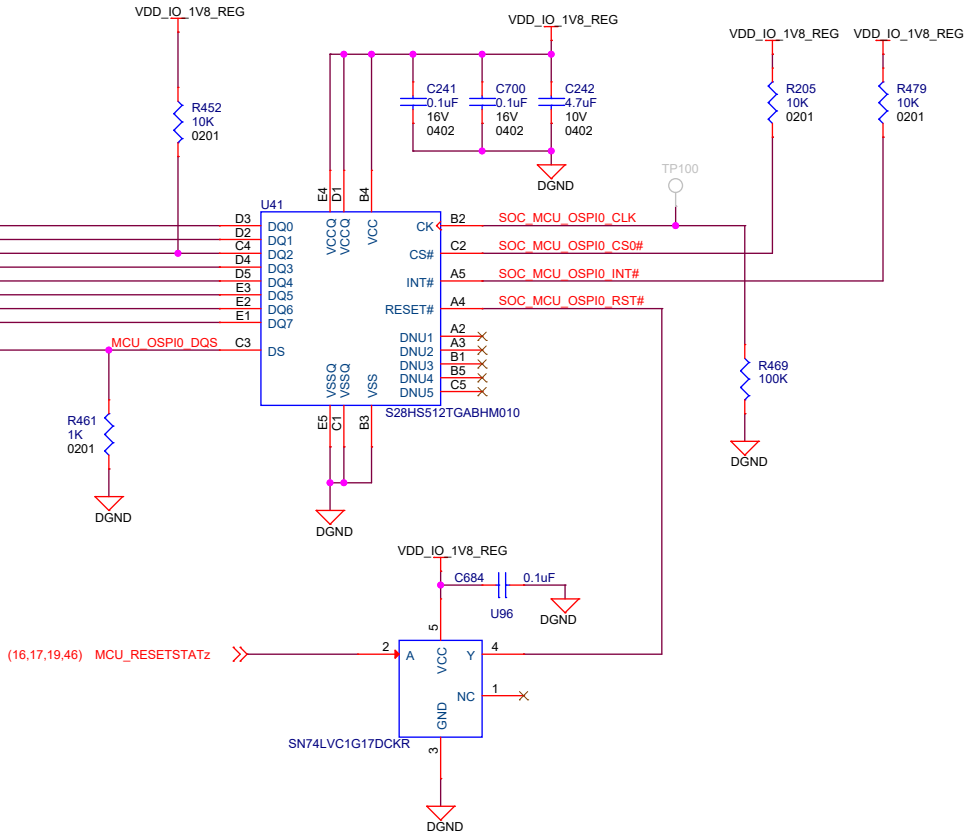
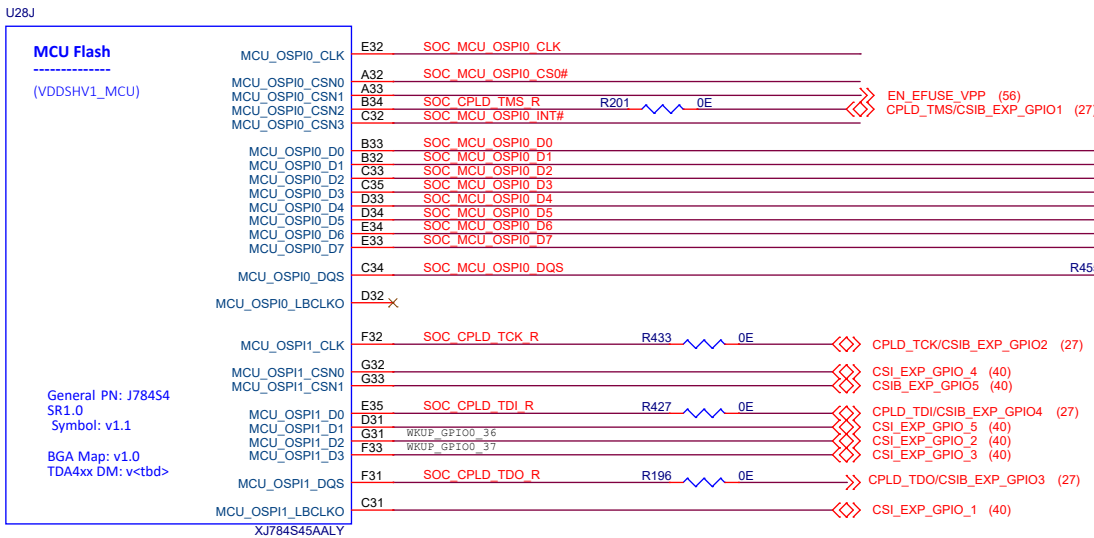


LPDDR4 - DDR3



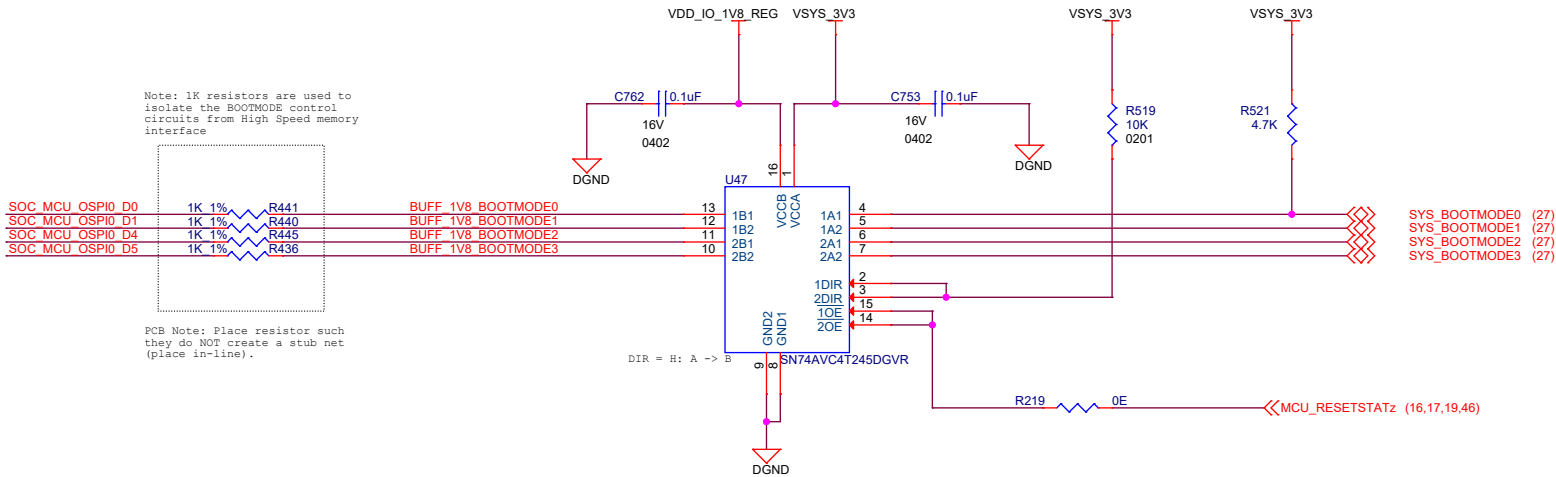
MCU FLASH

OSPI FLASH

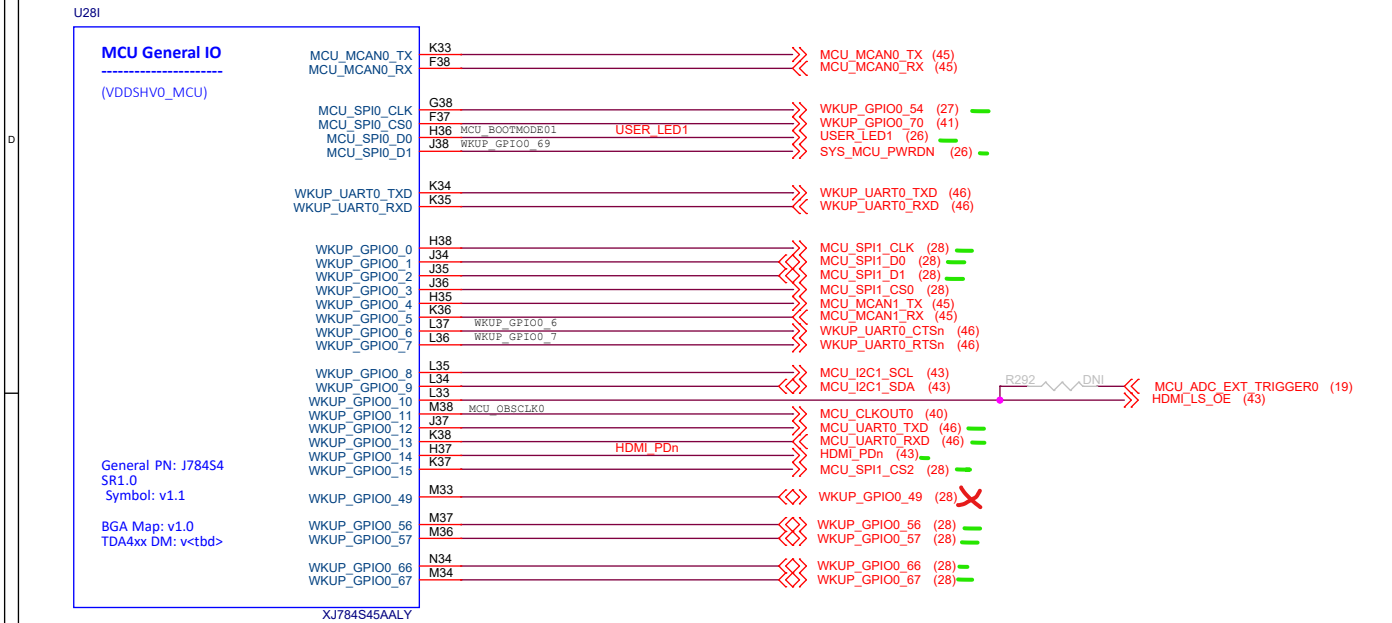


BOOTMODE Control Logic

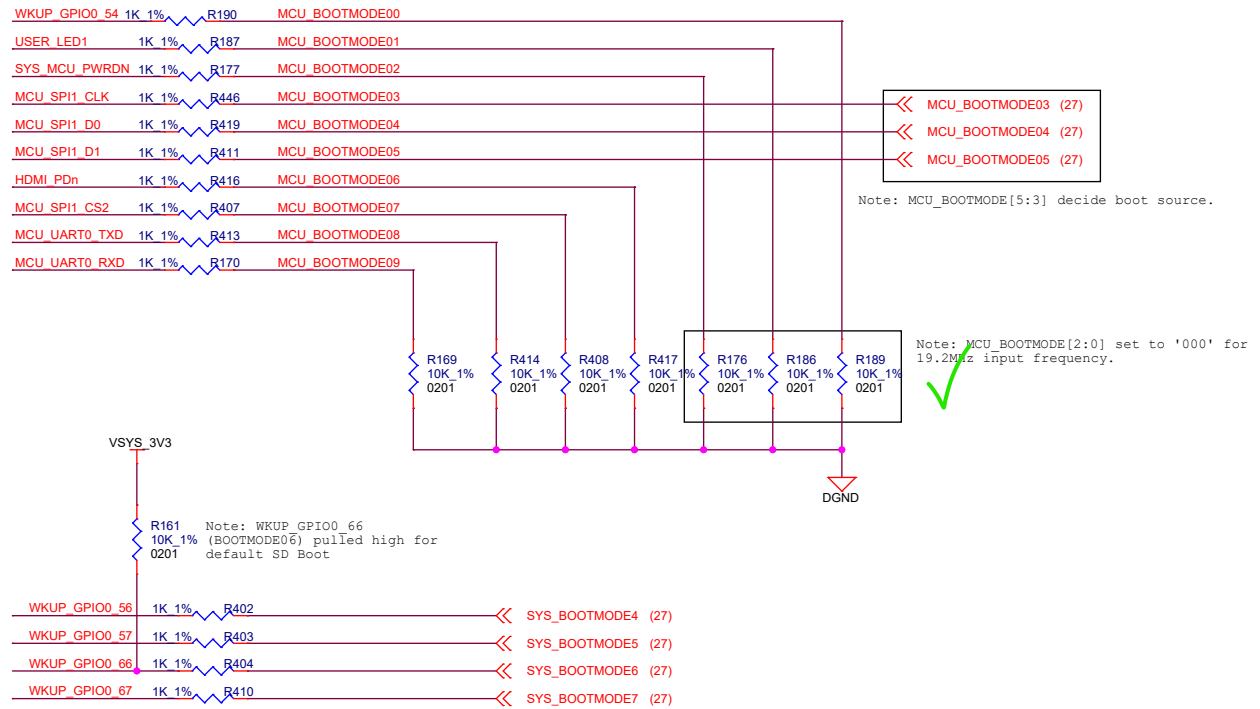
Note: Logic used to configure BOOTMODE settings during reset. This is four (4) of a total of eighteen (18) boot pins. Specific value is user configured (dip switch).



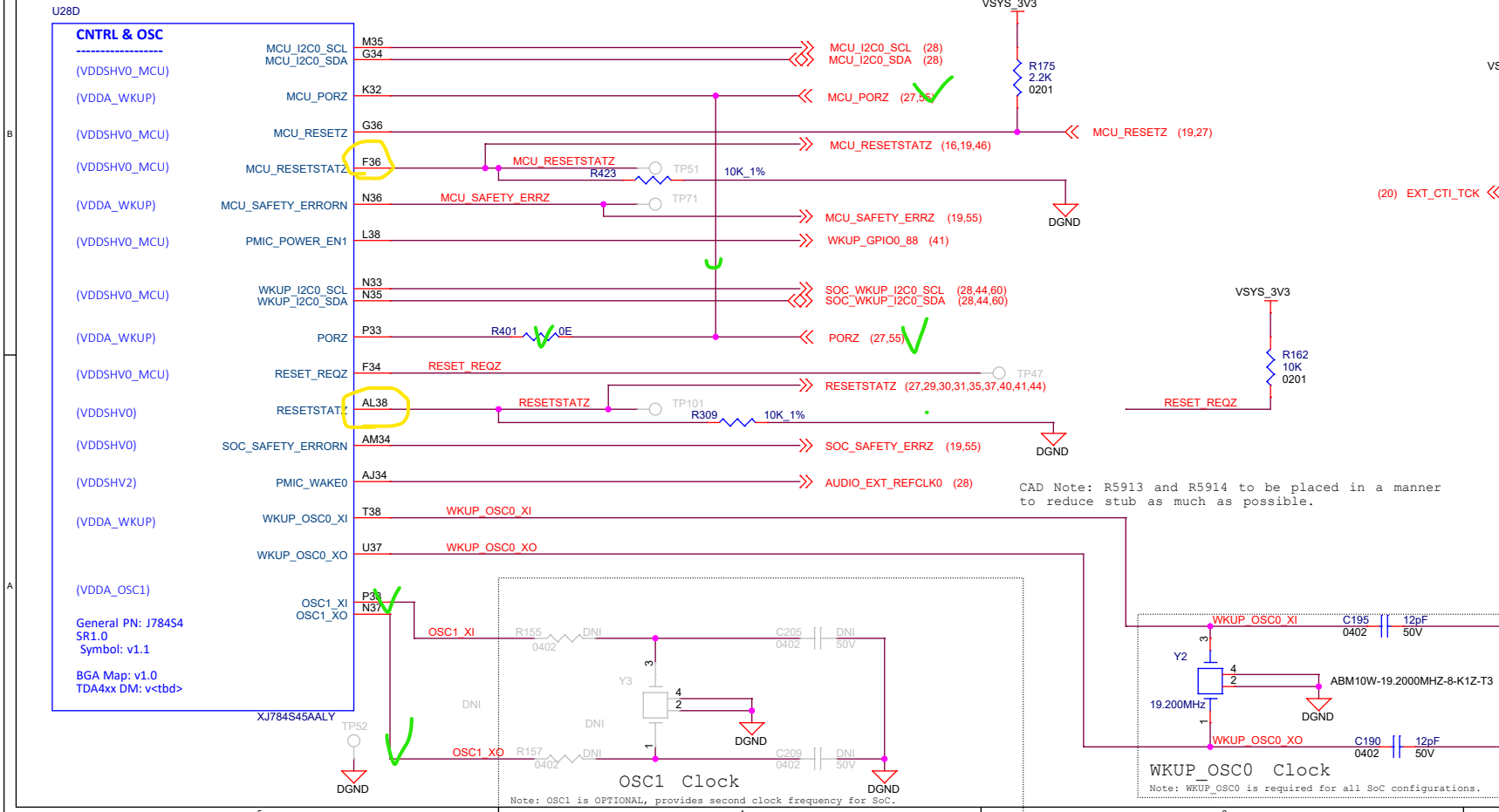
MCU & MAIN GENERAL IO, OSC CLKS



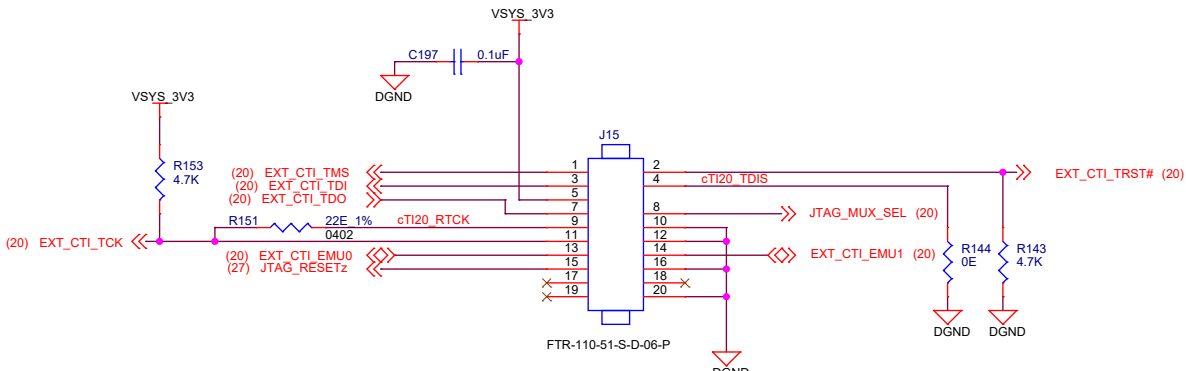
BOOTMODE



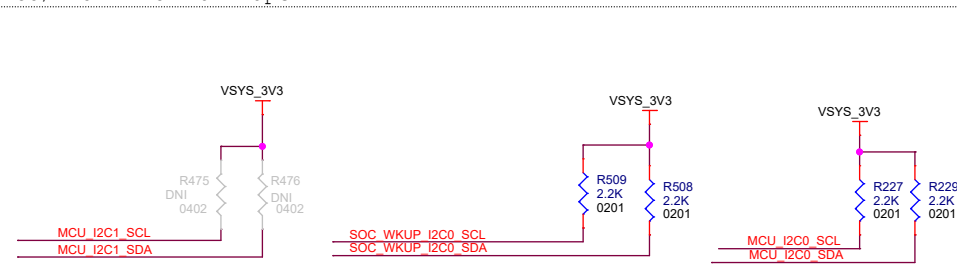
CONTROL & OSC



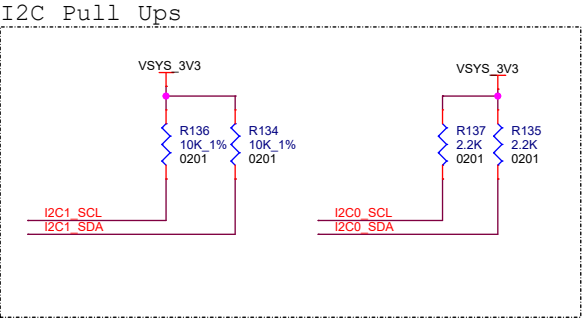
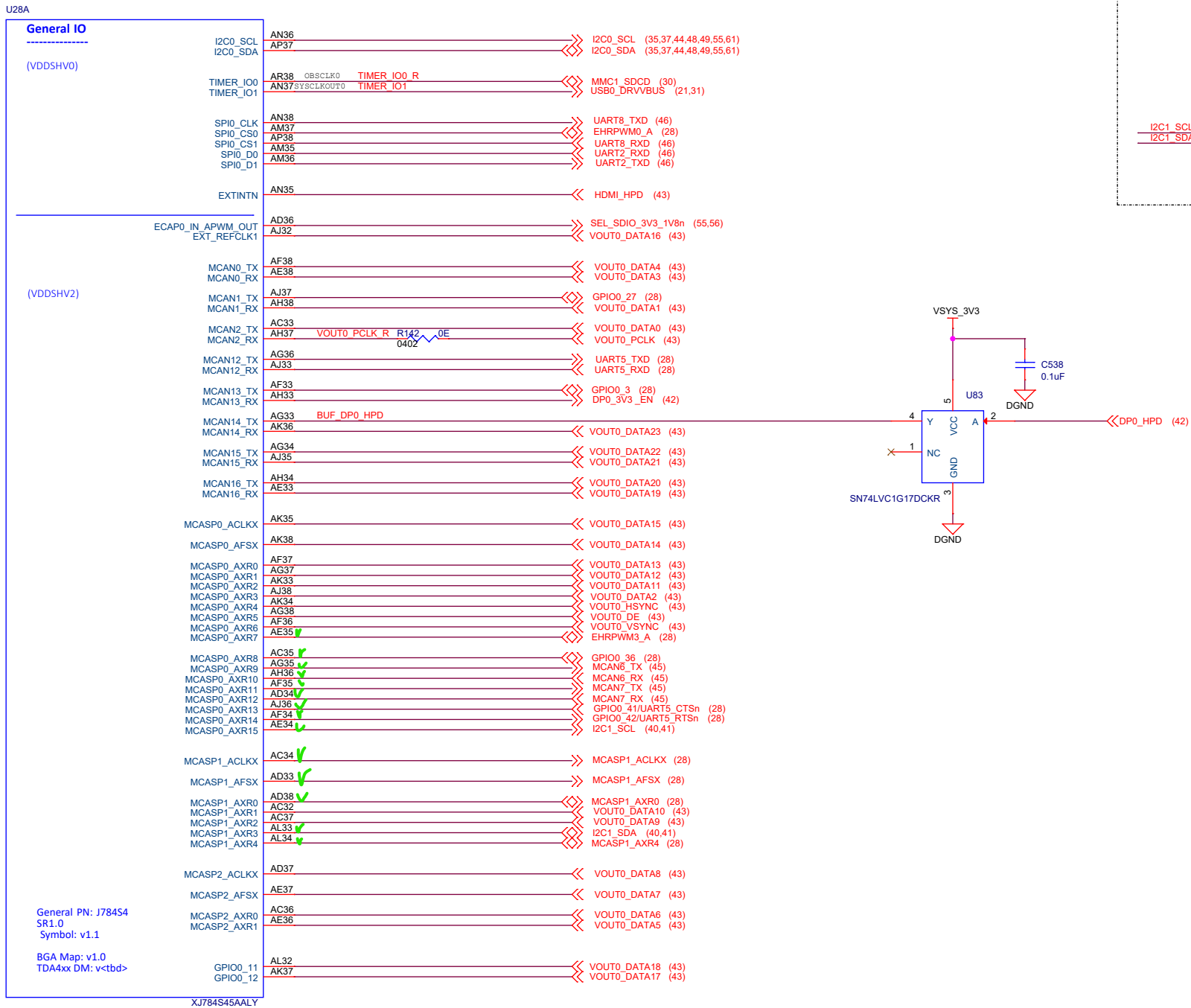
cTI 20PIN JTAG HEADER



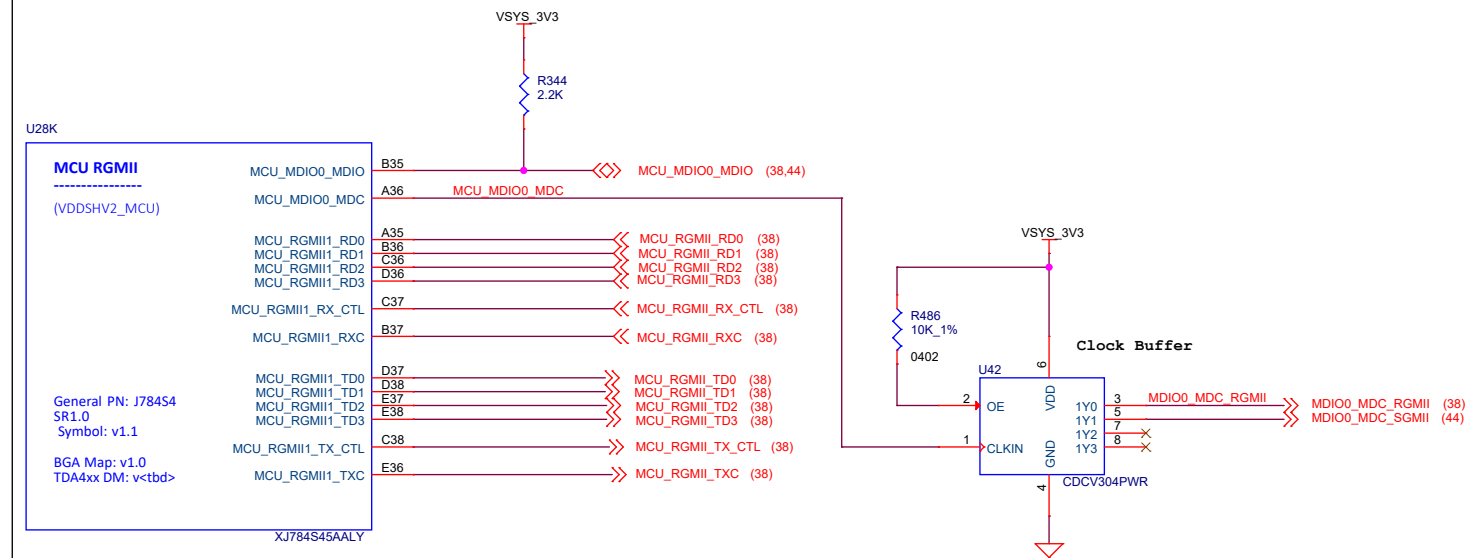
MCU/WKUP I2C Pull-ups



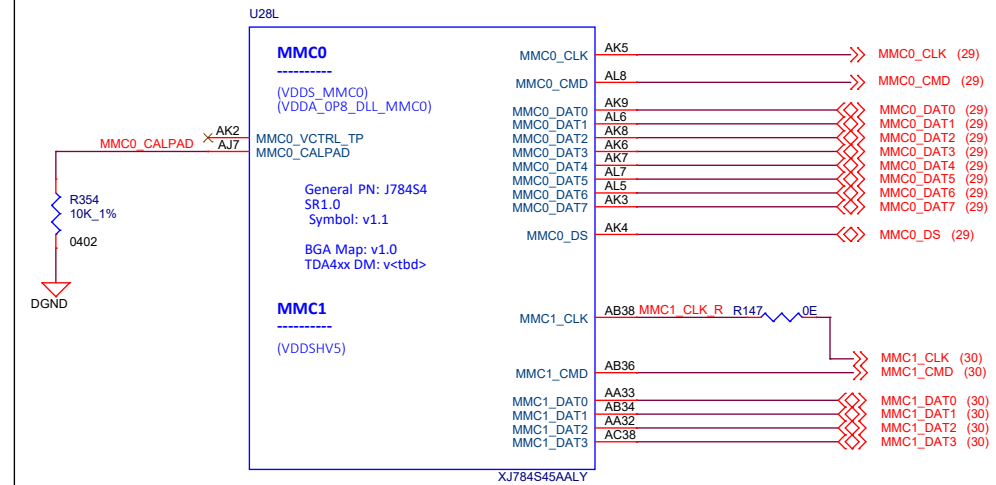
GENERAL IO



MCU_RGMI



MMC0 and MMC1

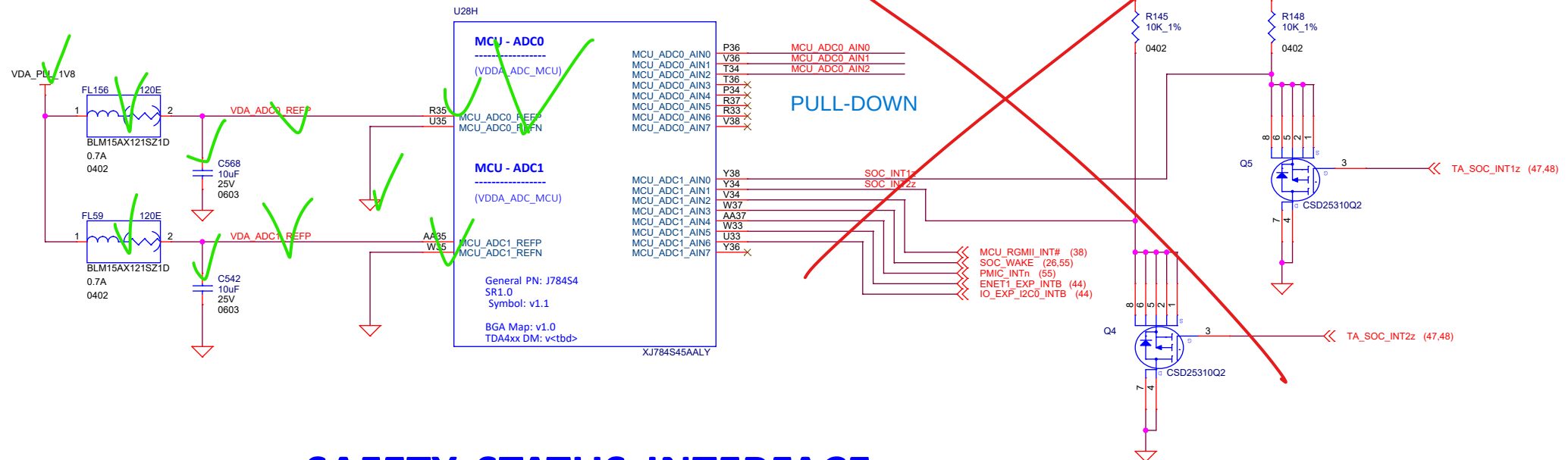


Resistor option to bypass clock buffer.

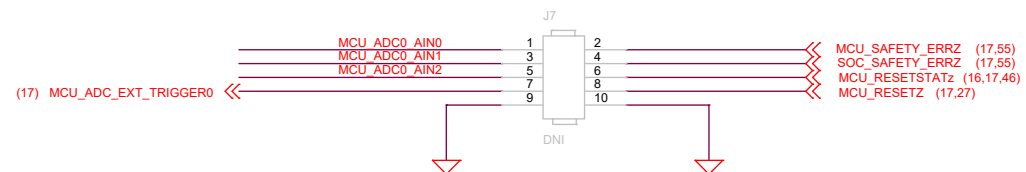


CAD Note: Place resistors in such a way as to avoid any stubs.

MCU_ADC



SAFETY STATUS INTERFACE



Project :
AM69 Edge AI Kit



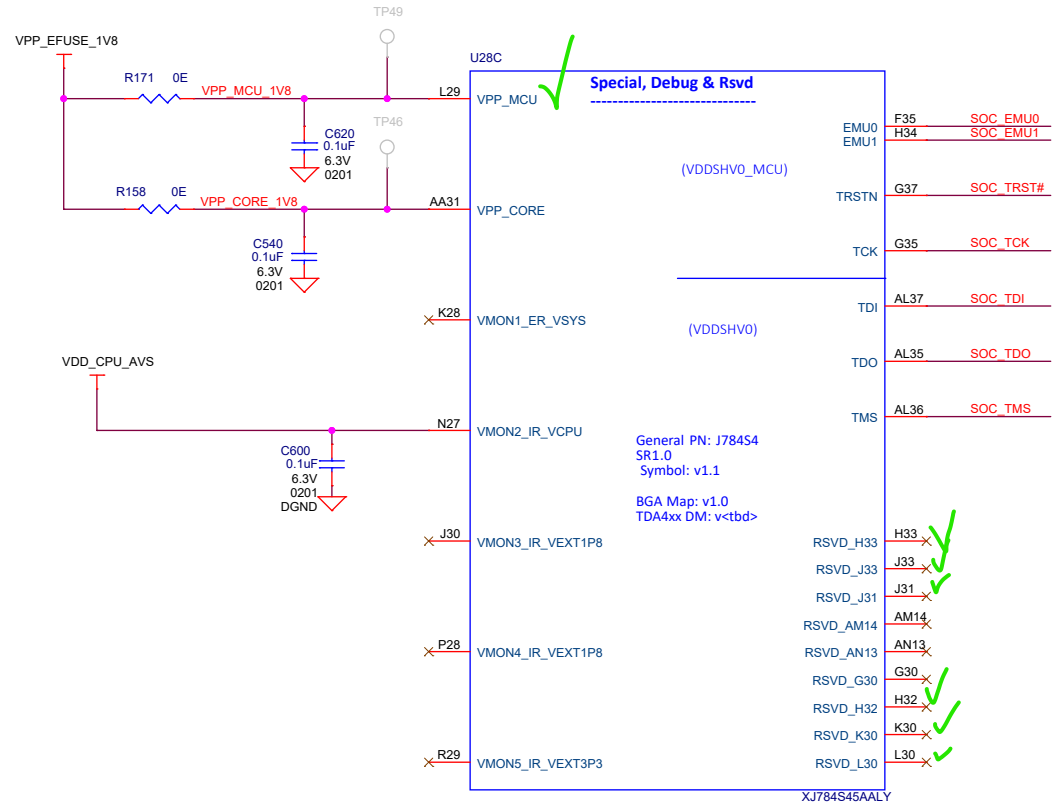
Title	SOC - MCU RGMII / MMC[0:1] / MCU ADC
-------	--------------------------------------

Size	PROC154E3 001 SK AM69
C	

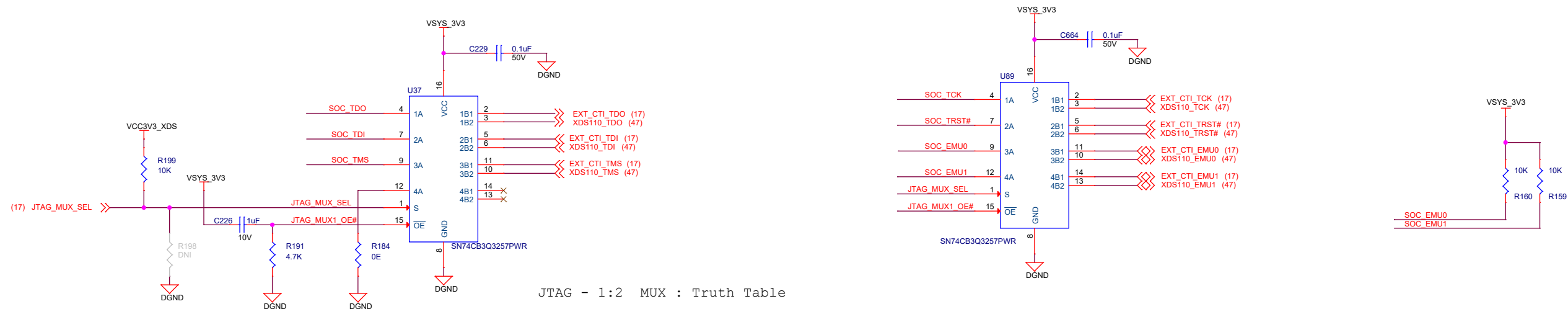
Rev
E3

Date: Monday, March 20, 2023	Sheet 19 of 62
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SPECIAL, DEBUG & RSVD



JTAG CONNECTOR AND XDS110 MUX

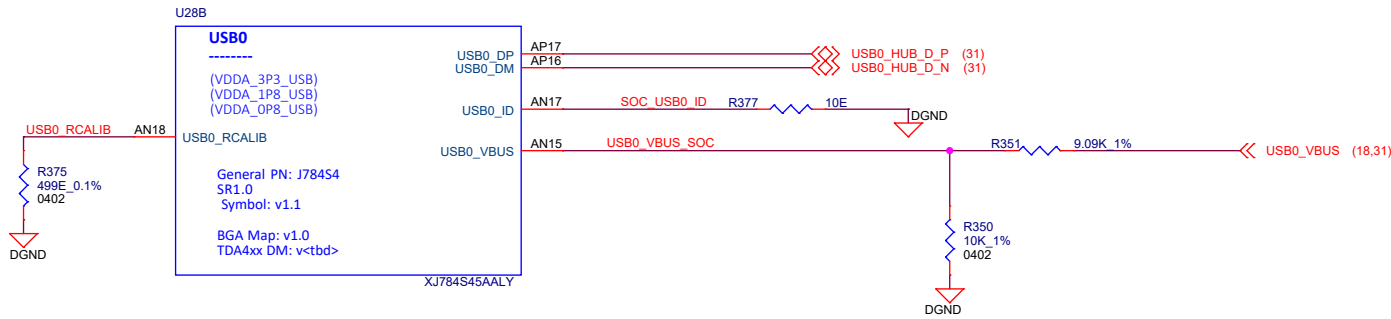


JTAG - 1:2 MUX : Truth Table

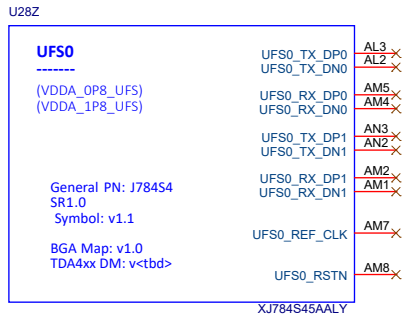
MUX_SEL	CONDITION	FUNCTION
LOW	External Emulator attached & No Power to XDS110	A-->B1 port [EXTERNAL EMU] (default)
HIGH	No External Emulator attached & XDS110 Powered via USB	A-->B2 port [ON Board EMU]

USB0 2.0

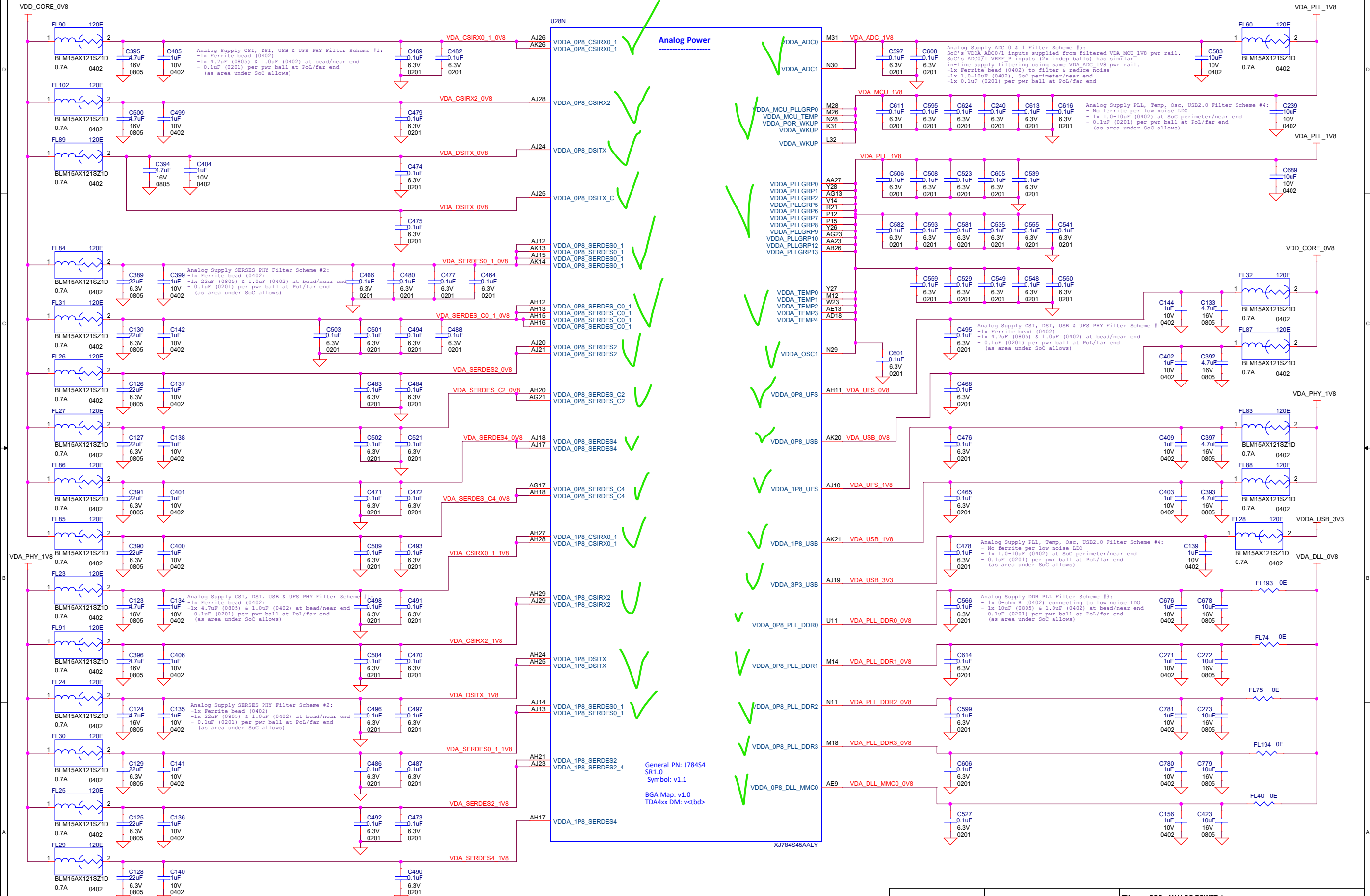
USB VBUS Resistor divider circuit



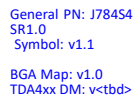
UFS FLASH



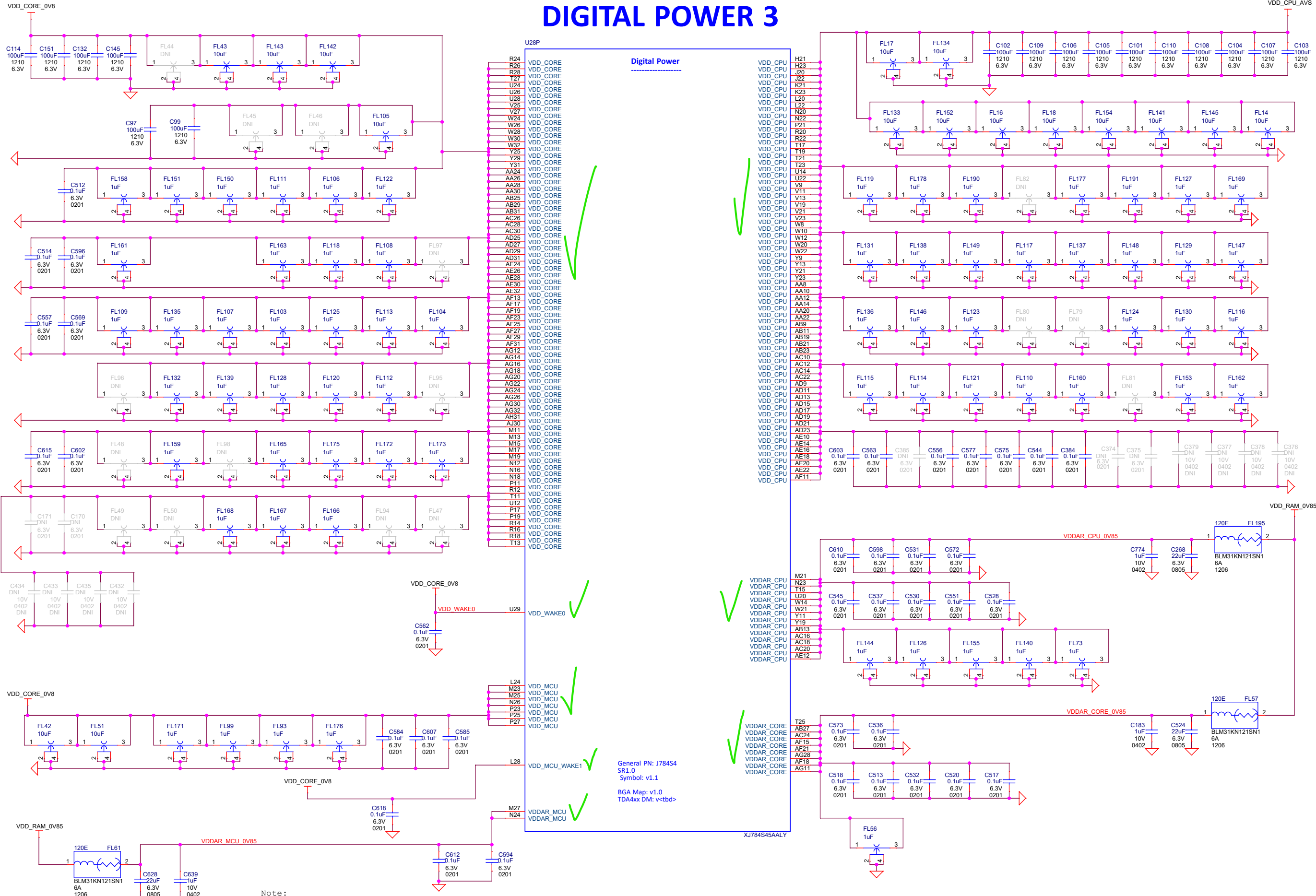
ANALOG POWER 1



IO Power & Support



DIGITAL POWER 3



Note:

A few Dcaps shown here have been provisioned on PCB layout underneath SoC at individual power ball vias & around perimeter in case additional high-freq decoupling might be needed.

Some Dcaps may be shown as "Do Not Install" (DNI) components if Power Integrity (PI) simulation results for a particular power rail on this SK PCB design combined with Dcap scheme (value, pkg type, ESL, Loop-Inductance, etc.) results in an impedance response below or equal to the desired target impedance (Zt).

Project :
AM69 Edge AI Kit



Title SOC - DIGITAL POWER 3

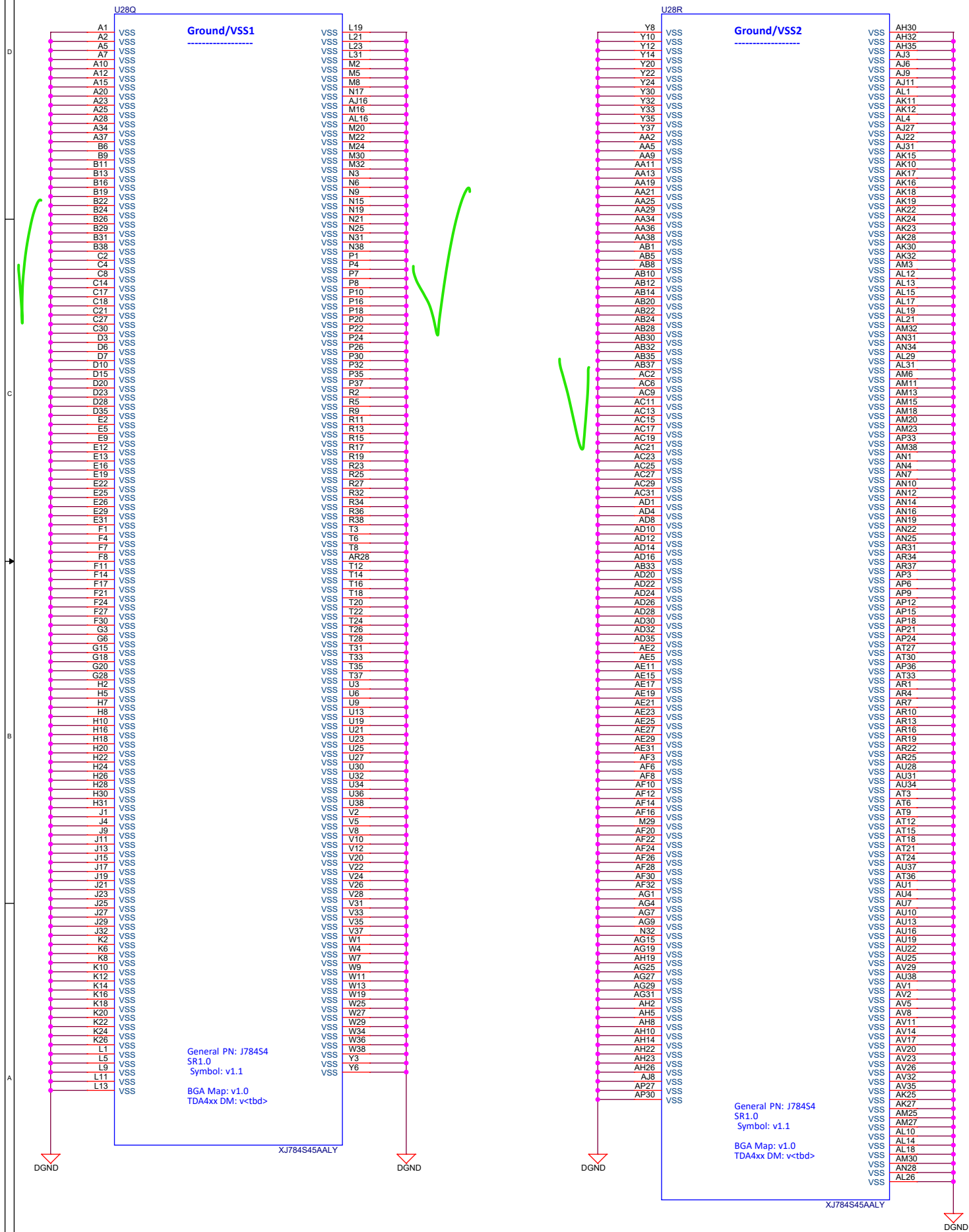
Size	PROC154E3 001 SK AM69
------	-----------------------

Date: Monday, March 20, 2023

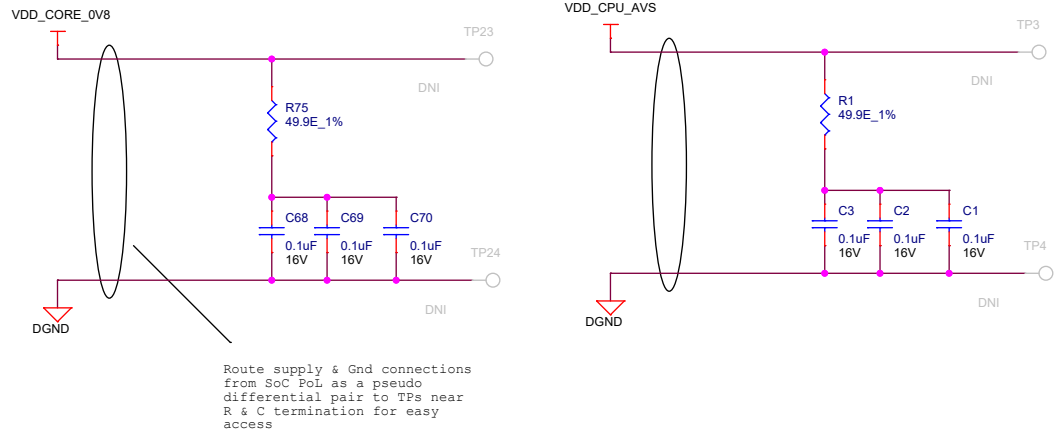
Re

62

SOC GROUND

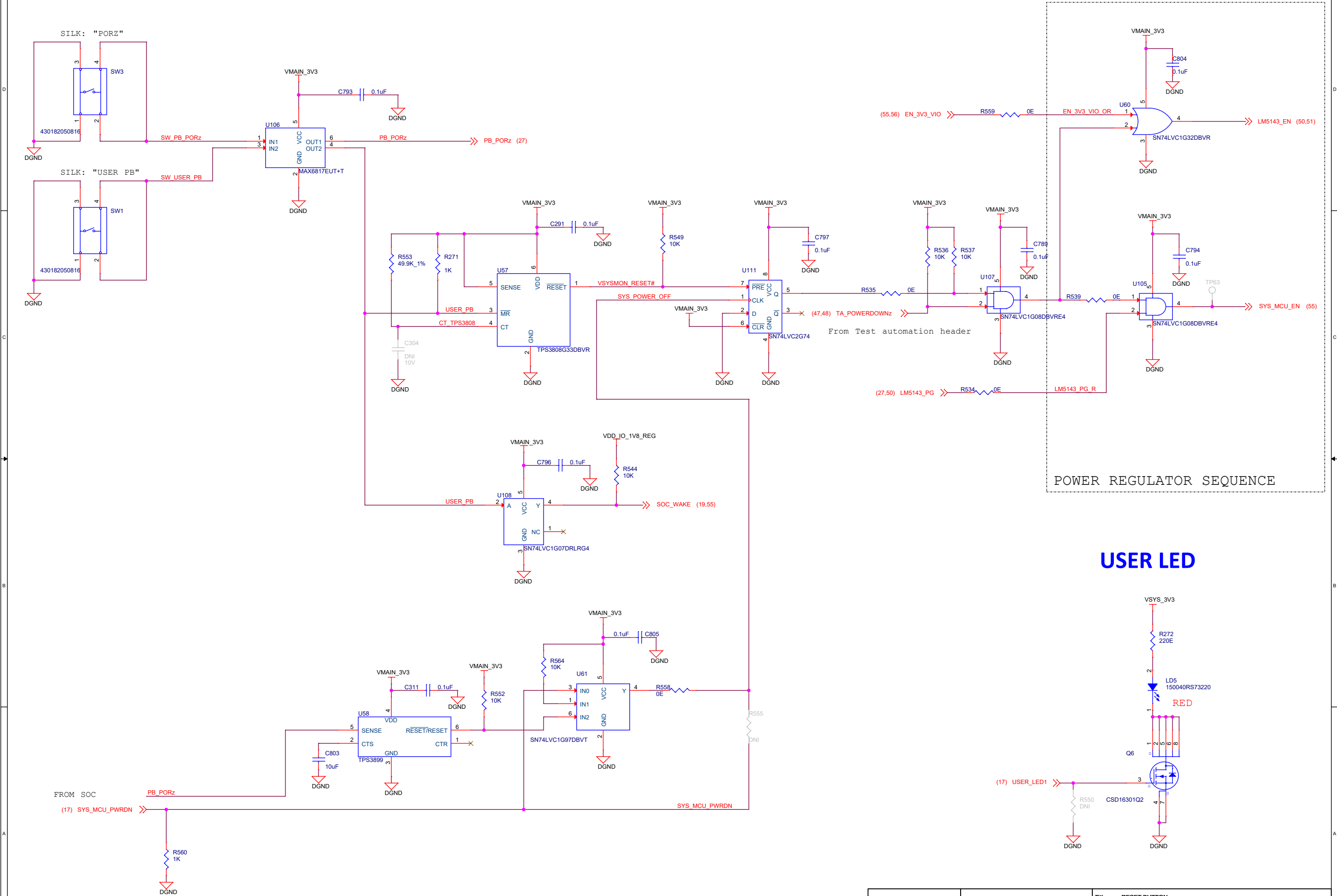


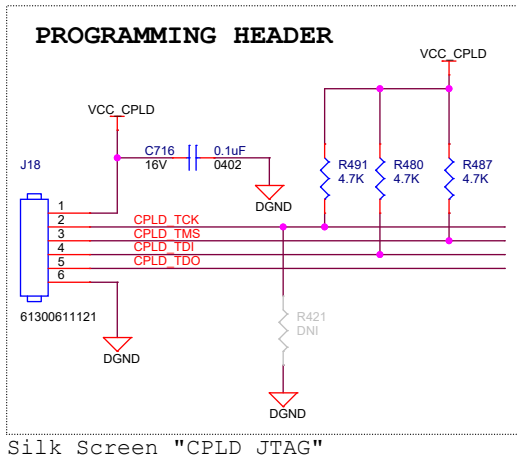
SoC Supply Noise Kelvin Sensing



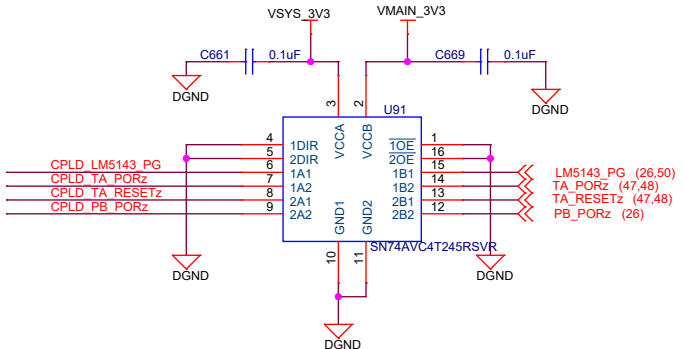
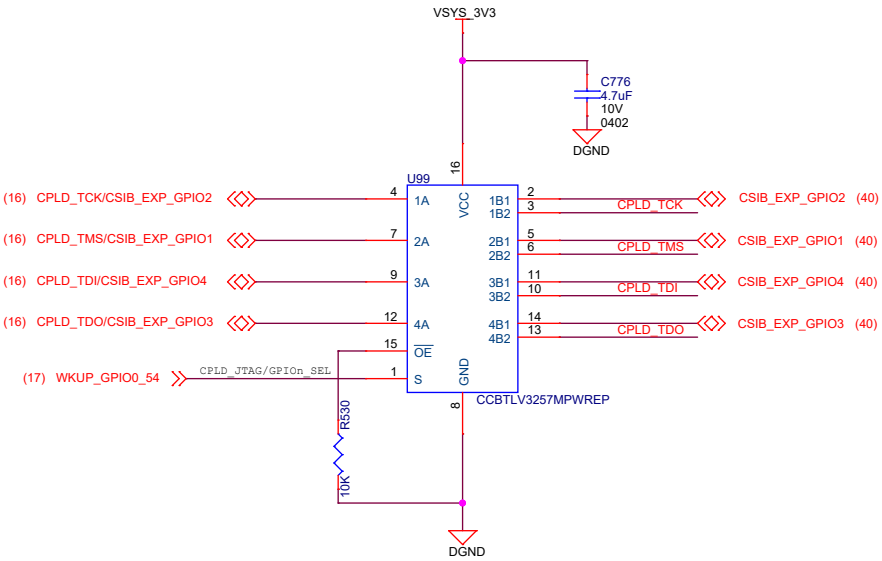
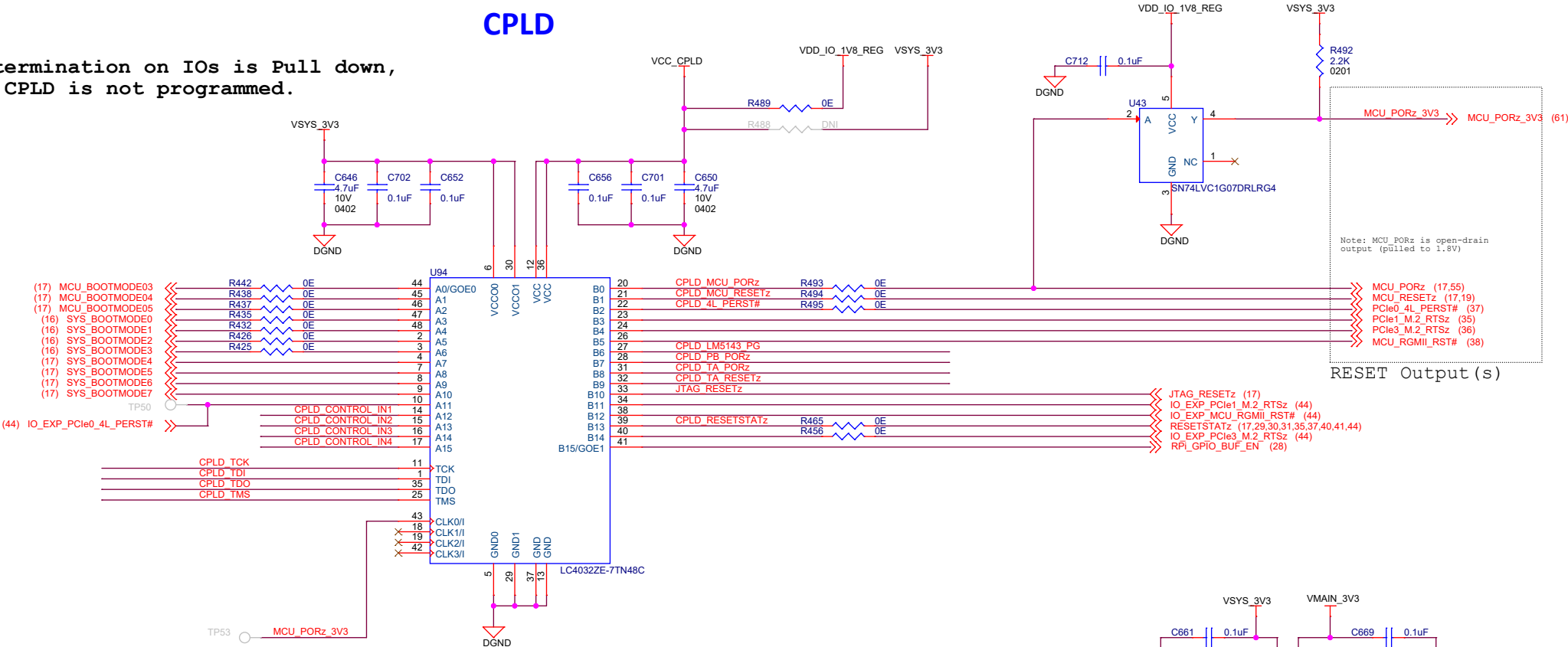
Project : AM69 Edge AI Kit		Title SOC - GROUND & KELVIN SENSING	
Size C		PROC154E3 001 SK AM69	Rev E3
Date: Monday, March 20, 2023		Sheet 25	of 62

RESET BUTTONS





Default termination on IOs is Pull down,
when the CPLD is not programmed.

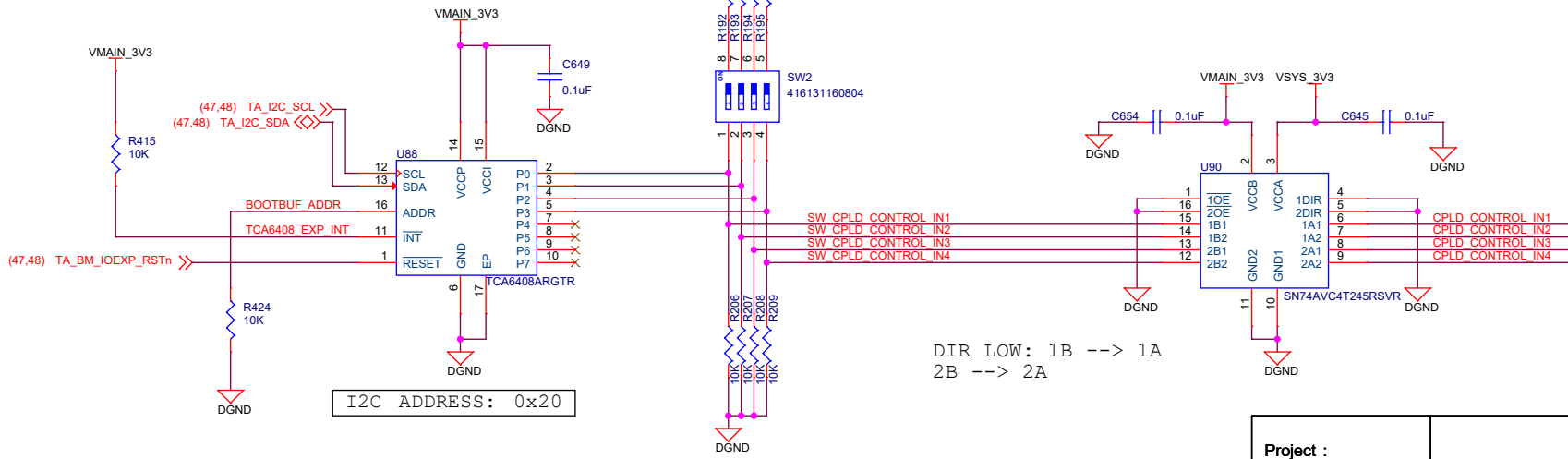


Test Automation BOOTMODE Logic

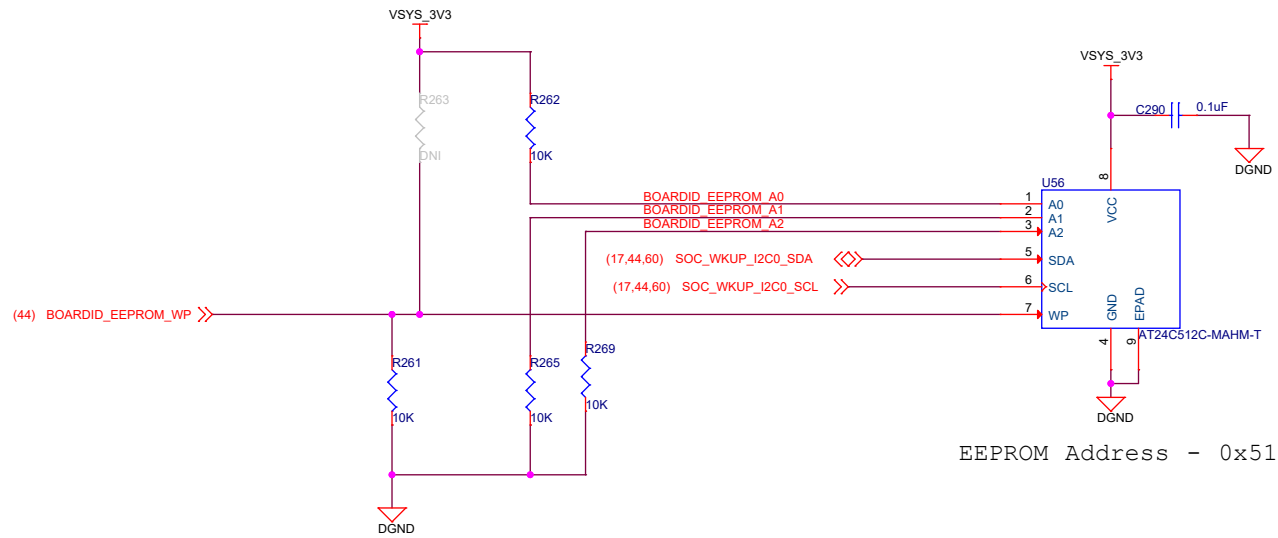
Note: Test Automation logic to set desired BOOTMODE.

Bootmode Table

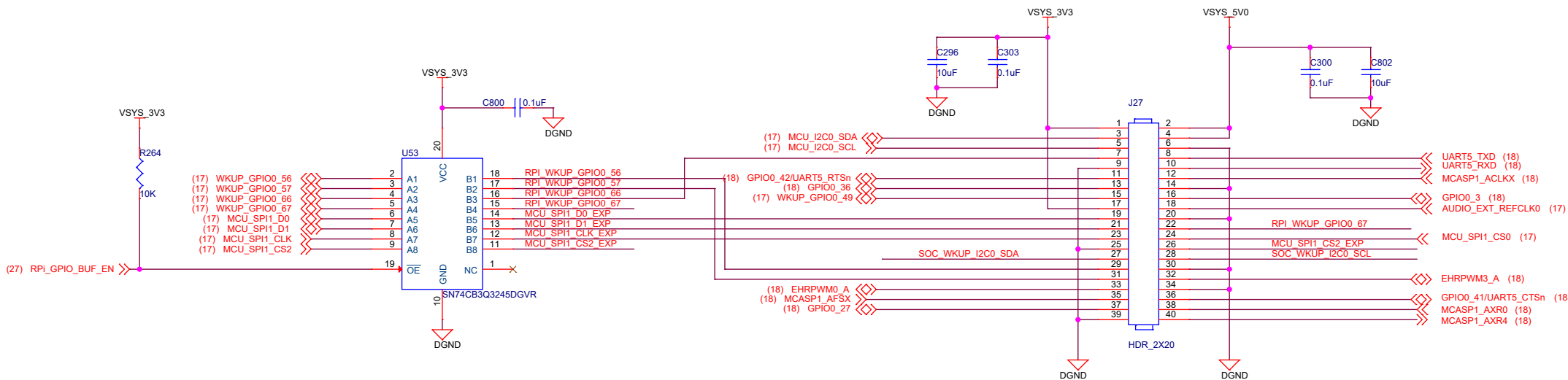
SW2.3	SW2.2	SW2.1	BOOTMODE
0	0	0	SD
0	0	1	NO Boot
0	1	0	Ethernet
0	1	1	USB
1	0	0	xSPI - 1S
1	0	1	UART
1	1	0	eMMC
1	1	1	xSPI SFDP



BOARD ID EEPROM

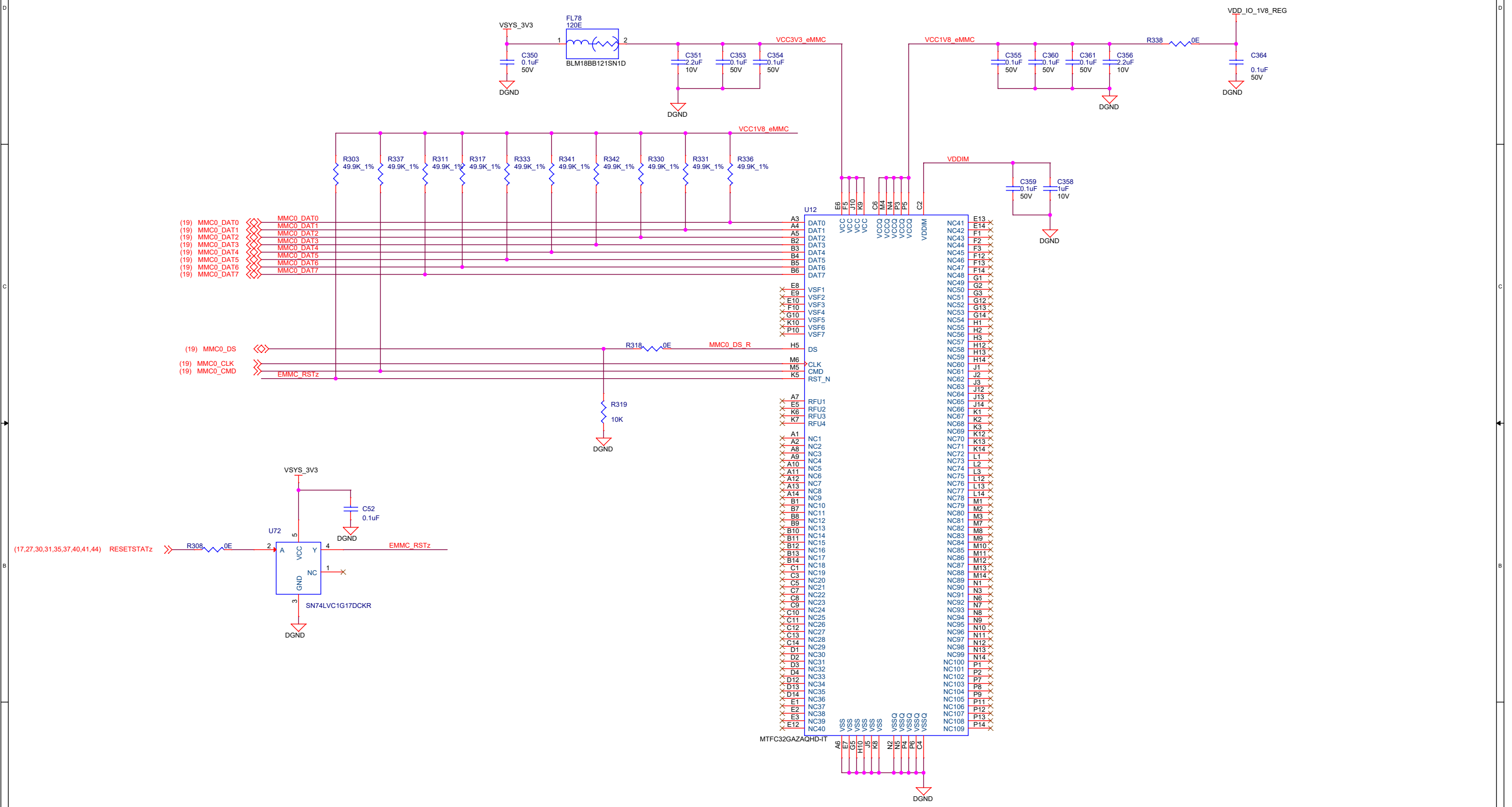


40Pin Expansion Header

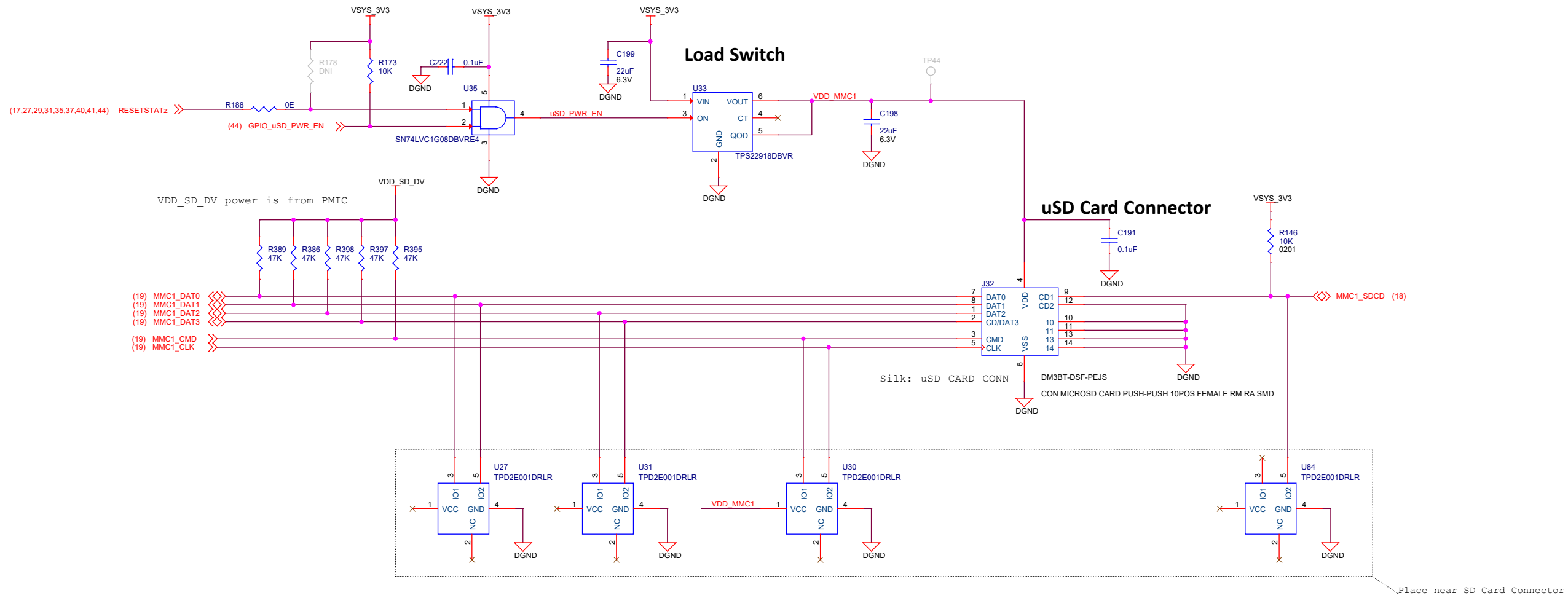


Silk Screen "40p EXP HDR"

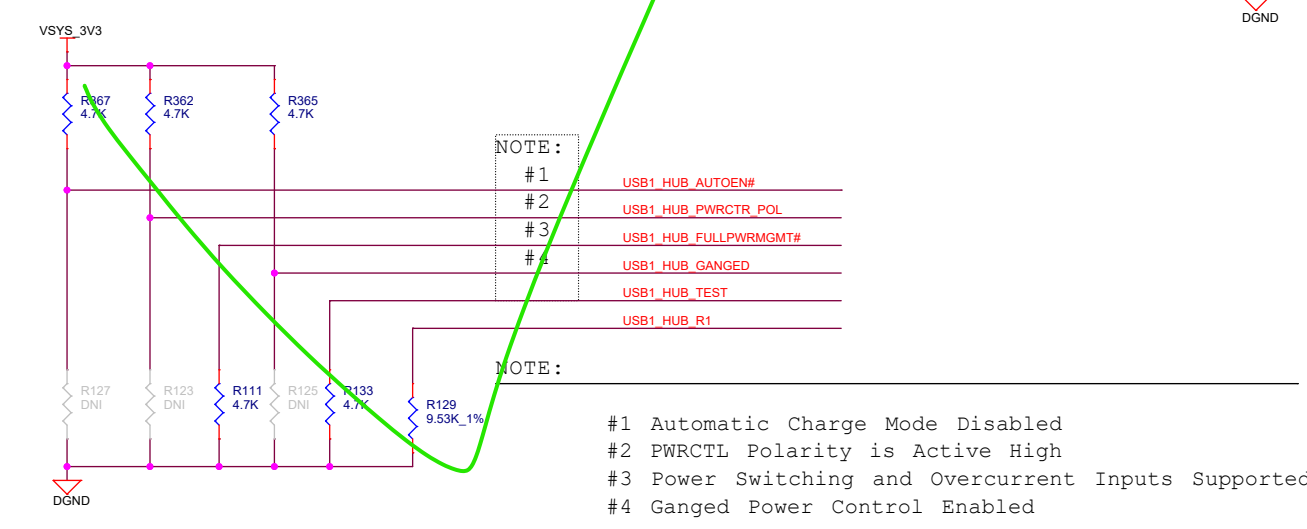
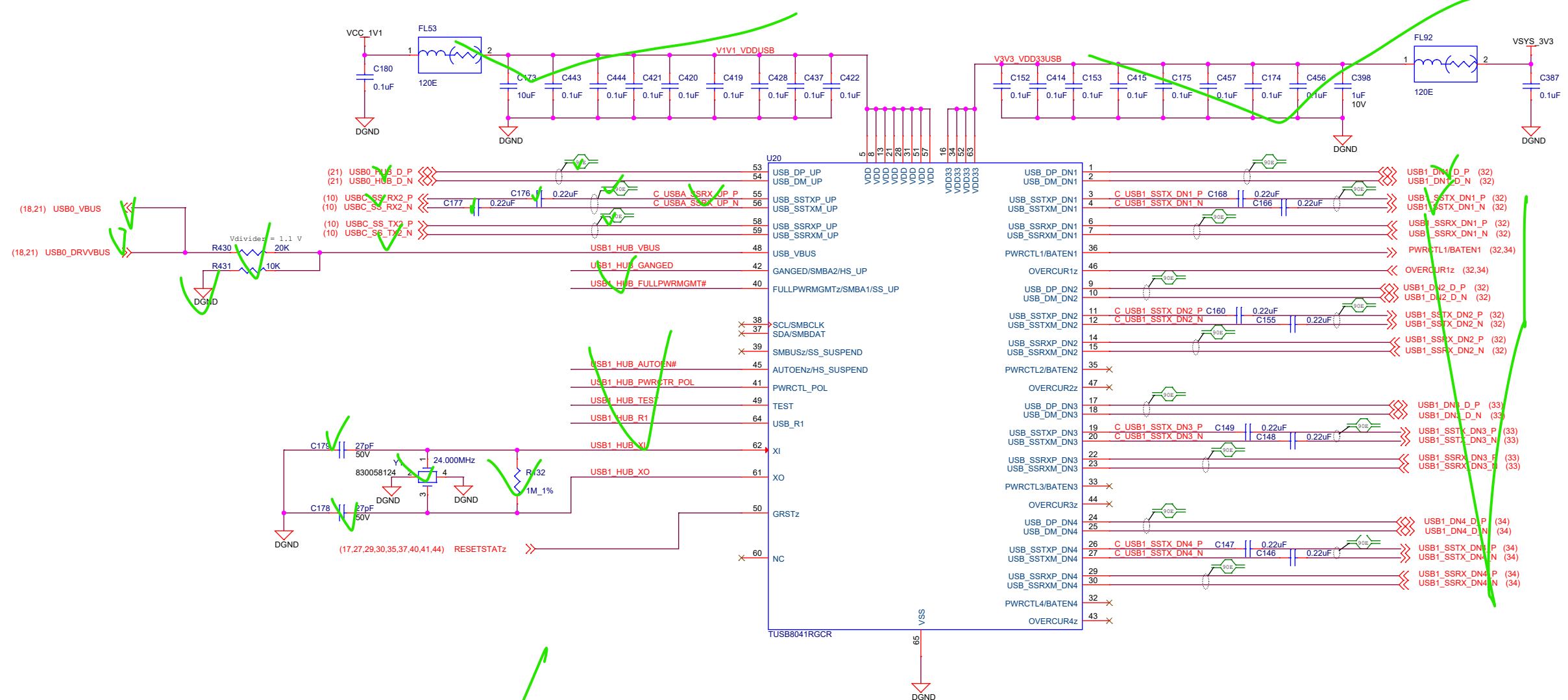
eMMC FLASH



Micro SD CARD INTERFACE



USB3.0 HUB

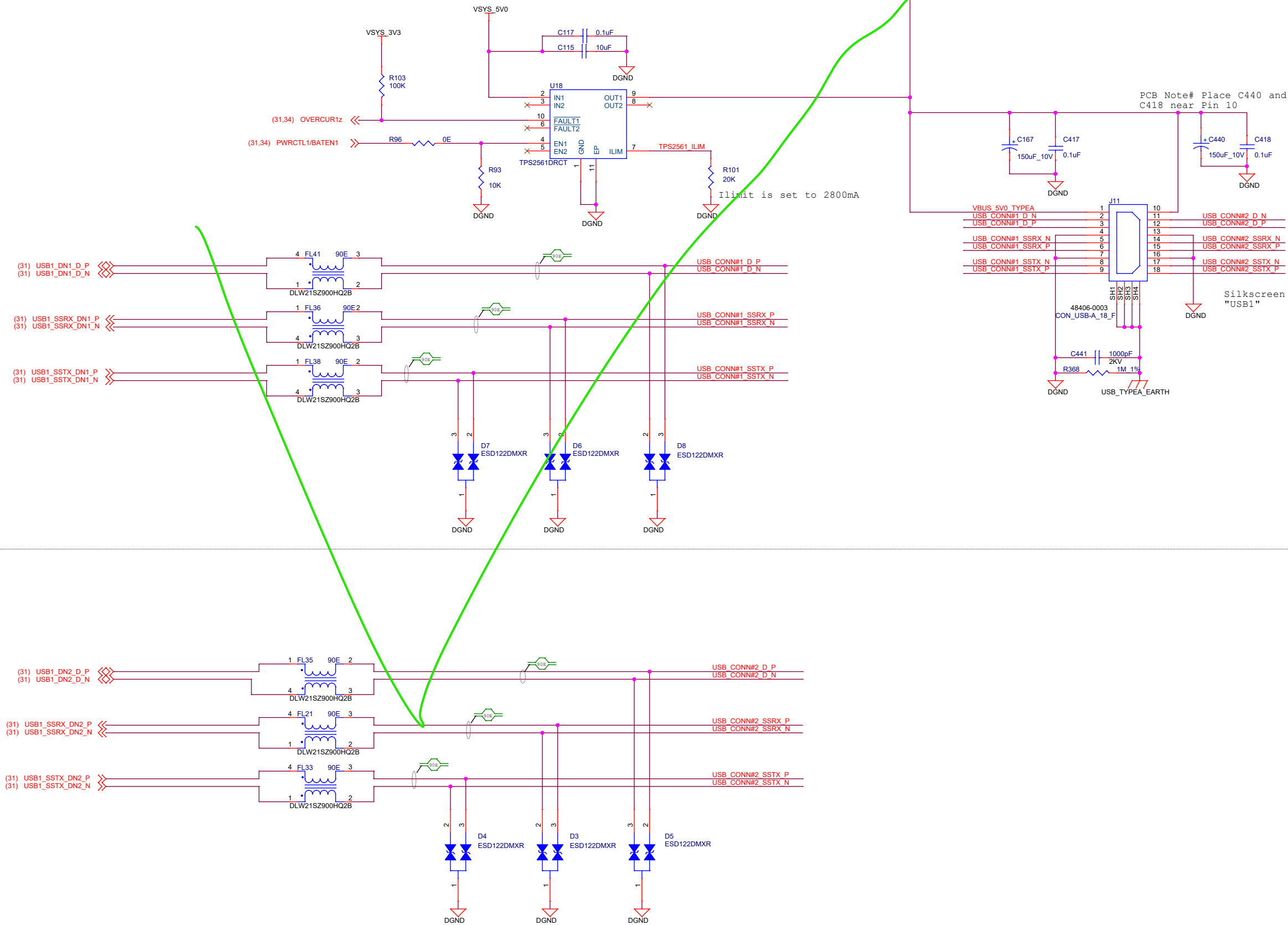


NOTE:

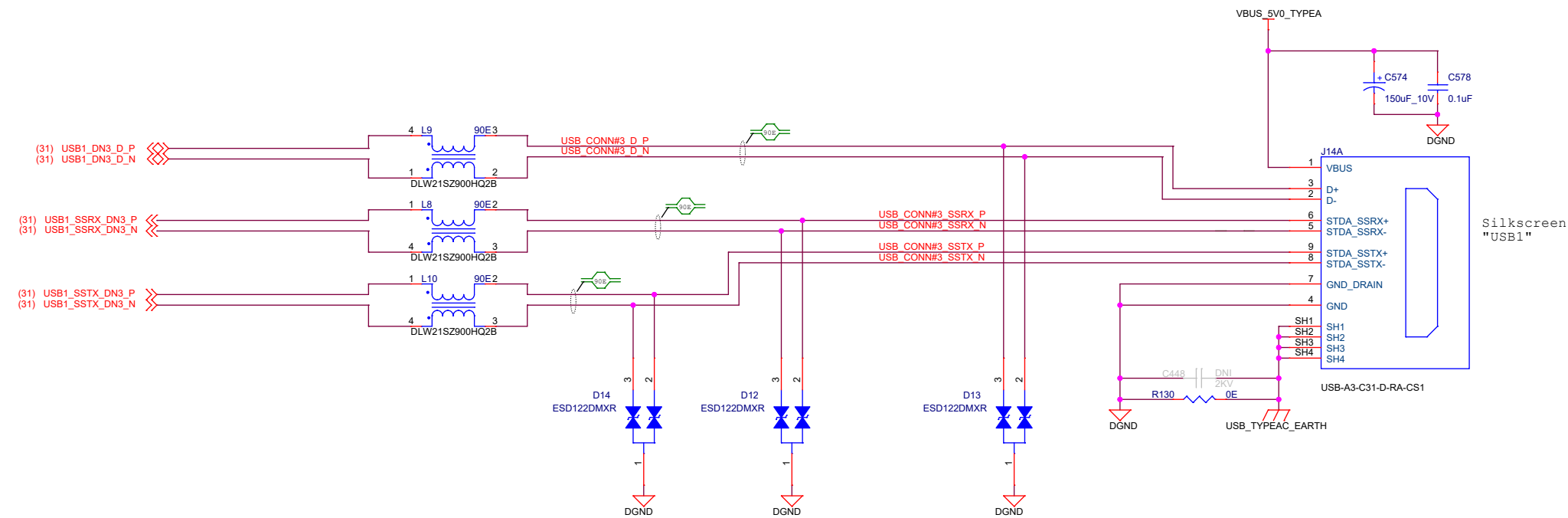
#1	USB1_HUB_AUTOEN#
#2	USB1_HUB_PWRCTR_POL
#3	USB1_HUB_FULLPWRMGMT#
#4	USB1_HUB_GANGED
	USB1_HUB_TEST
	USB1_HUB_R1

- NOTE:
- #1 Automatic Charge Mode Disabled
 - #2 PWRCTL Polarity is Active High
 - #3 Power Switching and Overcurrent Inputs Supported
 - #4 Ganged Power Control Enabled

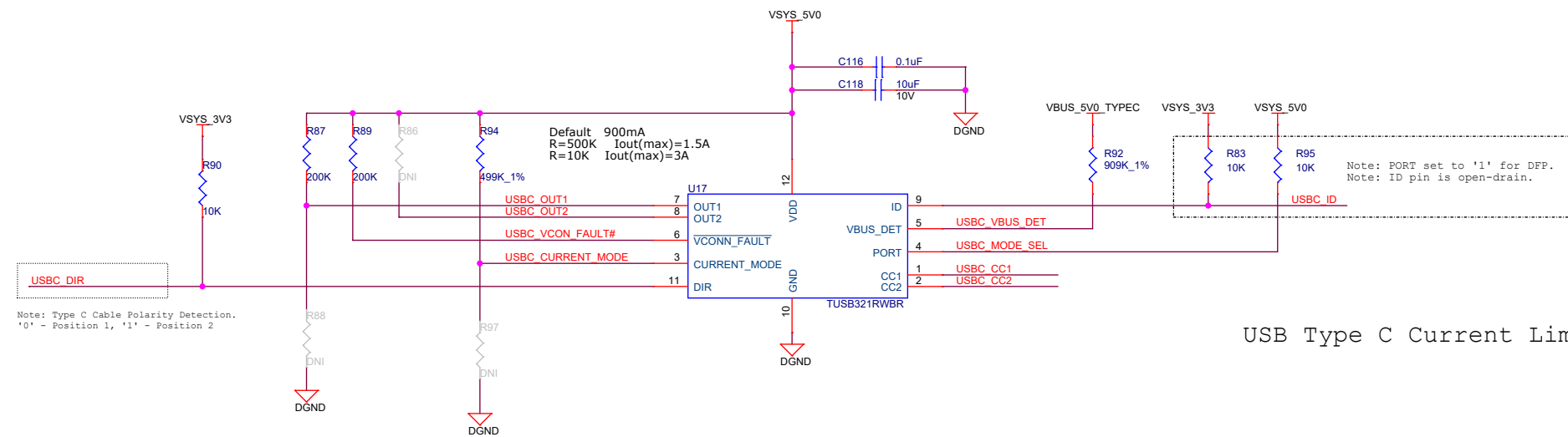
USB 3.0 TYPE-A CONNECTORS - 1



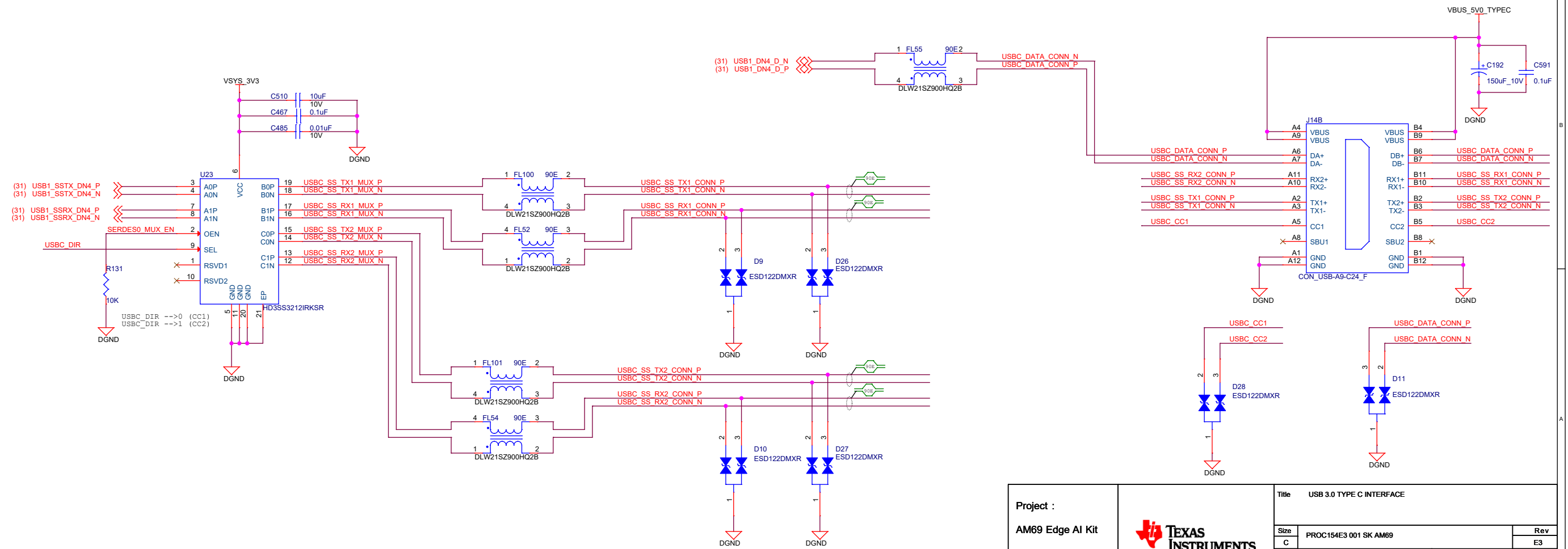
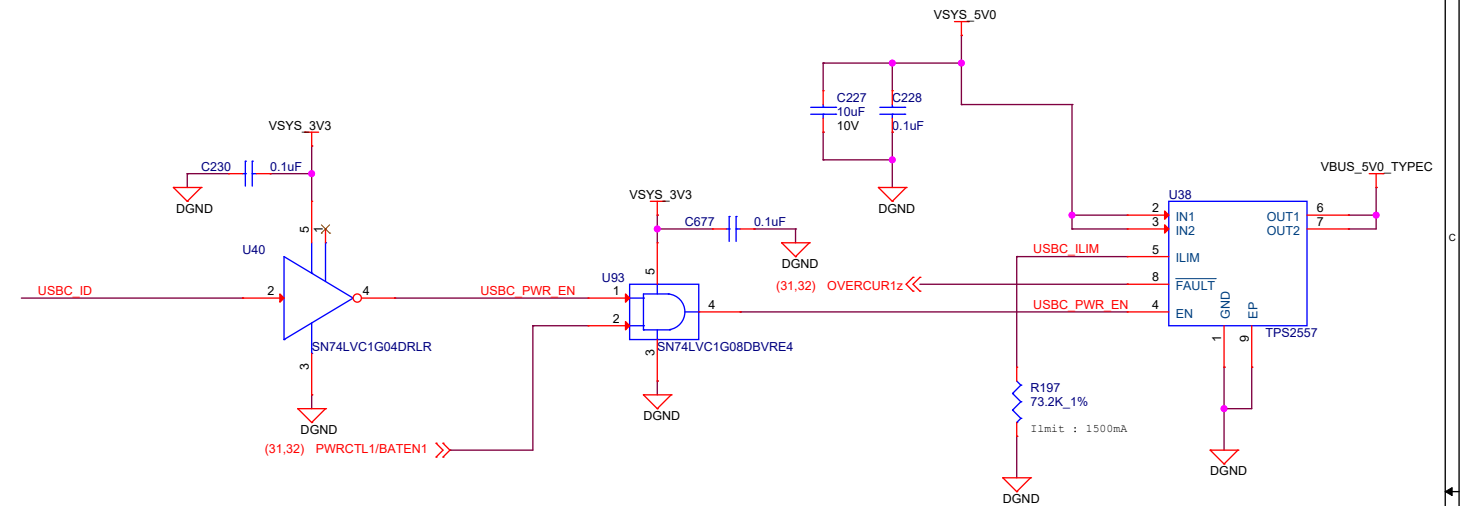
USB 3.0 TYPE-A CONNECTORS - 2



USB 3.0 TYPE C INTERFACE

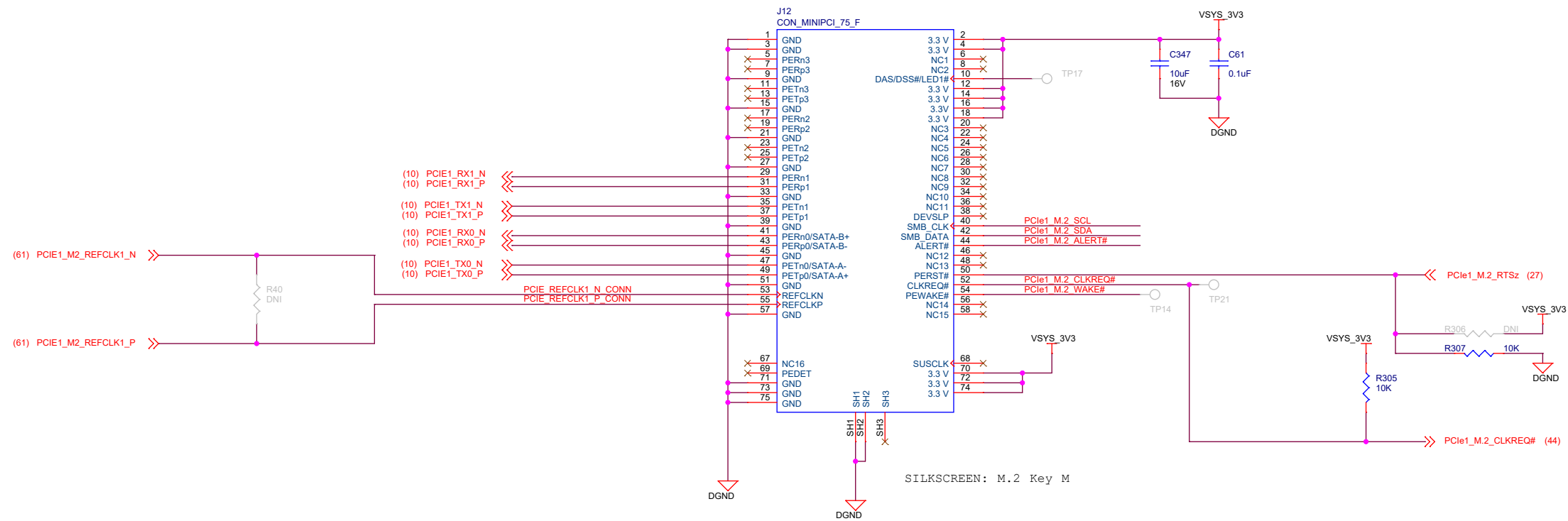


USB Type C Current Limit

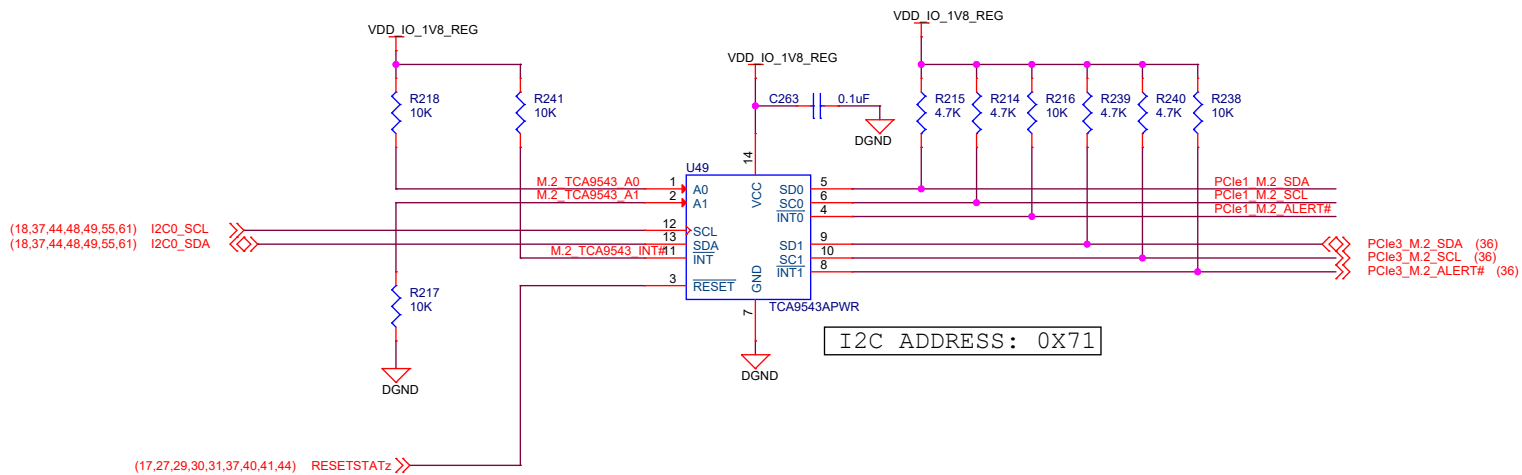


PCIE_M.2_INTERFACE SSD

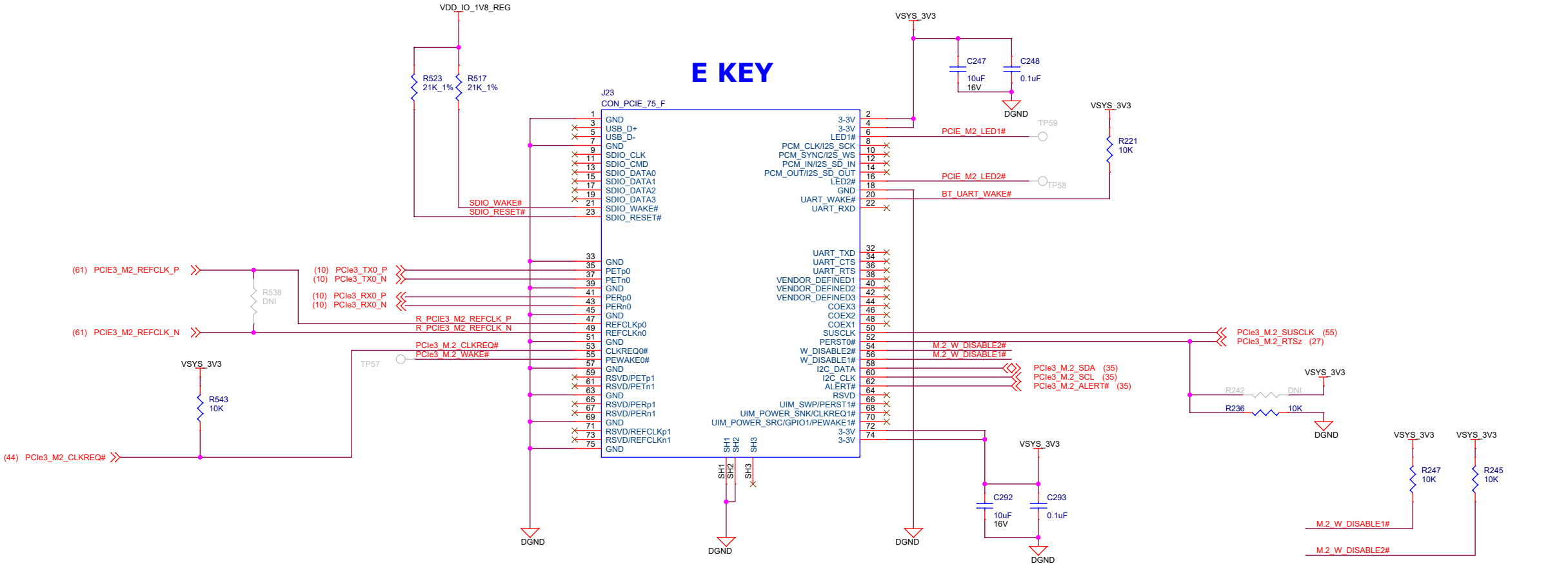
M KEY



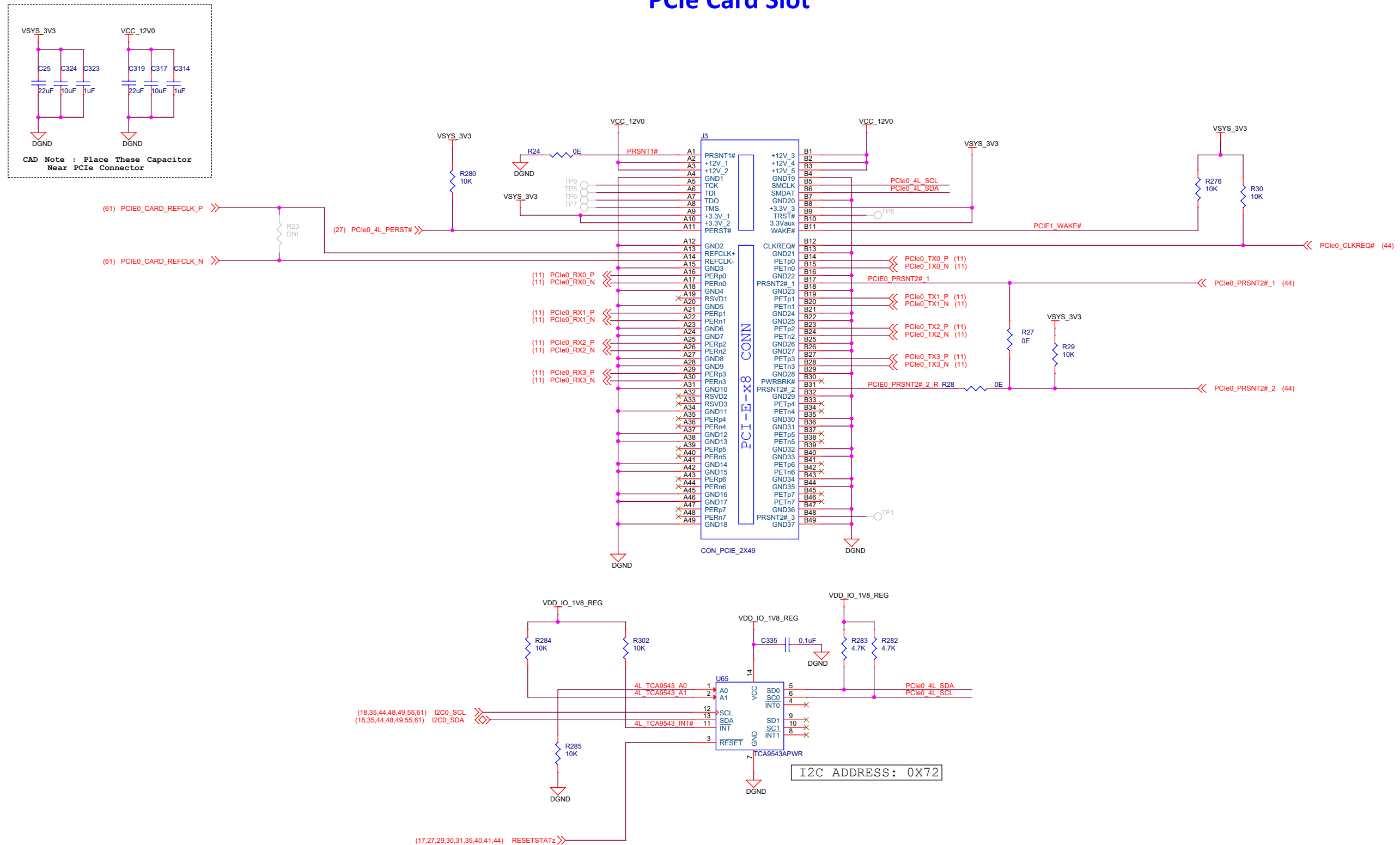
3.3V To 1V8 Level translator



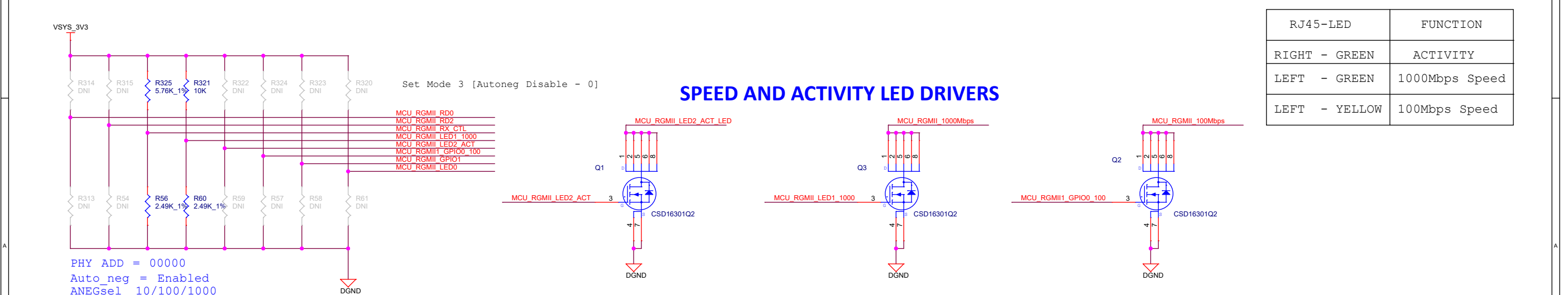
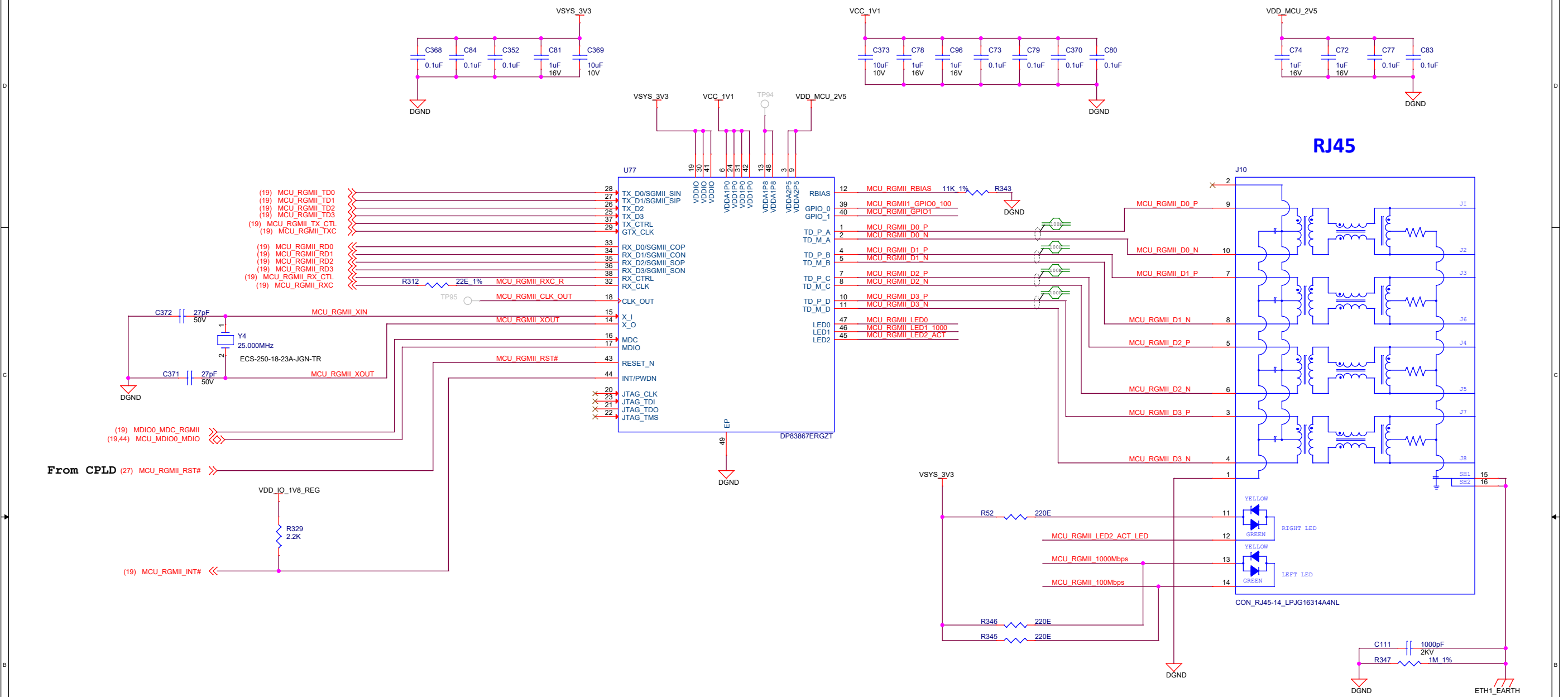
PCIE_M.2_INTERFACE - SDIO



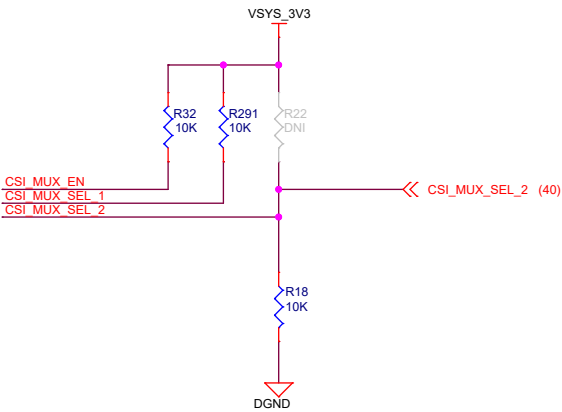
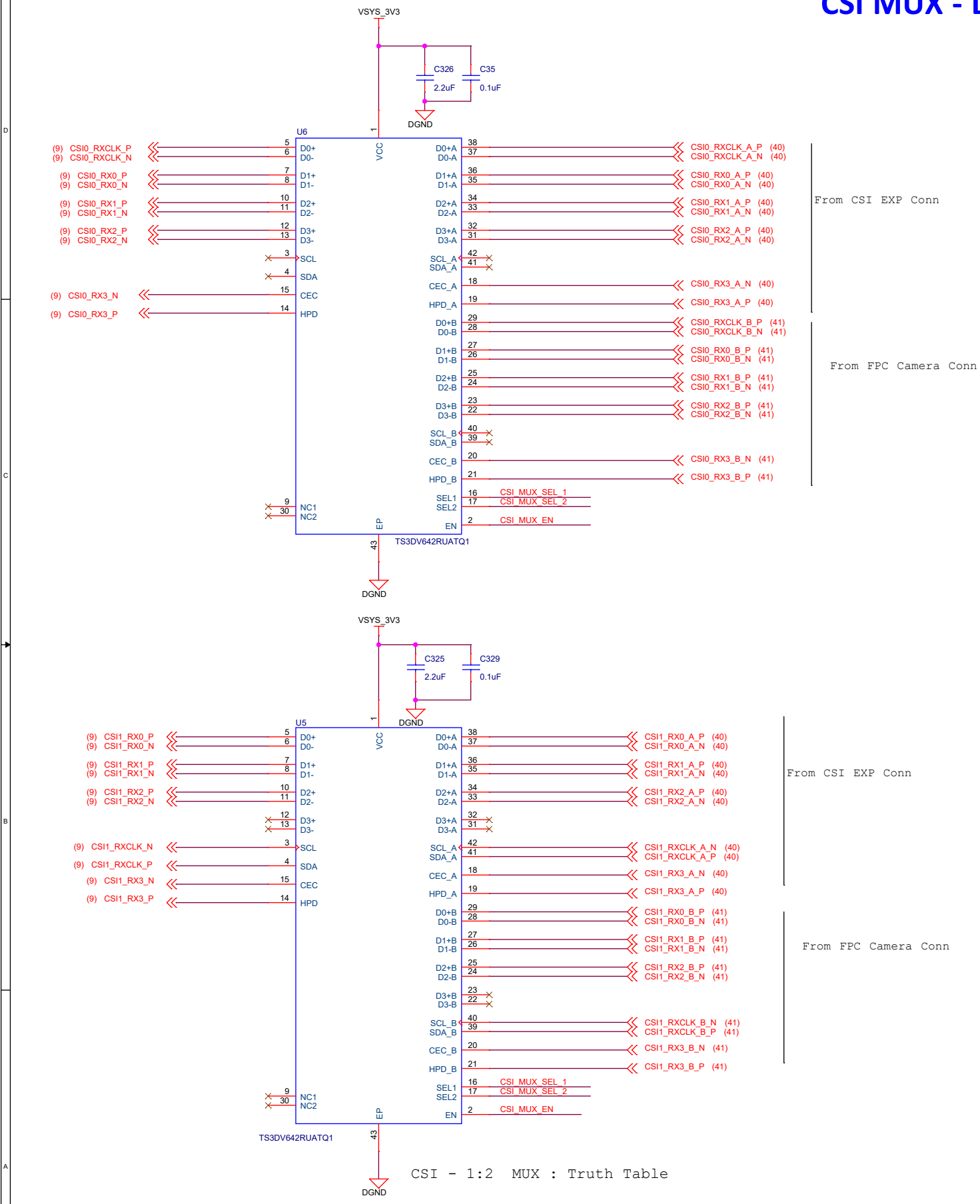
PCIe Card Slot



MCU GB ETHERNET

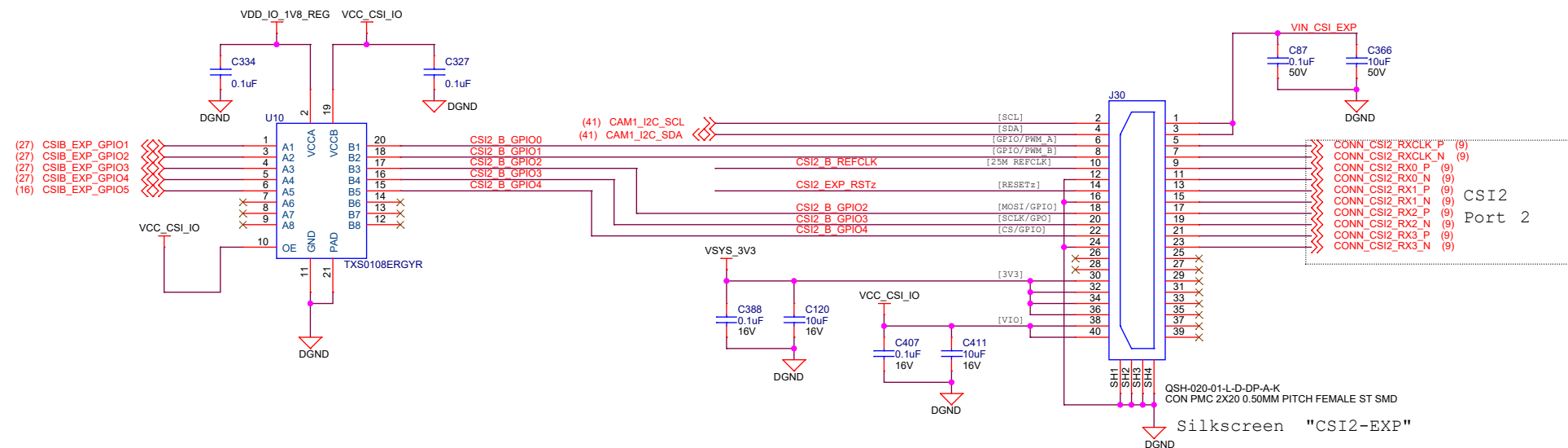
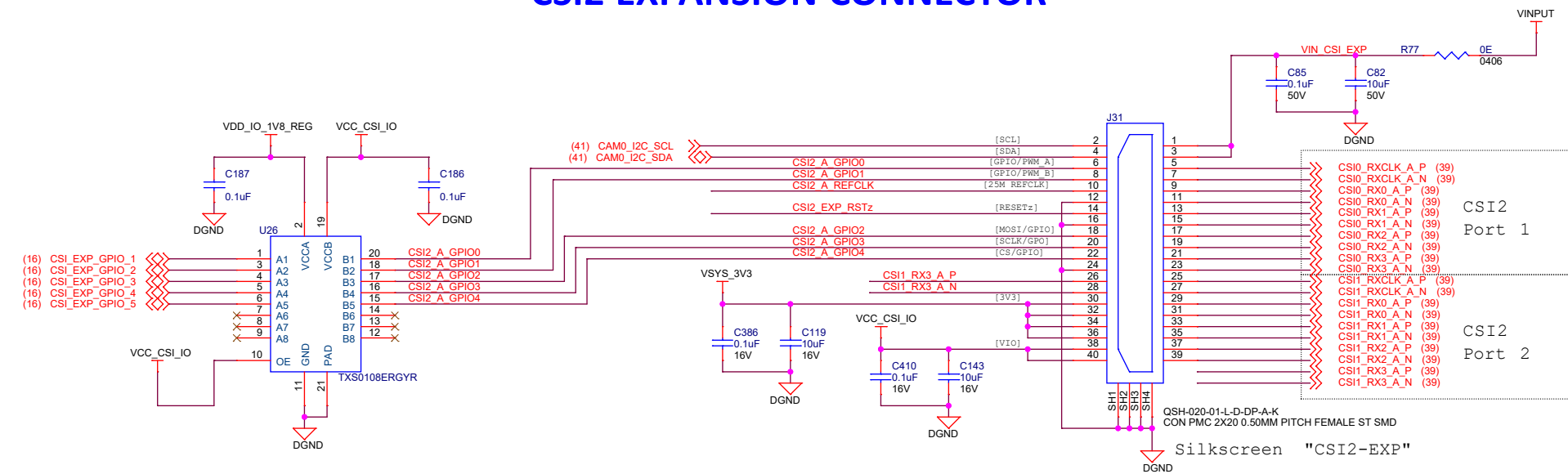


CSI MUX - DATA

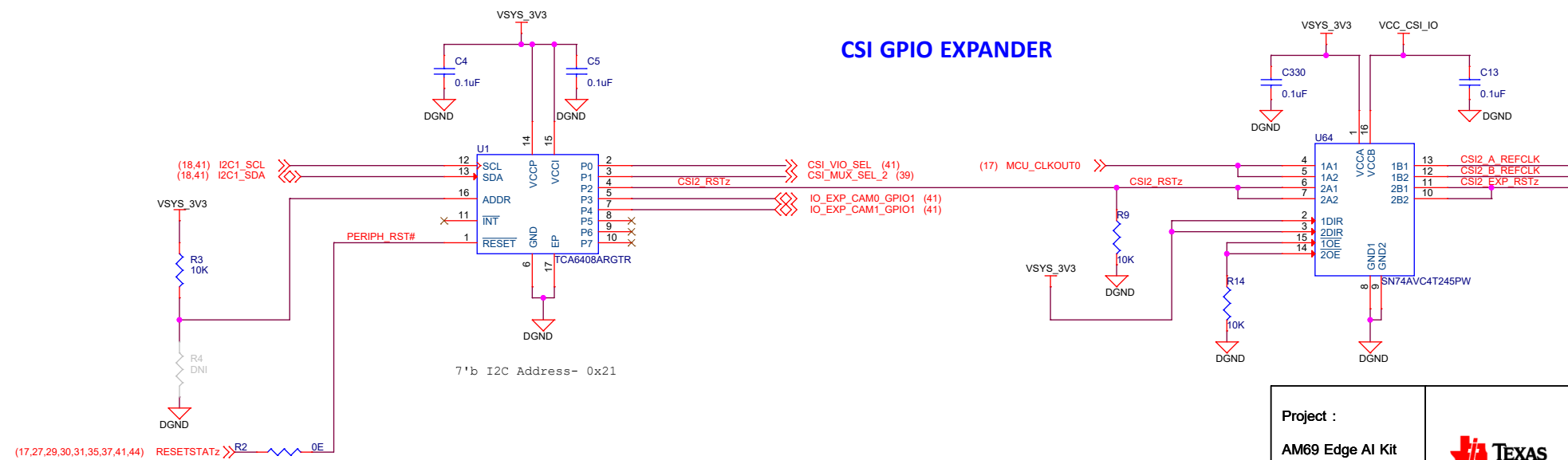


MUX_SEL_2	FUNCTION	
LOW	INPUT<-- A Port [CSI2 Connector]	(default)
HIGH	INPUT<--B port [FPC Camera Connector]	

CSI2 EXPANSION CONNECTOR

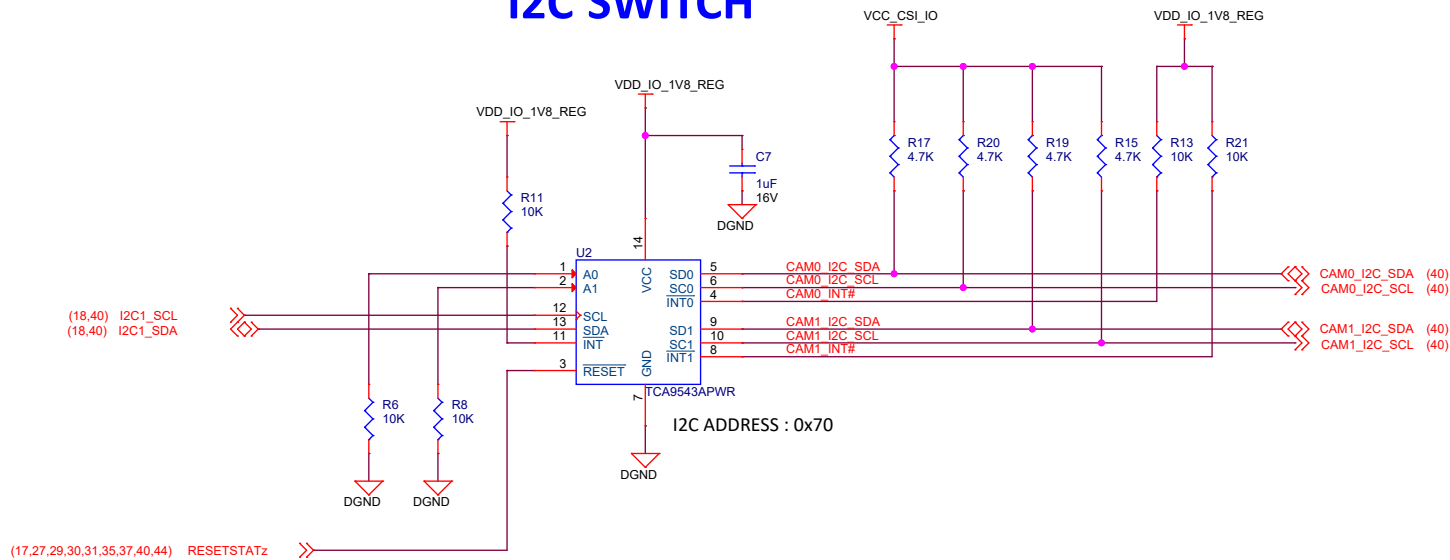


CSI GPIO EXPANDER

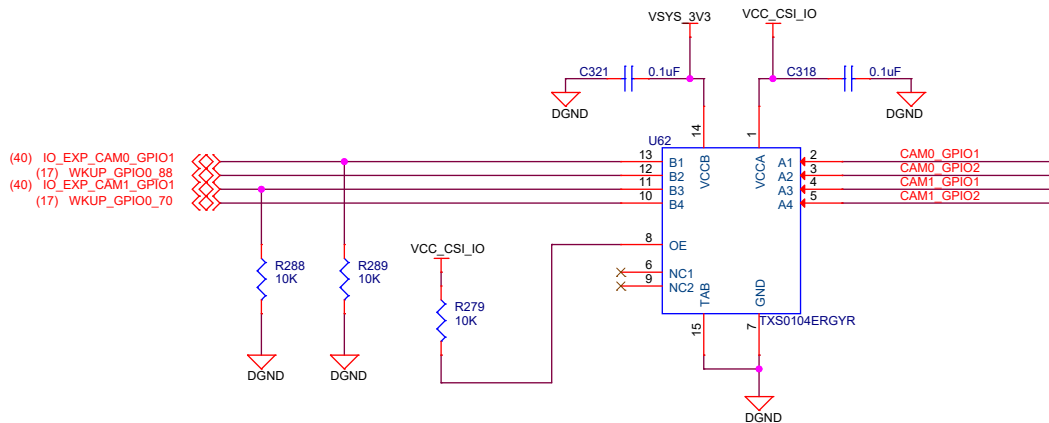


CSI FPC CAMERA CONNECTORS

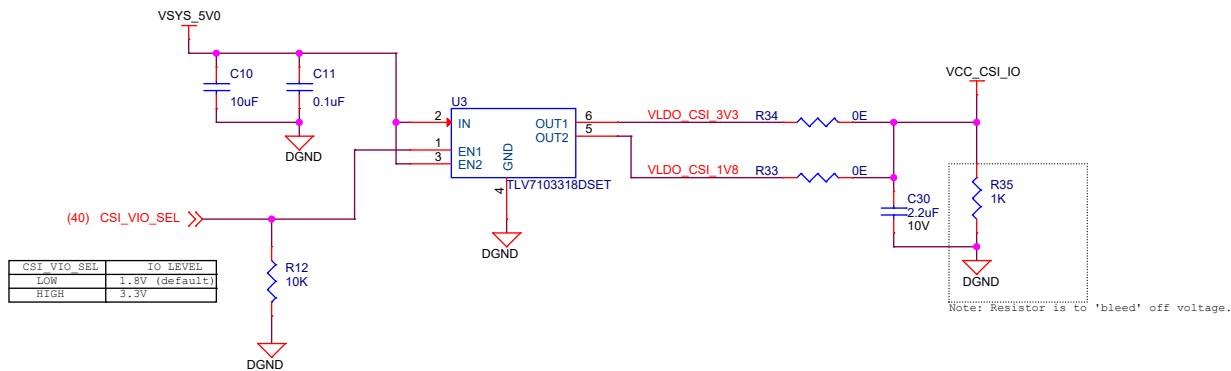
I2C SWITCH



GPIO LEVEL TRANSLATOR



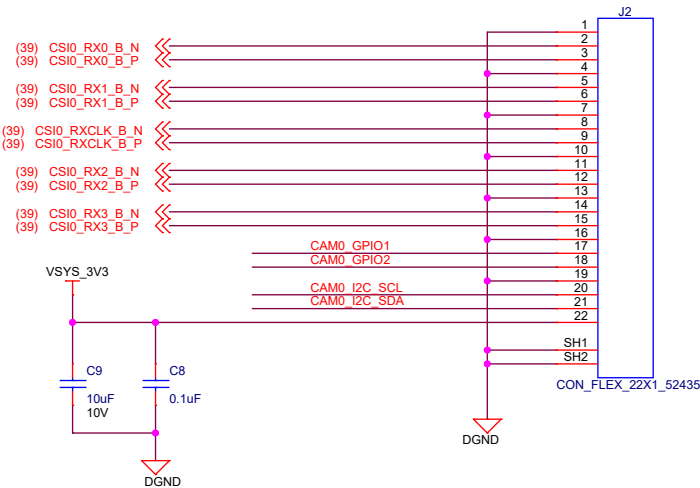
CAMERA IO SUPPLY



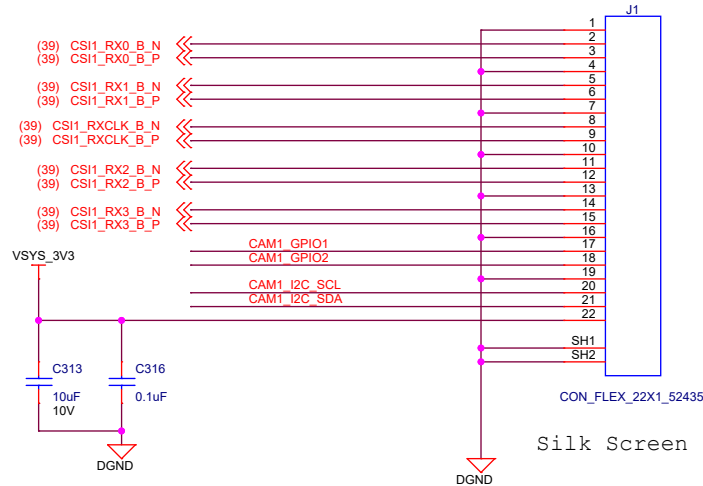
CSI_VIO_SEL	IO_LEVEL
LOW	1.8V (default)
HIGH	3.3V

Silk Screen "CAM1"

FPC Camera Connector -1

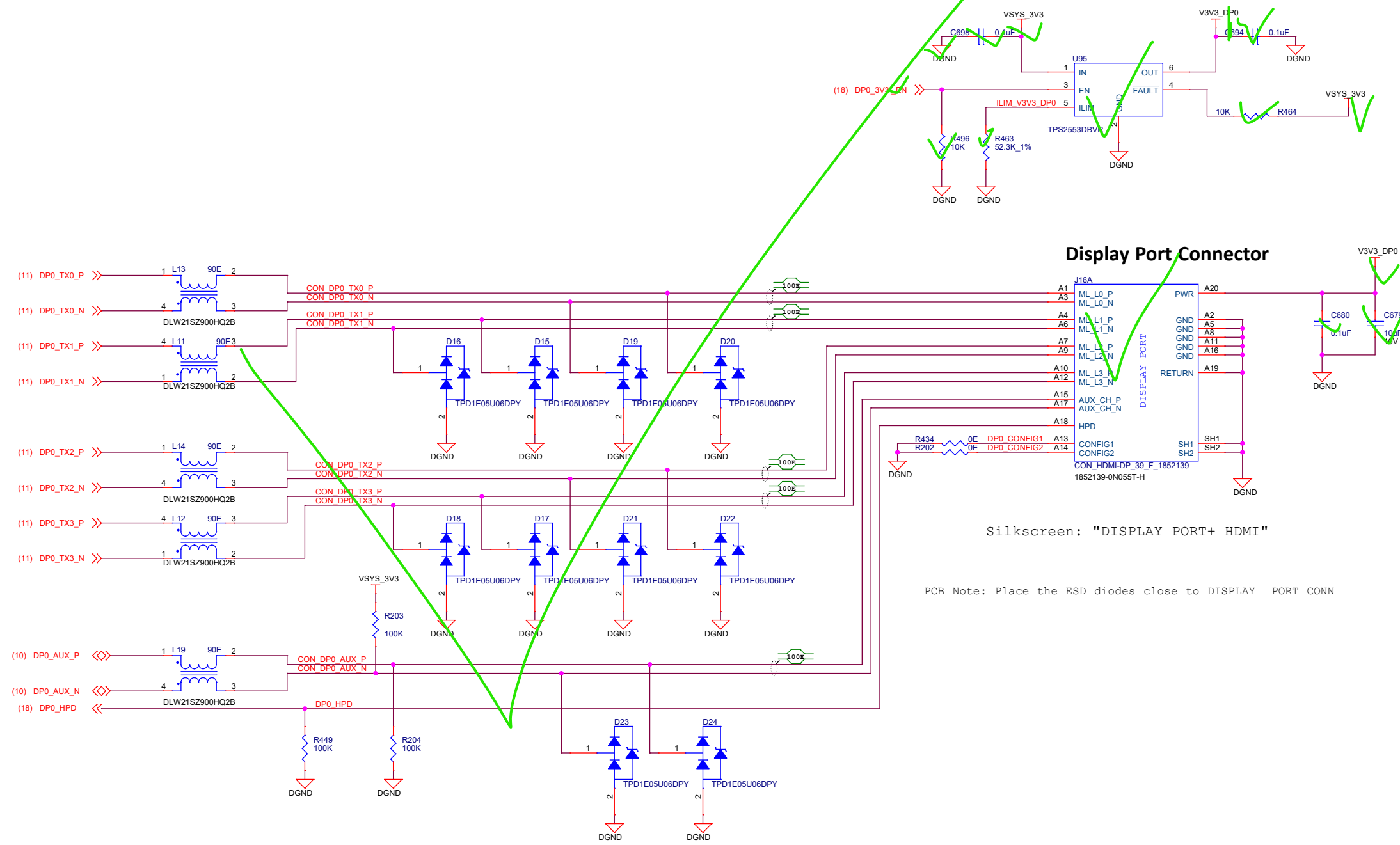


FPC Camera Connector -2



Silk Screen "CAM2"

DISPLAY PORT INTERFACE

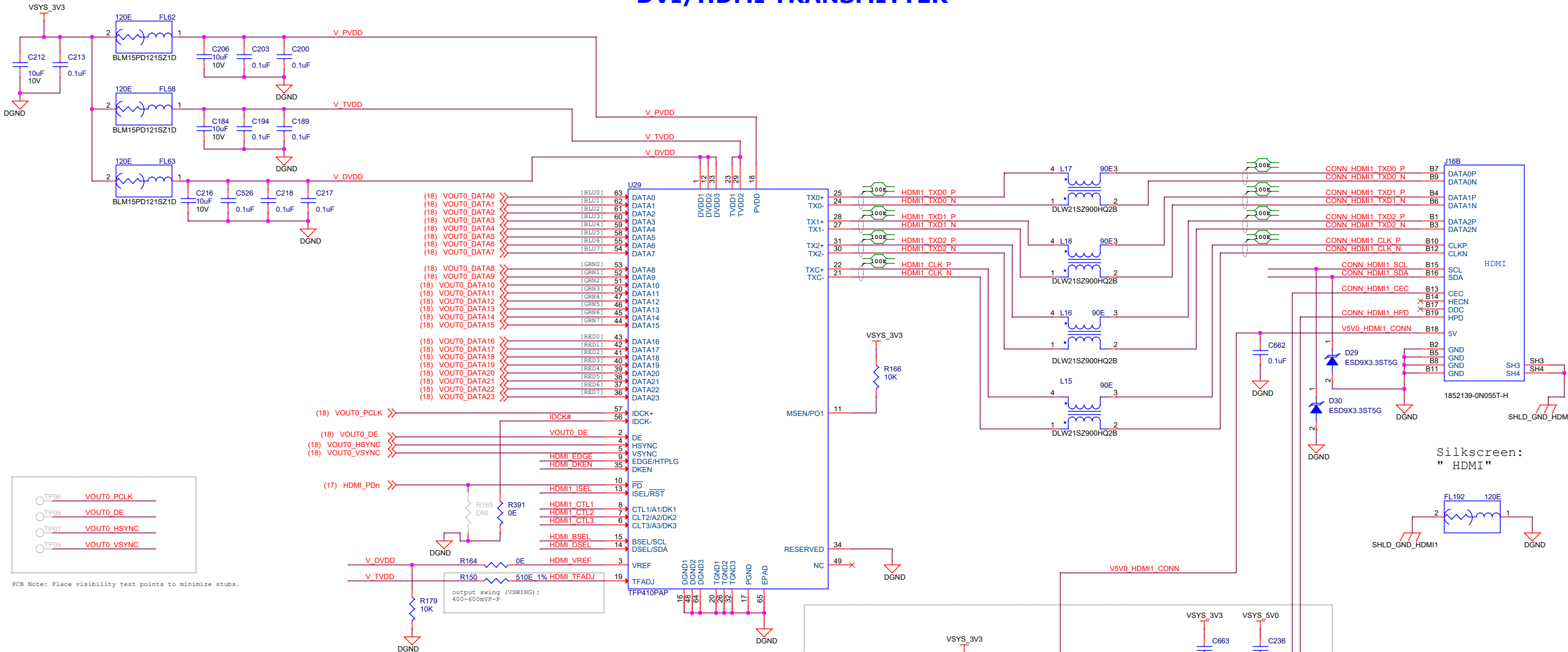


Display Port Connector

Silkscreen: "DISPLAY PORT+ HDMI"

PCB Note: Place the ESD diodes close to DISPLAY PORT CONN

DVI/HDMI TRANSMITTER



DVI Configuration Settings

VSYS_3V3

R140 10K

R183 10K

R168 10K

HDMI_BSEL

HDMI_EDGE

HDMI_DKEN

HDMI_DSEL

DGND

VSYS_3V3

R163 10K

HDMI1_CTL3

HDMI1_CTL1

HDMI1_CTL2

DGND

VSYS_3V3

R174 DNI

HDMI1_ISEL

R185 10K

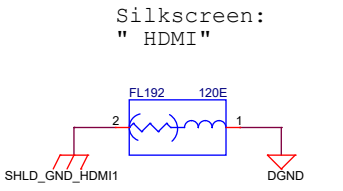
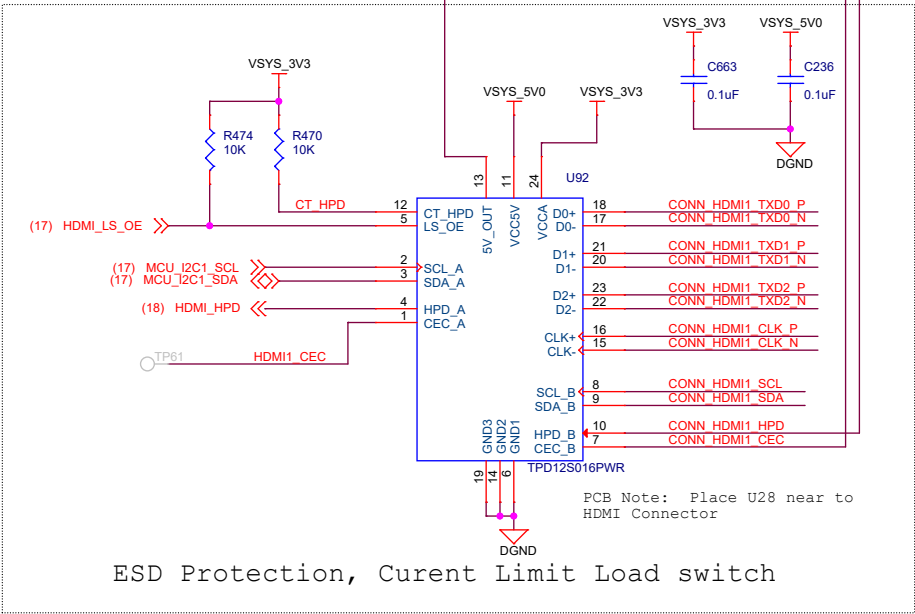
C223 1uF

DGND

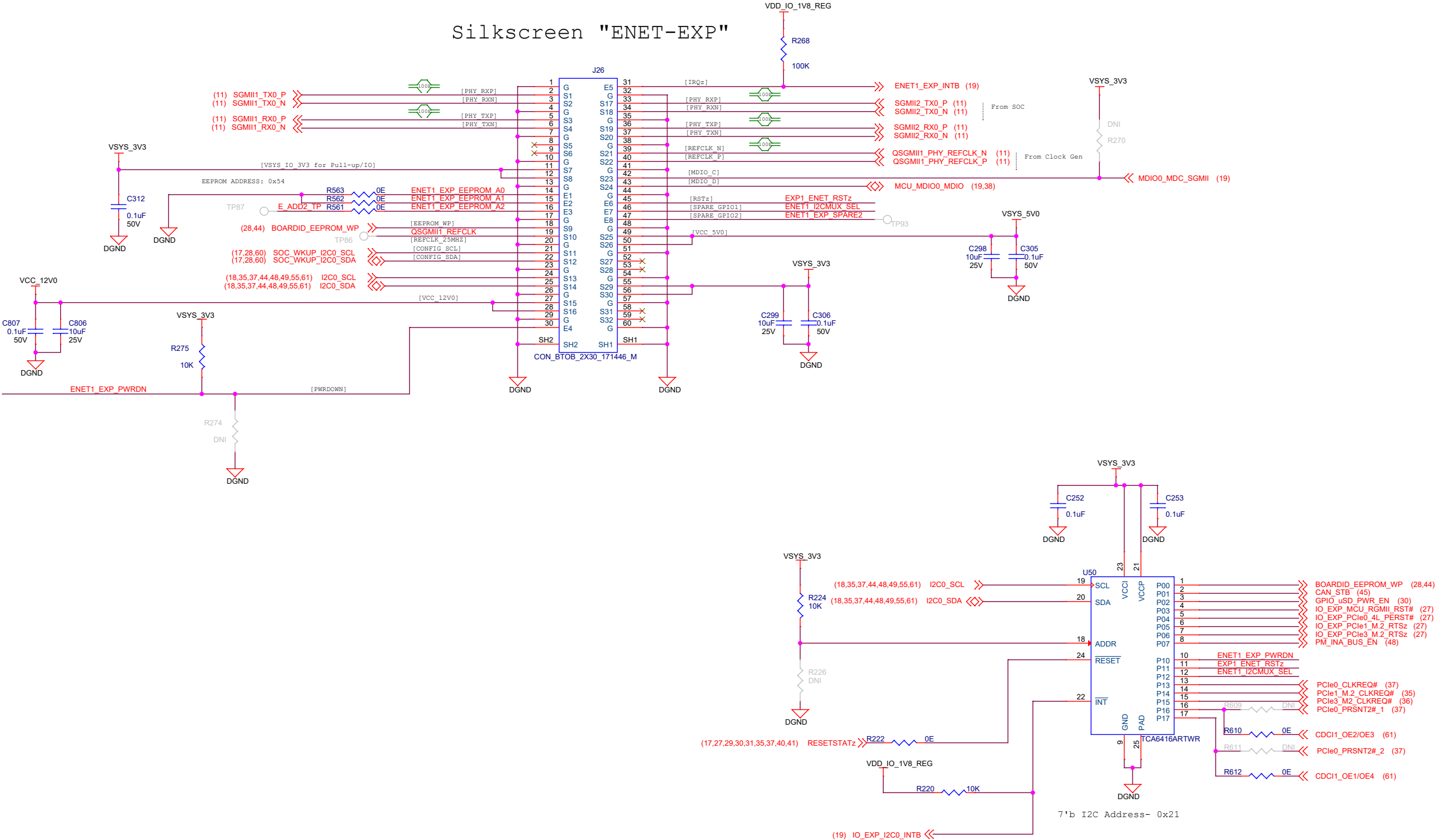
VREF	BSEL	EDGE	DSEL	BUS WIDTH	LATCH MODE	CLOCK MODE	CLOCK EDGE
0.55V-0.9V	1	0	0	24-bit	Single-ended	Falling	Single-ended
Default	1	1	0	24-bit	Single-ended	Raising	Single-ended

ISEL:- Low (default): I2C interface is disabled and chip configuration is specified by BSEL, DSEL, EDGE, VREF pins

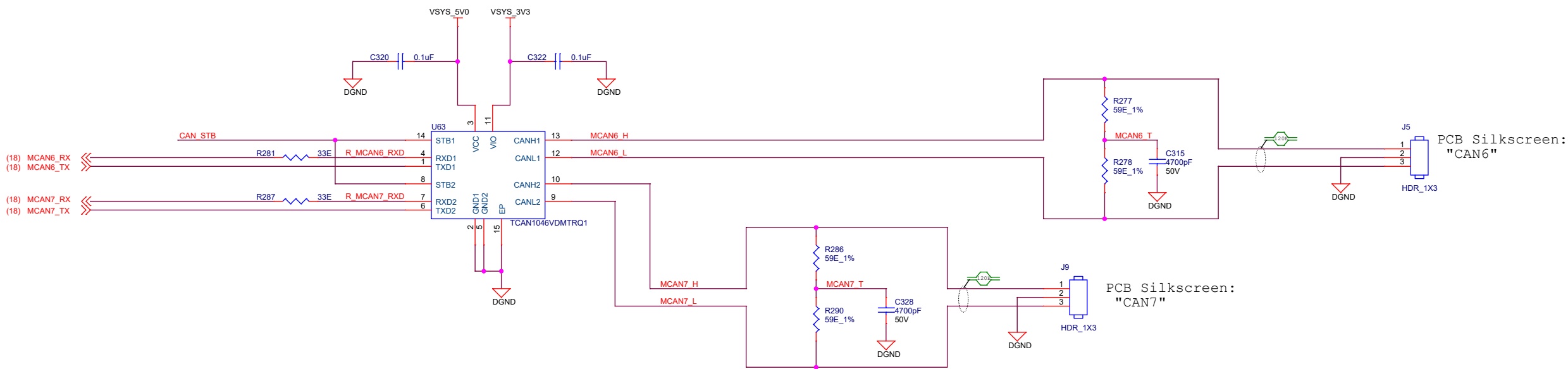
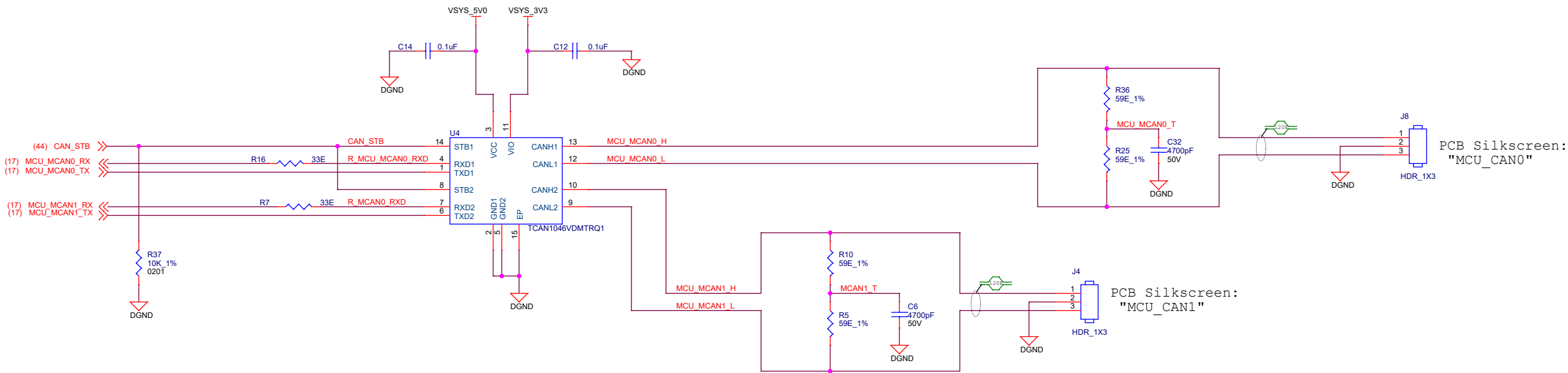
When ISEL: L, DSEL-H- enables de-skew function (default)



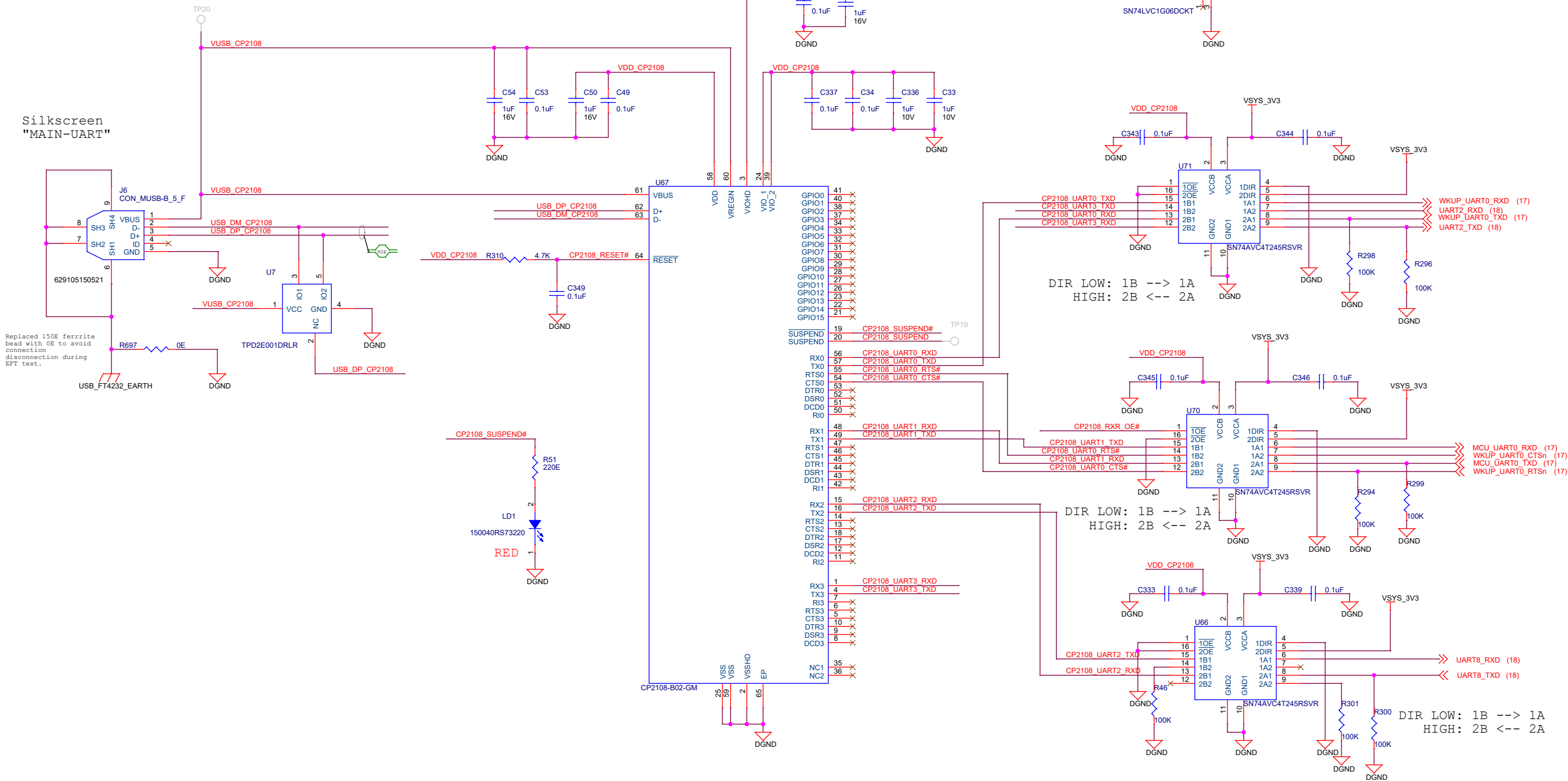
ENET EXPANSION CONNECTOR



CAN TRANSCEIVERS



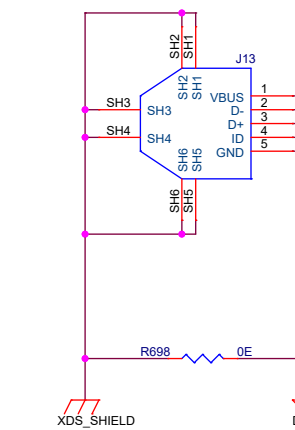
QUAD PORT CONSOLE



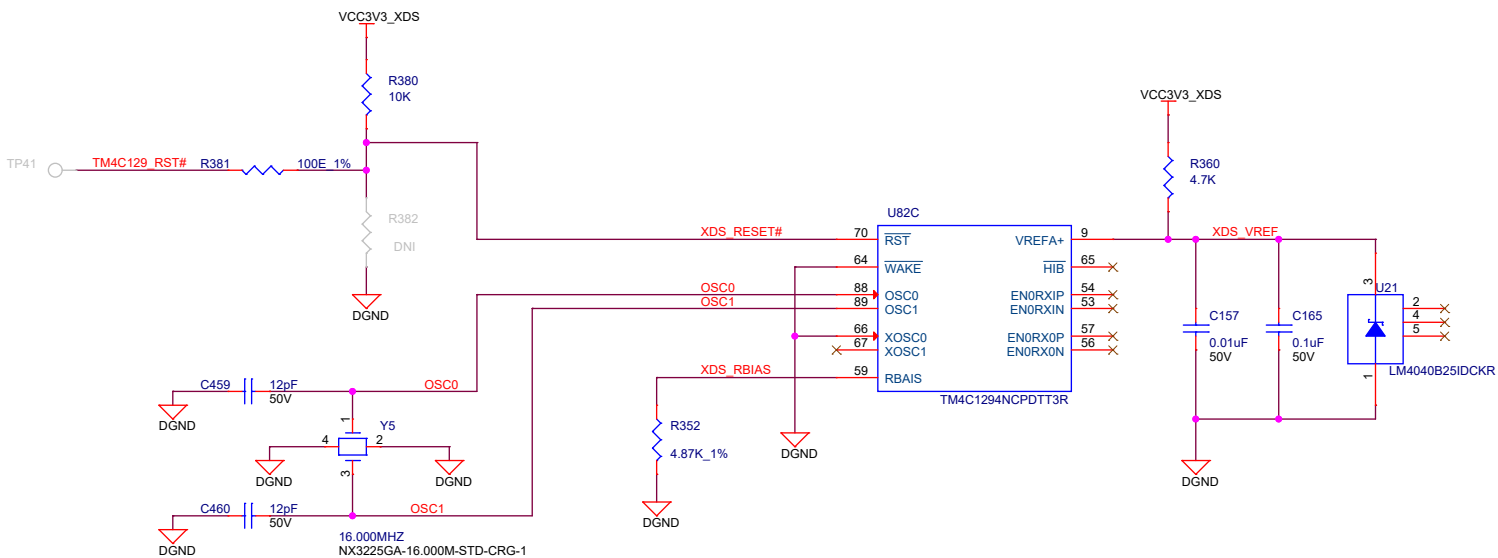
XDS110 DEBUGGER

Silkscreen "XDS110"

1051640001
CON MICRO USB-B TYPE 5POS FEMALE RT SMD



- | | | | | |
|--------------------------|-------|------|------|-----------------------|
| (26,48) TA_POWERDOWNz | << 0E | R439 | 0201 | XDS TA_POWERDOWNz |
| (27,48) TA_PORz | << 0E | R444 | 0201 | XDS TA_PORz |
| (27,48) TA_RESETz | << 0E | R448 | 0201 | XDS TA_RESETz |
| (19,48) TA_SOC_INT1z | << 0E | R454 | 0201 | XDS TA_SOC_INT1z |
| (19,48) TA_SOC_INT2z | << 0E | R460 | 0201 | XDS TA_SOC_INT2z |
| (48) TA_BOOTMODE_CNTL# | << 0E | R467 | 0201 | XDS TA_BOOTMODE_CNTL# |
| (27,48) TA_BM_IOEXP_RSTn | << 0E | R473 | 0201 | XDS TA_BM_IOEXP_RSTn |



- | | | |
|-------------------|-------|---------------|
| (20) XDS110_TCK | << 33 | PA0/U0RX |
| (20) XDS110_TMS | << 34 | PA1/U0TX |
| (20) XDS110_TDO | << 36 | PA2/SSIOCLK |
| (20) XDS110_TDI | << 37 | PA3/SSIOFSS |
| (20) XDS110_TRST# | << 40 | PA4/SSIOXDAT0 |

- | | | |
|------|-----|---------------|
| TP34 | 100 | PC0/TCK/SWCLK |
| TP35 | 99 | PC1/TMS/SWDIO |
| TP36 | 98 | PC2/TDI |
| TP38 | 97 | PC3/TDO/SWO |

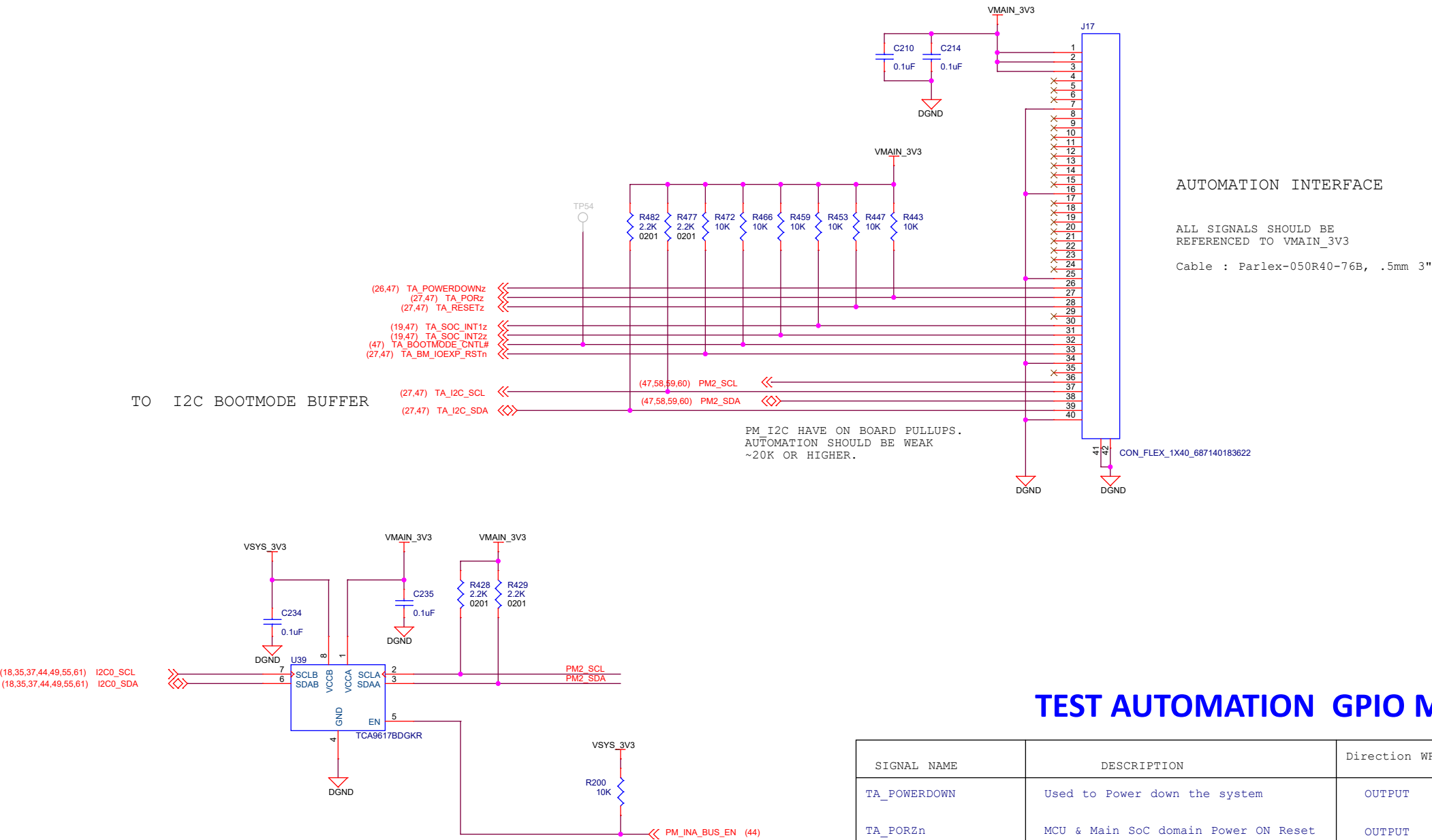
- | | | |
|-------------|----|-----|
| XDS PM1_SCL | 49 | PG0 |
| XDS PM1_SDA | 50 | PG1 |
| XDS PM2_SDA | 81 | PL0 |
| XDS PM2_SCL | 82 | PL1 |

- | | | |
|-------------|----|------------|
| XDS USB_D_P | 94 | PL6/USB0DP |
| XDS USB_D_N | 93 | PL7/USB0DM |

- | | | |
|-------------------|----|-----------|
| XDS110_VBUS_DET | 1 | PB5/AIN11 |
| XDS110_PROG_STA22 | 2 | PD0/AIN15 |
| XDS110_PROG_STA21 | 3 | PD1/AIN14 |
| XDS110_EMU0 | 42 | PF0 |
| XDS110_EMU1 | 43 | PF1 |

Set the unique ID
of the debugger

TEST AUTOMATION HEADER

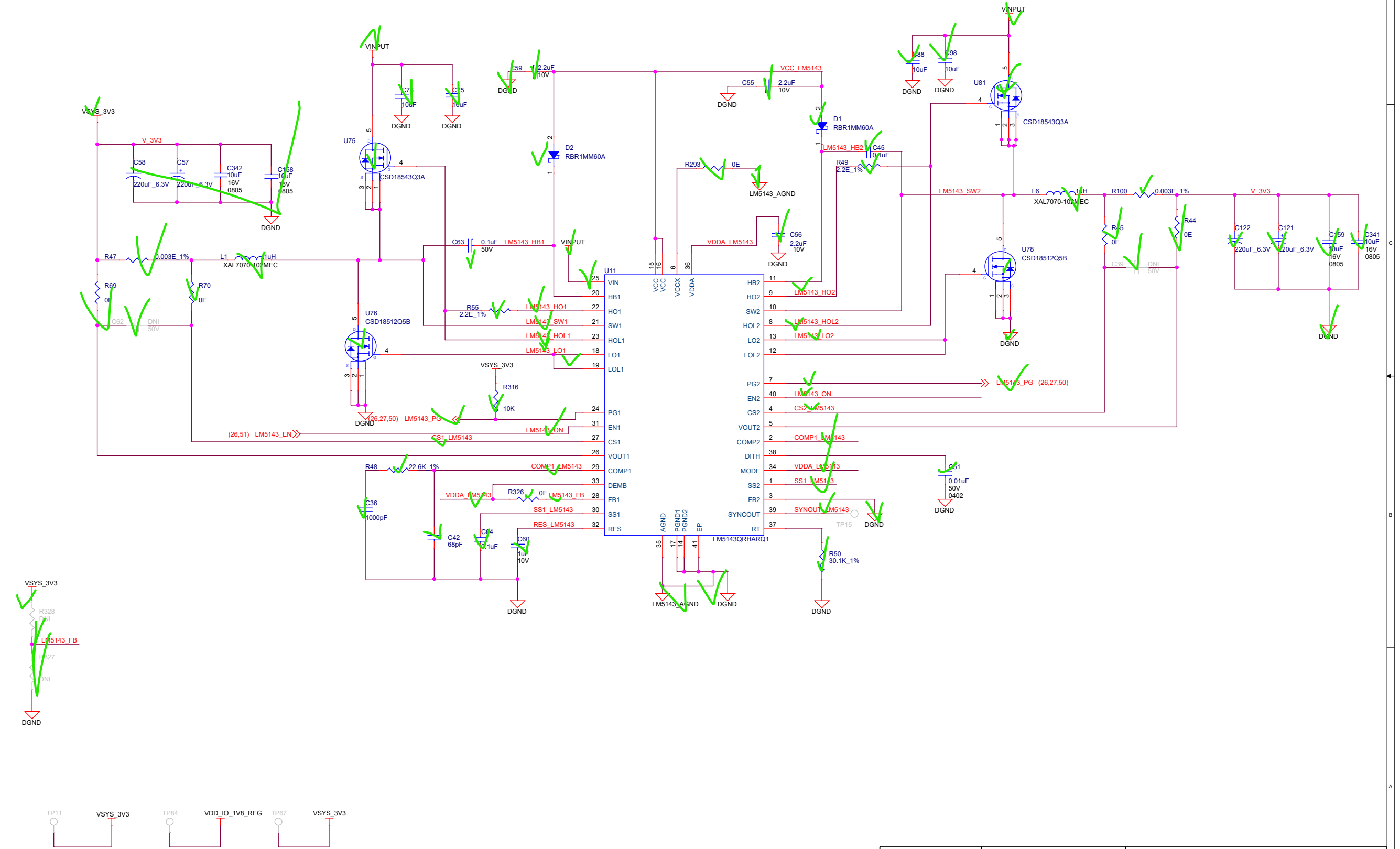


TEST AUTOMATION GPIO MAPPING

SIGNAL NAME	DESCRIPTION	Direction WRT CTRL	Internal/ External PU/PD states
TA_POWERDOWN	Used to Power down the system	OUTPUT	External Pullup
TA_PORZn	MCU & Main SoC domain Power ON Reset	OUTPUT	External Pullup
TA_RESETz	SoC Warmreset	OUTPUT	External Pullup
TA_SOC_INT1z	Interrupt to SOC	OUTPUT	External Pullup
TA_SOC_INT2z	Interrupt to SOC	OUTPUT	External Pullup
TA_BM_IOEXP_RSTn	Used to Reset the Bootmode IO Expander	OUTPUT	External Pullup

POWER SUPPLY #1

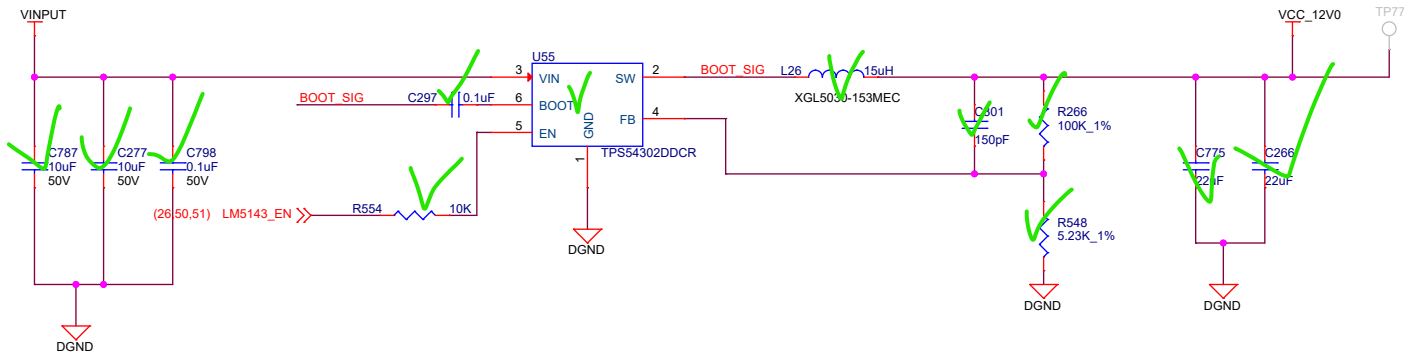
TI WEBENCH Simulation Inputs:
Vin (min) = 15V Vin (max) = 25V
Vout = 3.3V@30A
Ta = 25 deg



POWER SUPPLY #2

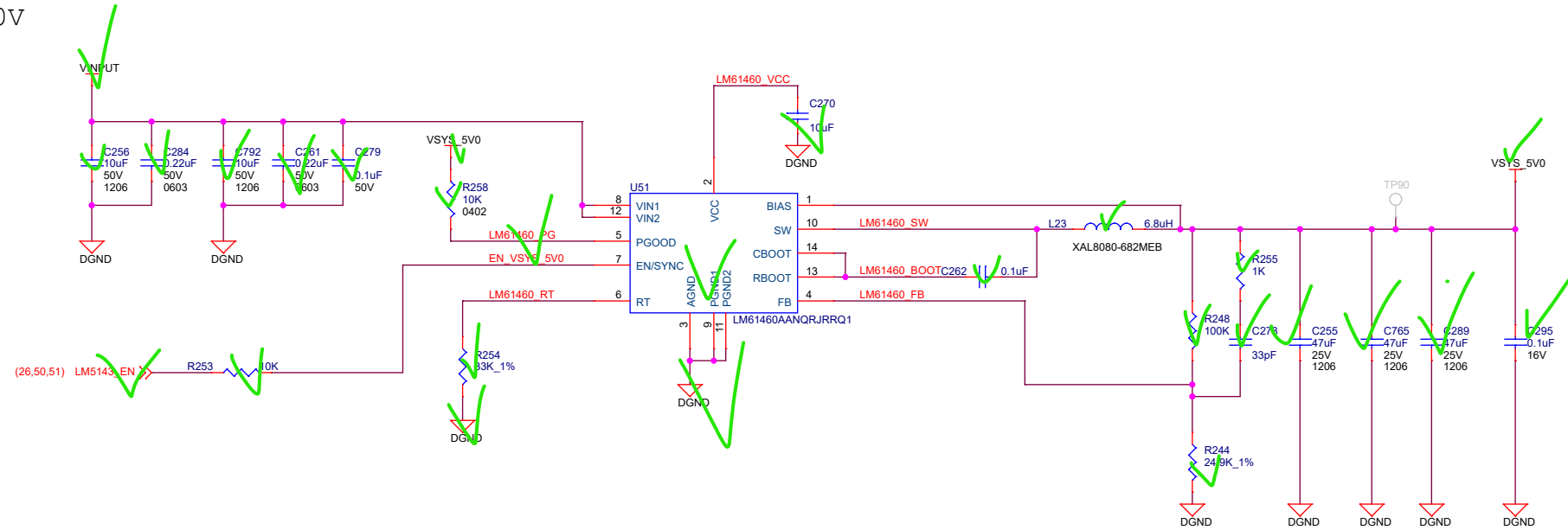
TI WEBENCH Simulation Inputs:
Vin (min) = 15V Vin (max) = 25V
Vout = 12V@3A
Ta = 25 deg

12V GENERATION



LM61460 5V BUCK REGULATOR
VinMin = 12V
VinMax = 25V
Vout = 5.0V
Iout = 6A

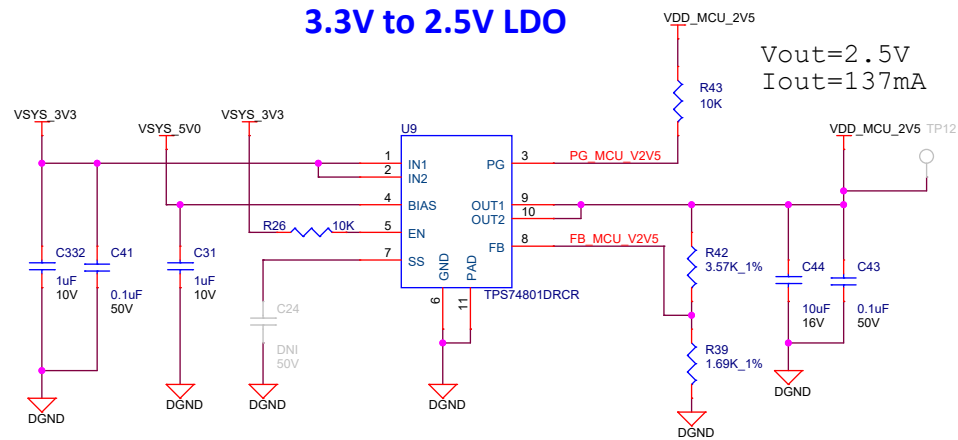
5V GENERATION



POWER SUPPLY #3

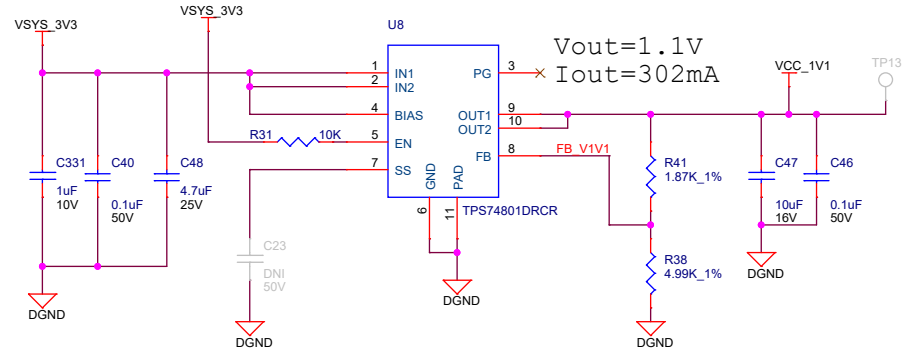
ETHERNET POWER- MCU RGMII

3.3V to 2.5V LDO



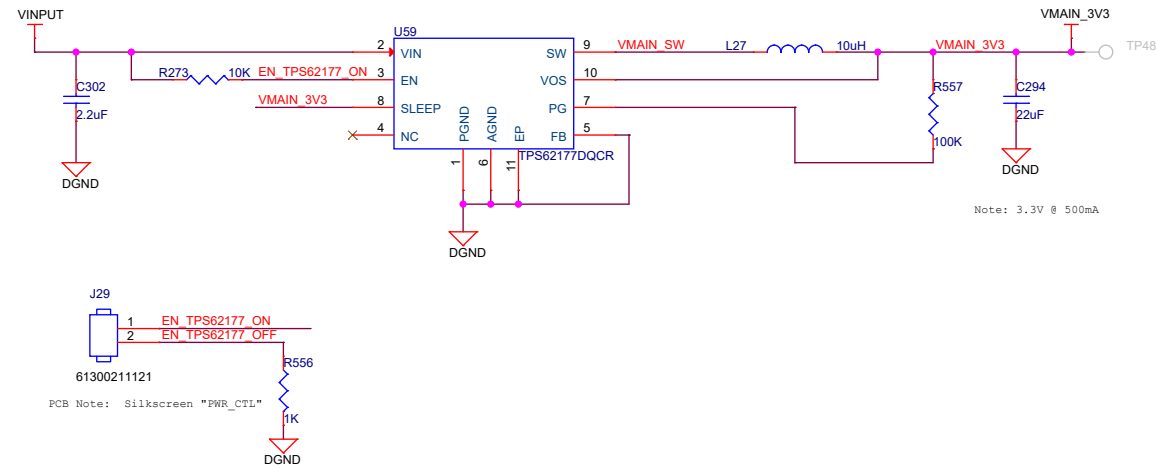
USB HUB POWER & ETHERNET POWER - RGMII1

3.3V to 1.1V LDO

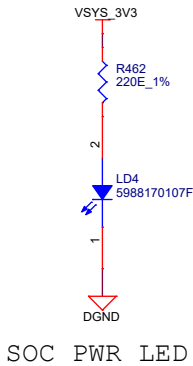


PCB NOTE:Keep 4.7uF capacitor close to BIAS pin

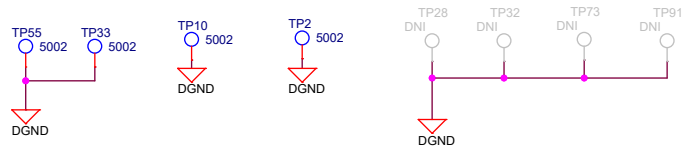
SYSTEM MANAGEMENT 3.3V REGULATOR



POWER INDICATION LED's



GROUND TEST POINTS



PCB NOTE: Spread the SMD test points Top and Bottom Side of PCB

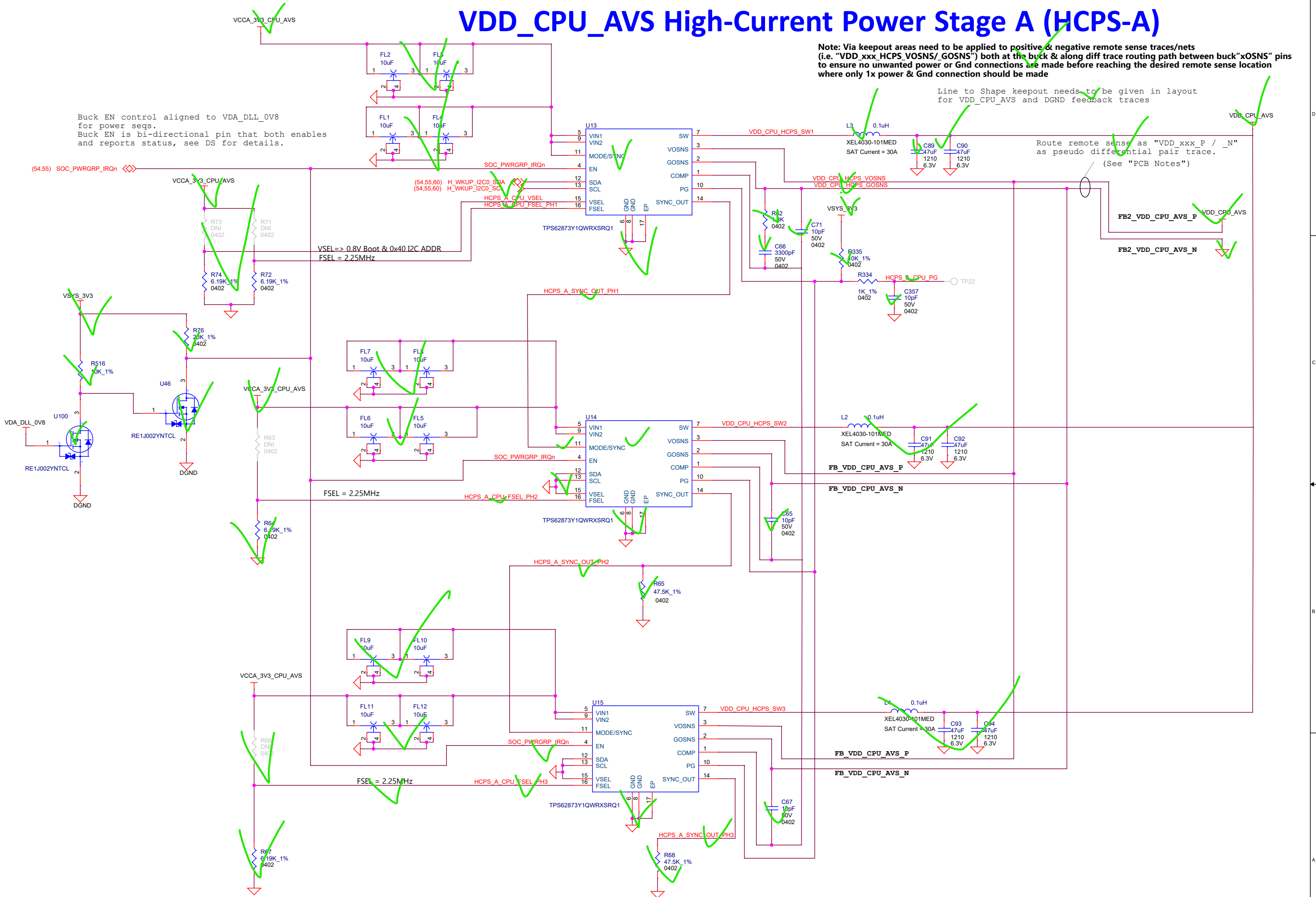
VDD_CPU_AVS High-Current Power Stage A (HCPS-A)

Note: Via keepout areas need to be applied to positive & negative remote sense traces/nets (i.e. "VDD_XXX_HCPS_VOSNS/"_GOSNS") both at the buck & along diff trace routing path between buck "xOSNS" pins to ensure no unwanted power or Gnd connections are made before reaching the desired remote sense location where only 1x power & Gnd connection should be made

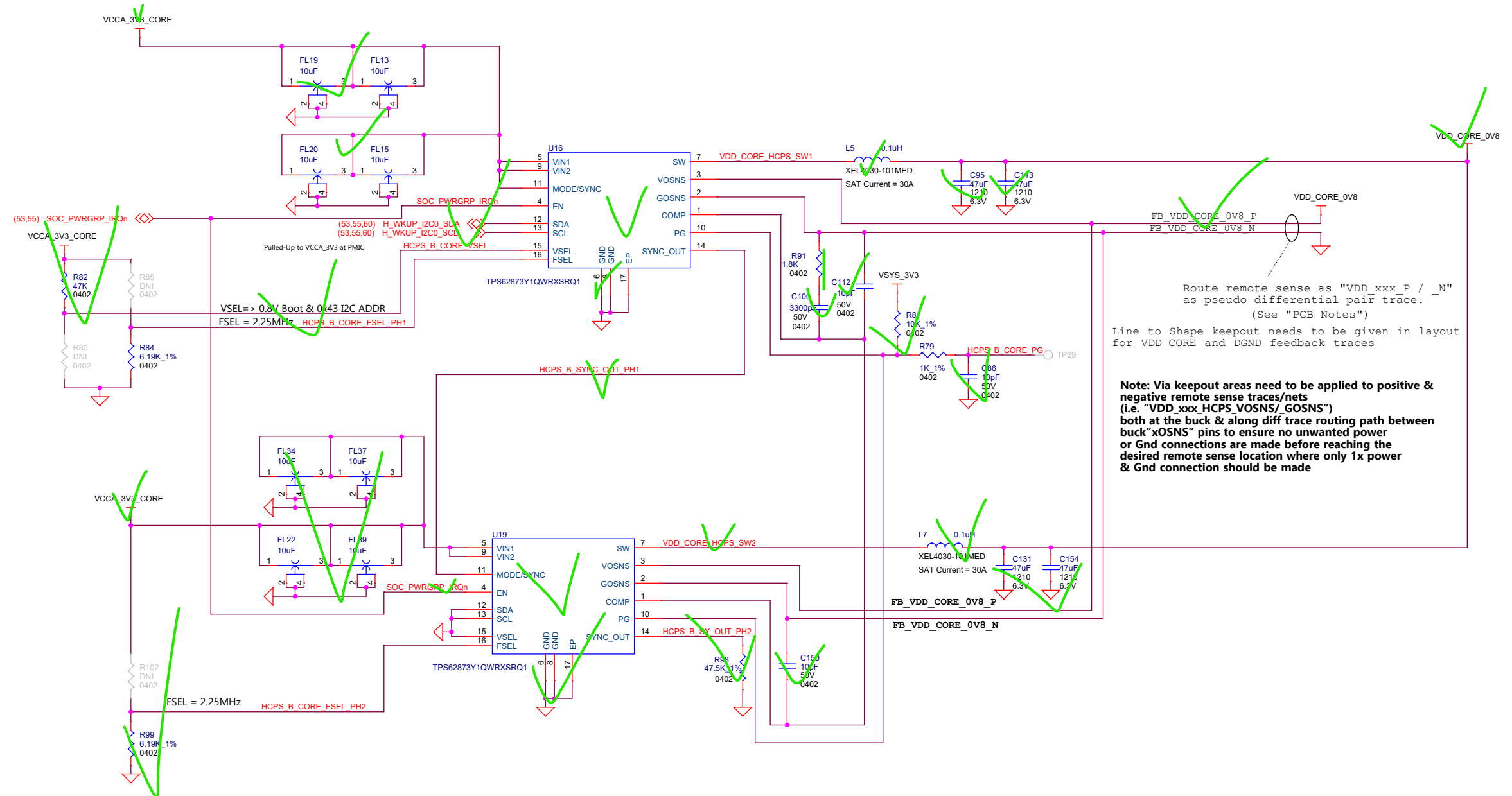
Buck EN control aligned to VDA_DLL_0V8 for power seqs.
Buck EN is bi-directional pin that both enables and reports status, see DS for details.

Line to Shape keepout needs to be given in layout for VDD_CPU_AVS and DGND feedback traces

Route remote sense as "VDD_XXX_P / _N" as pseudo differential pair trace. (See "PCB Notes")



VDD_CORE_0V8 High-Current Power Stage A (HCPS-B)

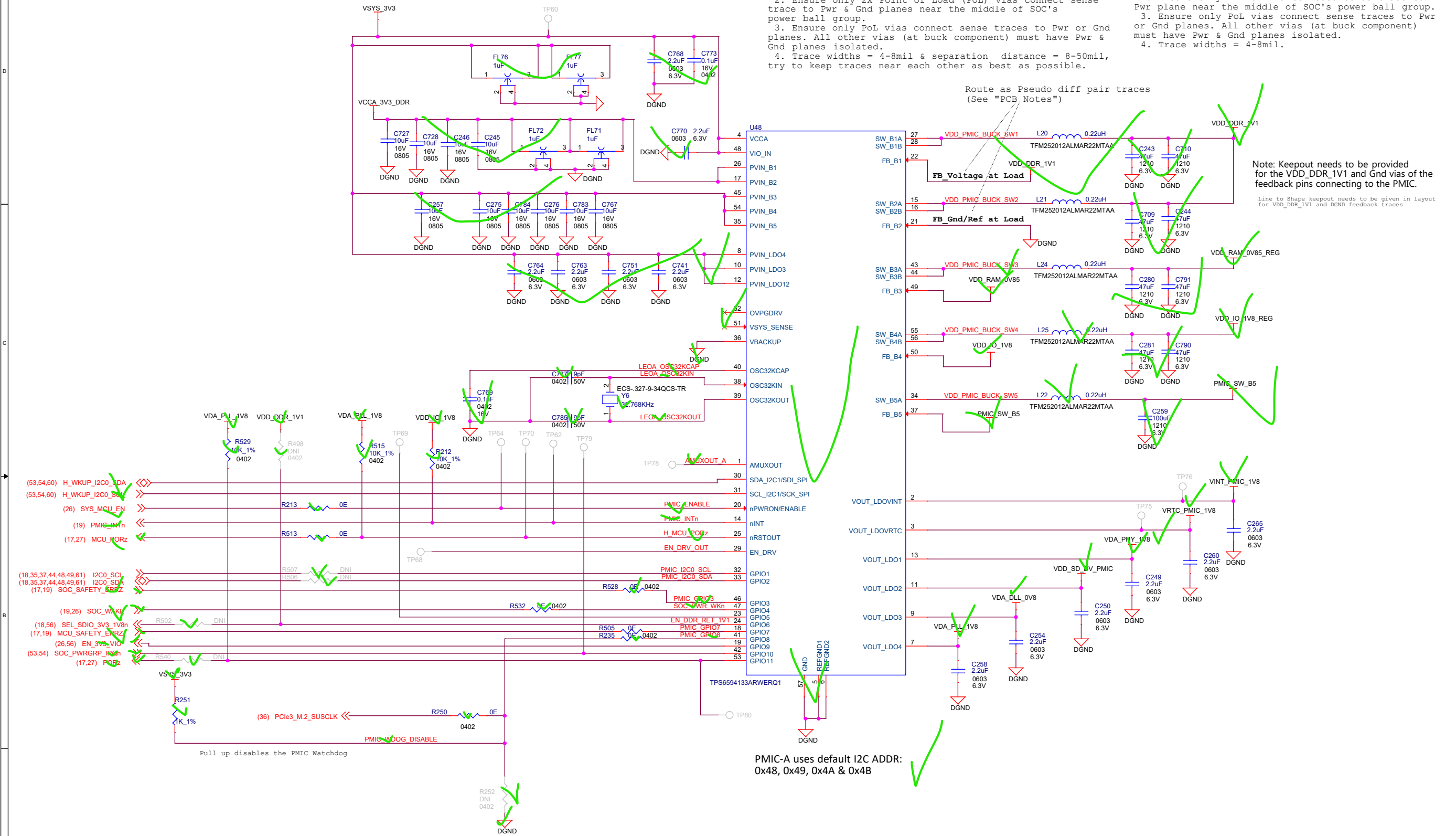


Note: Via knockout areas need to be applied to positive & negative remote sense traces/nets (i.e. "VDD_XXX_HPCS_VOSNS"/_GOSNS") both at the buck & along diff trace routing path between buck"xOSNS" pins to ensure no unwanted power or Gnd connections are made before reaching the desired remote sense location where only 1x power & Gnd connection should be made

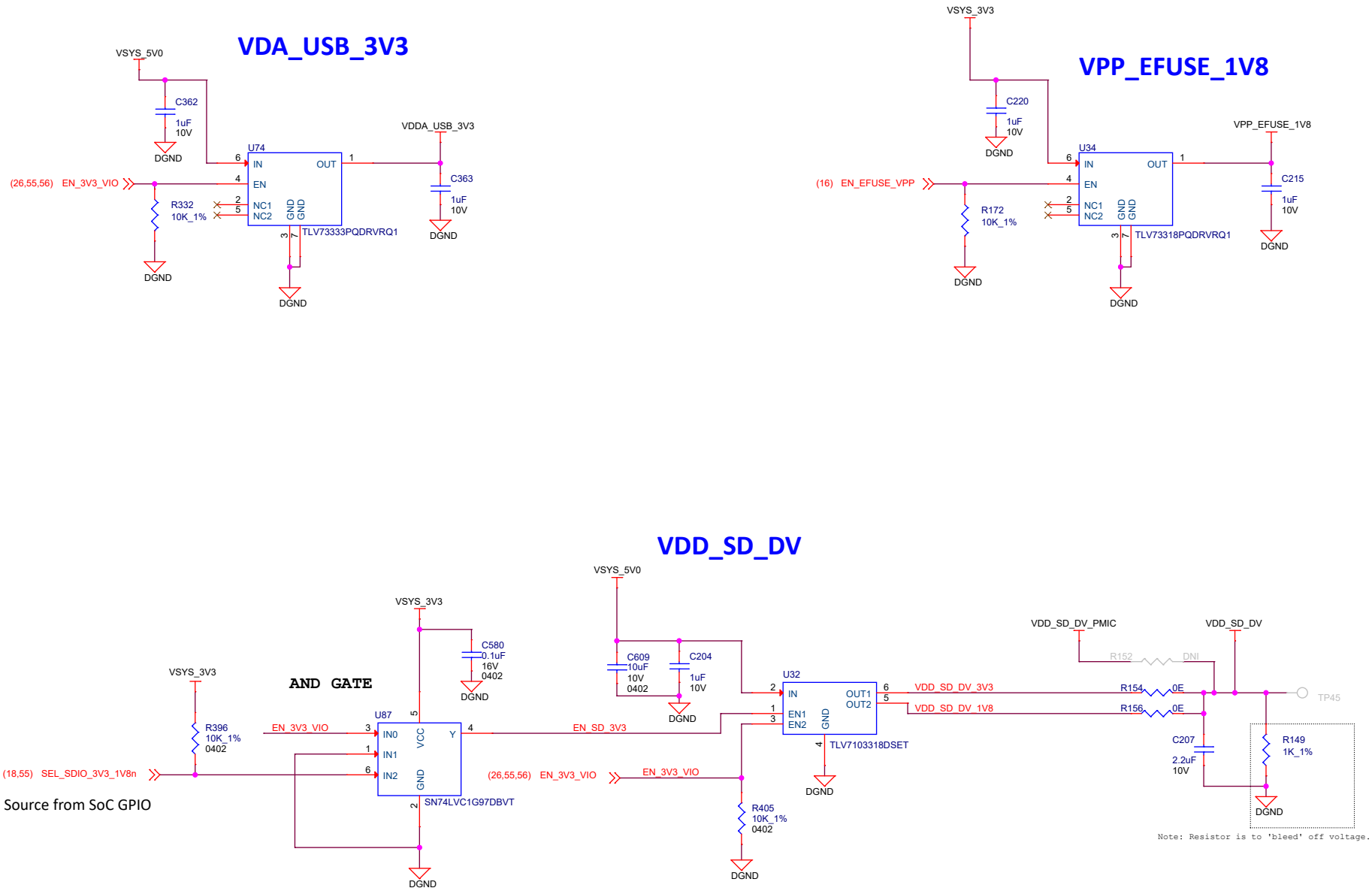
PMIC

"PCB Notes":
For multi-phase Buck converter configs, route remote sense feedback as follows:
1. Use pseudo differential pair traces on same layer & next to primarily power plane segment. Avoid routing near to any noisy/switching signals.
2. Ensure only 2x Point of Load (PoL) vias connect sense trace to Pwr & Gnd planes near the middle of SOC's power ball group.
3. Ensure only PoL vias connect sense traces to Pwr or Gnd planes. All other vias (at buck component) must have Pwr & Gnd planes isolated.
4. Trace widths = 4-8mil & separation distance = 8-50mil, try to keep traces near each other as best as possible.

For single-phase Buck converter configs, route remote sense feedback as follows:
1. Use single-ended traces on same layer & next to primarily power plane segment as best as possible. Avoid routing near to any noisy/switching signals.
2. Ensure only 1x PoL via connects sense trace to Pwr plane near the middle of SOC's power ball group.
3. Ensure only PoL vias connect sense traces to Pwr or Gnd planes. All other vias (at buck component) must have Pwr & Gnd planes isolated.
4. Trace widths = 4-8mil.

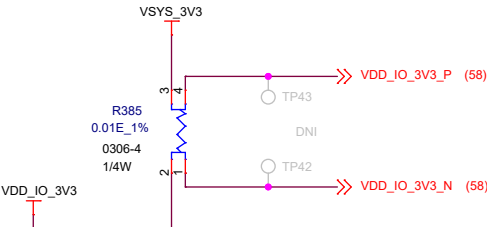
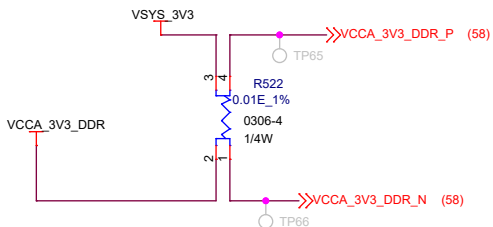
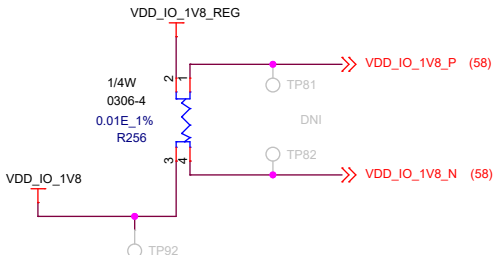
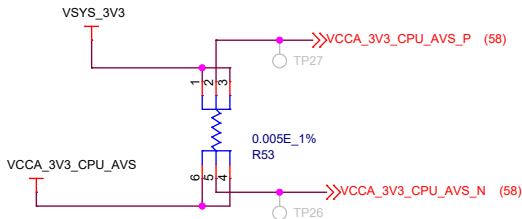
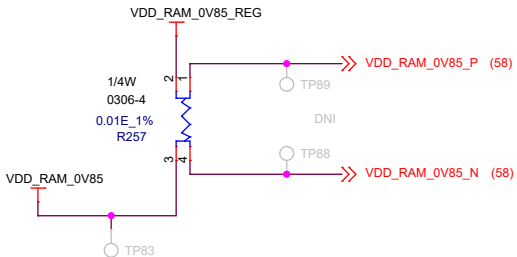
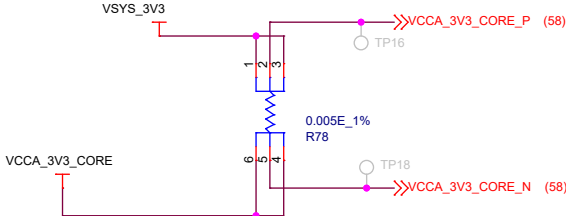


LDOs

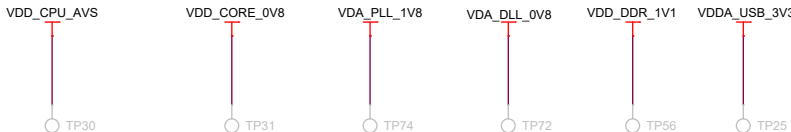


SOC Current Sense Resistors

CORE, AVS and DDR input supply sense resistors



PCB Note: Place all SMT TPs
on PCB top-side & on top of via at
Bd-to-Bd connector



Project :
AM69 Edge AI Kit



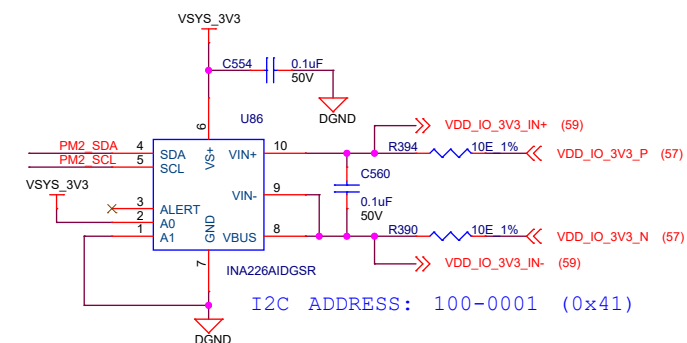
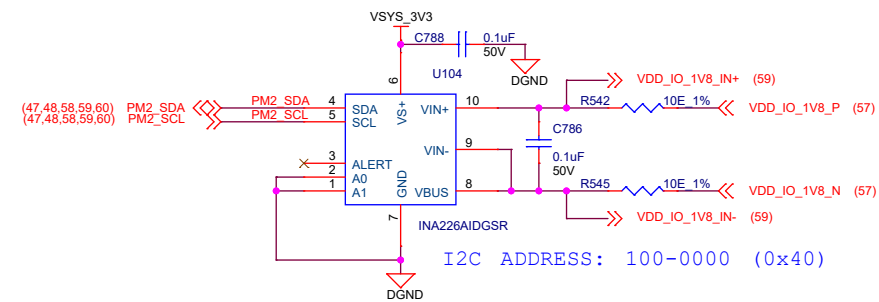
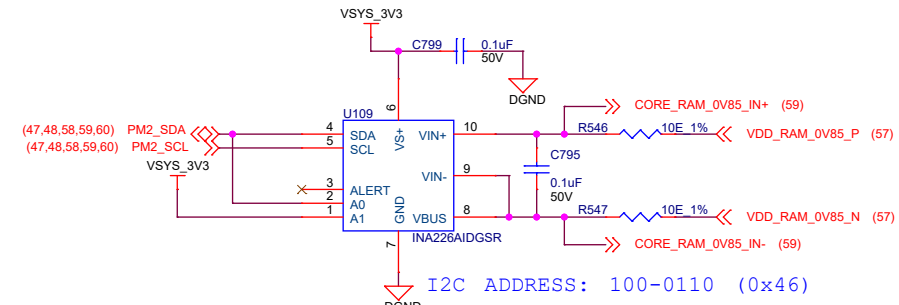
Title	SOC - CURRENT SENSE RESISTORS
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Size	PROC154E3 001 SK AM69
C	

Rev
E3

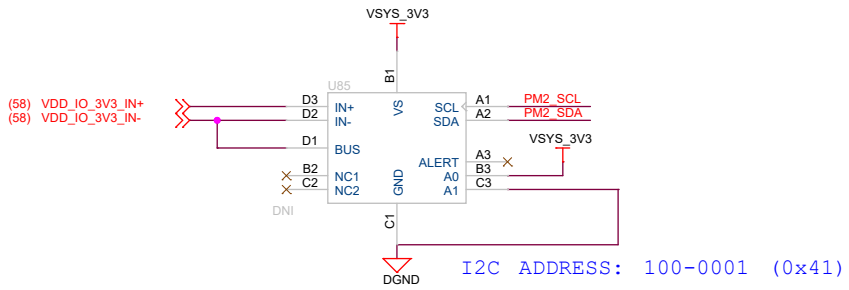
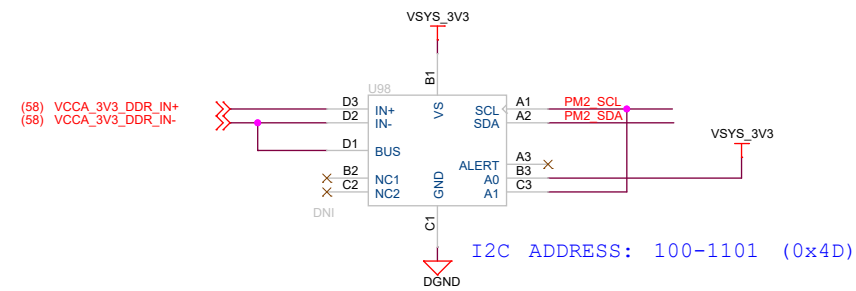
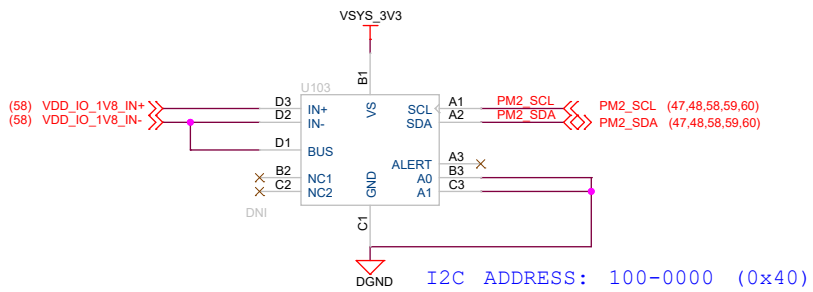
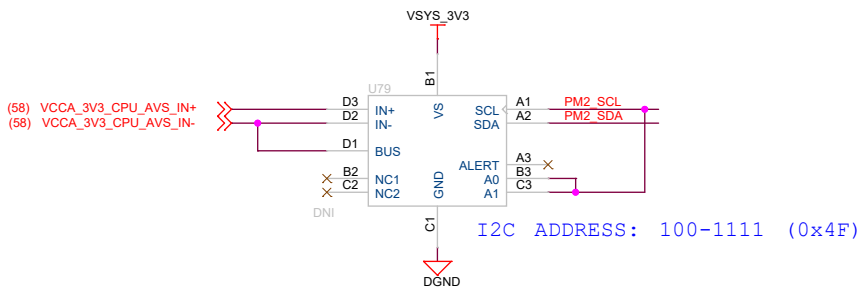
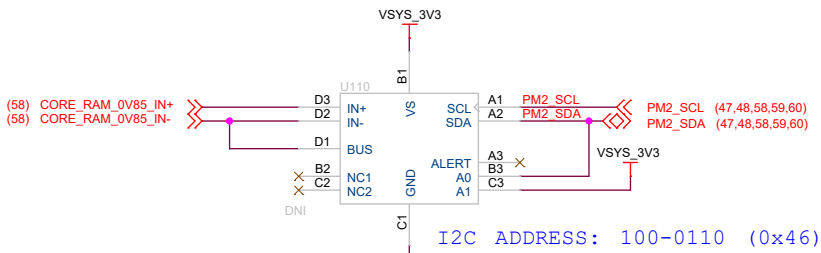
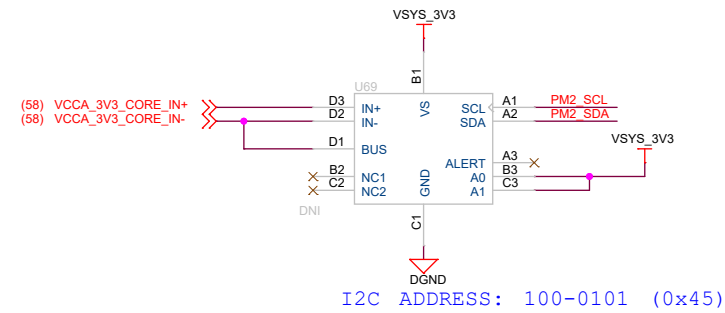
Date: Monday, March 20, 2023	Sheet 57 of 62
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CURRENT MONITORS - INA226

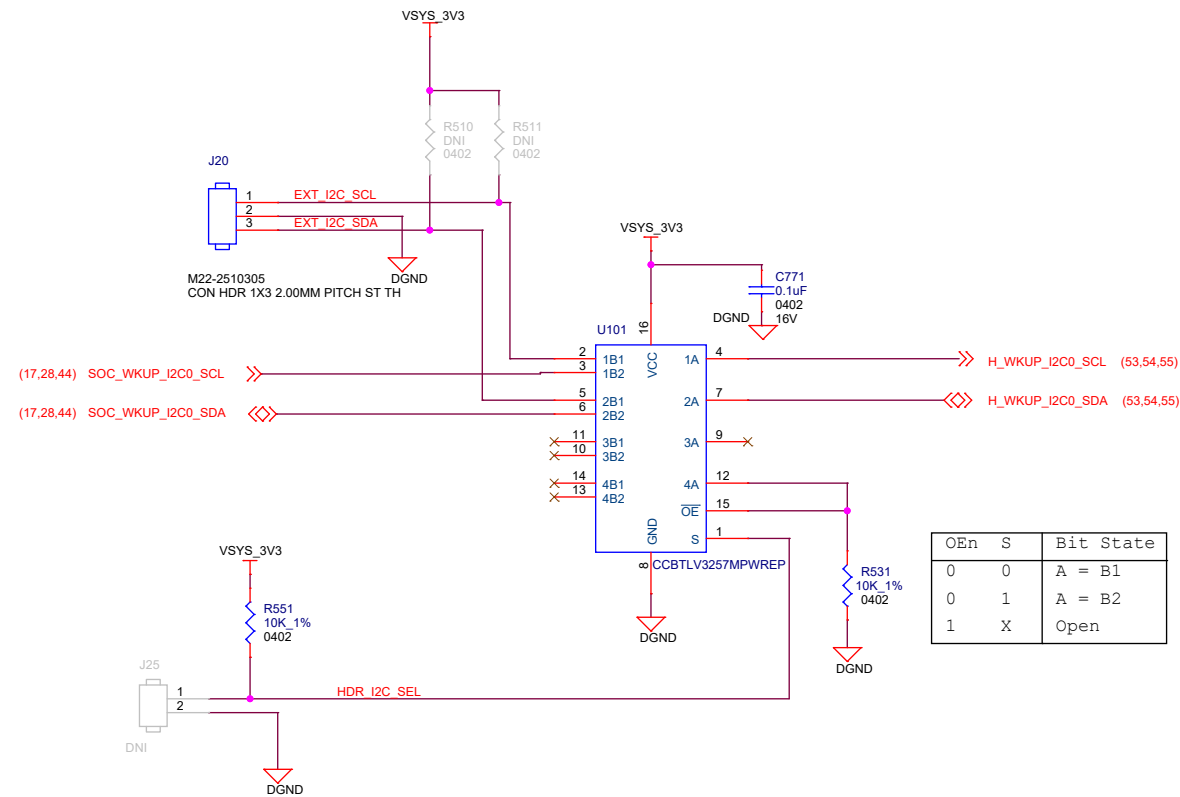


Note: The design supports current/voltage measurements using either INA226 or INA231. The SK will be assembled with either INA226 or INA231, but not both (implemented via dual or stacked PCB footprint). These two INA devices are register compatible- so functionality and performance should not be impacted with either INA

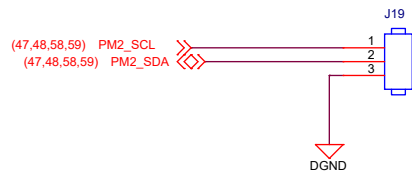
CURRENT MONITORS - INA231



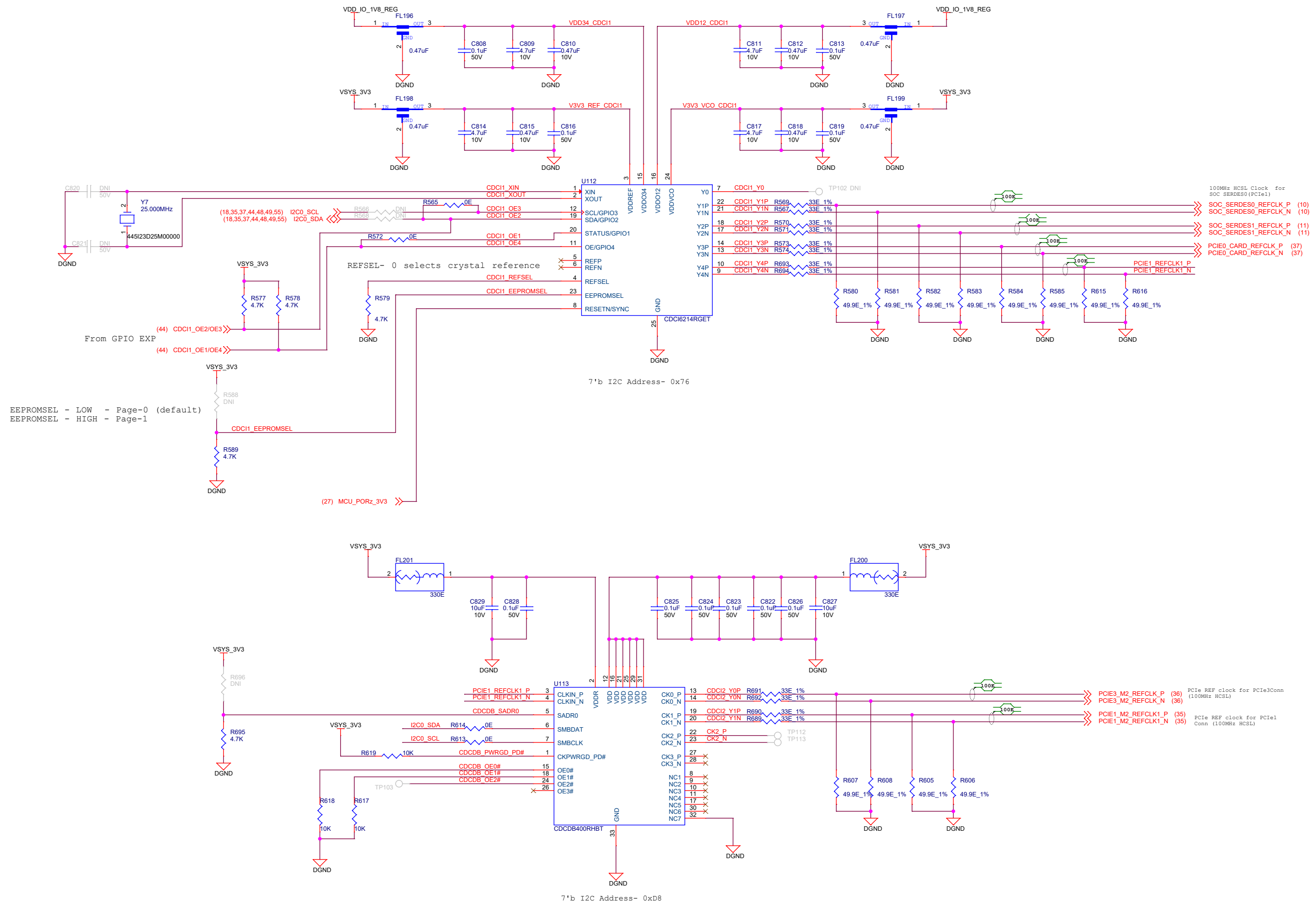
PMIC Support Circuit



EXT POWER MEASUREMENT



SERDES CLOCK GENERATORS



Project :
AM69 Edge AI Kit



Title	SERDES CLOCK GENERATORS
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Size	PROC154E3 001 SK AM69	
C		
Date:	Monday, March 20, 2023	Sheet 61 of 62

NOTES, HW & LABELS

ASSEMBLY NOTES

- 1. All MSL components should be baked as per JEDEC standard.
- 2. PCB should be baked at 120 degree for 8 hours.
- 3. Board assembly must comply with workmanship standards. IPC-A-610 Class 2, unless otherwise specified.
- 4. These assemblies are ESD sensitive, ESD precautions shall be observed.
- 5. These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.
- 6. Provide serial numbers to the assembled boards for identification.
- 7. The assembled board are wrapped in ESD Covers(individual) and packed securely before shipment.

LABELS

Board Serial No.



AM6-COMPROCEVM

Assembly Revision.

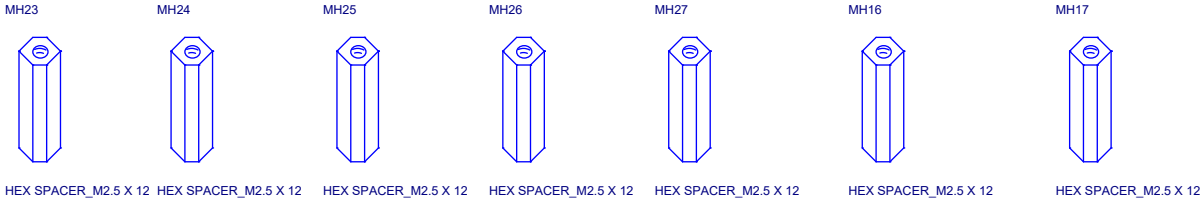


AM6-COMPROCEVM

SCREWS



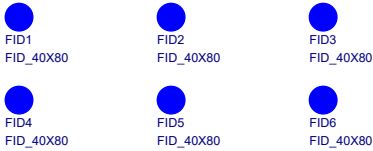
STANDOFFs



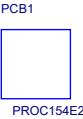
WASHER



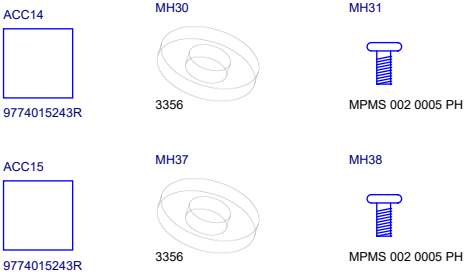
FIDUCIALS



BARE PCB



SCREW & WASHER FOR PCIe M.2



LOGOs



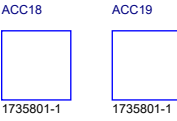
HEAT SINK



FAN



CRIMP PIN



CONN HOUSING



SCREW FOR FAN ASSEMBLY

